

United Nations Development Programme

Project Title: Enabling Zero Carbon Energy in Rural Towns and Villages in China (EZCERTV) Project		
Country(ies): P.R. China	**Implementing Partner (GEF Executing Entity):** In China – Ministry of Agriculture and Rural Affairs (MARA)	**Execution Modality**: Full National Implementation Modality (NIM)
Contributing Outcome (UNDAF/CPD, RPD, GPD): UN Development Assistance Framework China (2016-2020): More people enjoy a cleaner, healthier, and safer environment as a result of improved environmental protection and sustainable green growth (Outcome 2). **UNDP Strategic Plan Output:** *Output 1.5.1:* Solutions adopted to achieve universal access to clean, affordable and sustainable energy. *Output 2.5.1:* Solutions developed, financed, and applied at scale for energy efficiency and transformation to clean energy and zero-carbon development, for poverty eradication and structural transformation		
UNDP Social and Environmental Screening Category: Substantial	**UNDP Gender Marker: GEN2**	
Atlas Award ID: 00129191	**Atlas Project/Output ID: 00122943**	
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Project duration in months: 60		
Planned start date: 15 July 2022	**Planned end date:** 14 June 2027	
Expected date of Mid-Term Review: 15 January 2025	**Expected date of Terminal evaluation:** 15 March 2027	
Brief project description: The objective of the EZCERTV Project is the acceleration of low/zero-carbon transformation in China's rural areas to contribute to global climate change mitigation and the achievement of the United Nations sustainable development goals (SDG). The project's overall strategy is the removal of barriers to the widespread application of low/zero carbon energy technologies in China's rural towns and villages. Component 1 of the project is comprised of technical assistance activities addressing the policy/regulatory and institutional barriers, and the expected outcome is the effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China. The technical barriers are addressed in Component 2 of the project, which is comprised of investment type activities that will demonstrate the application of feasible and cost-effective low/zero carbon energy technologies in selected rural		

towns/villages. The expected outcome from this is the widespread implementation of low/zero carbon energy technology applications in rural towns and villages. Component 3 addresses the barriers related to capacity development in the area of low/zero carbon energy technologies/techniques, as well as information gathering, management and dissemination in the design, engineering, installation, operation, repair, and maintenance of low/zero carbon energy technology applications. The enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China is the expected outcome from this.	
FINANCING PLAN	
GEF Trust Fund grant	USD 8,932,420
UNDP TRAC resources[1]	USD 0
Confirmed cash co-financing to be administered by UNDP	USD 0
(1) Total Budget administered by UNDP	**USD 8,932,420**
CO-FINANCIERS THAT WILL DELIVER PROJECT RESULTS INCLUDED IN THE PROJECT RESULTS FRAMEWORK (FUNDS NOT ADMINISTERED THROUGH UNDP ACCOUNTS)	
Ministry of Agriculture and Rural Affairs (Grant)	USD 13,200,000
Ministry of Agriculture and Rural Affairs (In-Kind)	USD 1,000,000
Department of Agricultural and Rural Affairs of Hebei Province (Grant)	USD 8,000,000
Department of Agricultural and Rural Affairs of Shanxi Province (Grant)	USD 5,000,000
Department of Agricultural and Rural Affairs of Liaoning Province (Grant)	USD 5,000,000
Department of Agricultural and Rural Affairs of Heilongjiang Province (Grant)	USD 5,000,000
Department of Agricultural and Rural Affairs of Hubei Province (Grant)	USD 5,000,000
Department of Agricultural and Rural Affairs of Yunnan Province (Grant)	USD 4,000,000
Rural Energy Workstations of Ningxia Hui Autonomous Region (Grant)	USD 4,950,000
Rural Energy Workstations of Ningxia Hui Autonomous Region (In-kind)	USD 50,000
China Association of Rural Energy Industry (Grant)	USD 5,000,000
Hebei Jing'an Bioenergy Technology Co., Ltd. (Grant)	USD 7,000,000
Shanxi Golden Cooperate DC Power Distribution Engineering Technology Co., Ltd. (Grant)	USD 10,000,000
Chaoyang Jiajiadian Farm Co., Ltd. (Grant)	USD 3,500,000
Hailun Limin Energy-saving Boiler Manufacturing Co., Ltd. (Grant)	USD 3,500,000
Songzi Senchun Agricultural Development Co., Ltd. (Grant)	USD 3,500,000
Yunan Fu Hui Technology Co., Ltd. (Grant)	USD 2,200,000
Ningxia Runwell Energy & Environment Engineering Co., Ltd. (Grant)	USD 3,500,000

[1] This is not a mandatory requirement.

United Nations Development Programme (In-kind)		USD 200,000
(2) Total confirmed co-financing		**USD 89,600,000**
(3) Grand-Total Project Financing (1)+(2)		**USD 98,532,420**
SIGNATURES:		
Signature: print name below	**Agreed by Government Development Coordination Authority**	**Date/Month/Year:** *within 25 days of GEF CEO endorsement*
Signature: print name below	**Agreed by Implementing Partner**	**Date/Month/Year:** *within 25 days of GEF CEO endorsement*
Signature: print name below	**Agreed by UNDP**	**Date/Month/Year:** *within 25 days of GEF CEO endorsement*

List of Abbreviations & Acronyms

ACWF	All China Women's Federation
BAU	Business As Usual
BPPS	Bureau for Policy and Programme Support
CFL	Compact Fluorescent Light
CO	Country Office
CO_2	Carbon Dioxide
CPC	Communist Party of China
CTA	Chief Technical Advisor
DSTE	Department of Science Technology and Education
EC	Energy Conservation
EE	Energy Efficiency
EIA	Energy Information Administration
ERC	Evaluation Resource Center
ESMF	Environmental Social Management Framework
EZCERTV	Enabling Zero Carbon Energy in Rural Towns and Villages in China
FL	Fluorescent Light
GDP	Gross Domestic Product
GEF	Global Environment Facility
GEFSEC	Global Environment Facility Secretariat
GHG	Greenhouse Gas
GOC	Government of China
ICE	Information Communication and Education
IMF	International Monetary Fund
IRR	Implementing Rules and Regulations
ktoe	kilo tons of oil equivalent
LED	Light Emitting Diode
LFA	Logical Framework Analysis
MARA	Ministry of Agriculture and Rural Affairs
MEE	Ministry of Ecology and Environment
MEPS	Minimum Energy Performance Standards
MOF	Ministry of Finance
MTR	Mid-Term Review
MW	Mega Watts
M&E	Monitoring and Evaluation
NDC	Nationally Determined Contributions
NDRC	National Development and Reform Commission
NGO	Non-Governmental Organization
NIM	National Implementation Modality
PIF	Project Identification Form
PIR	GEF Project Implementation Report
PM	Project Manager
PMO	Project Management Office
PMU	Project Management Unit
POPP	Programme and Operations Policies and Procedures

PPG	Project Preparation Grant
PV	Photo Voltaic
RE	Renewable Energy
REEA	Rural Energy and Environment Agency
RTA	Regional Technical Advisor
SDGs	Sustainable Development Goals
SESP	Social and Environmental Screening Procedure
tce	Tons of Coal Equivalent
tCO_2	Tons of Carbon Dioxide
TE	Terminal Evaluation
TJ	Tera Joule
ToC	Theory of Change
TOR	Terms of Reference
UNDP	United Nations Development Programme

I. Table of Contents

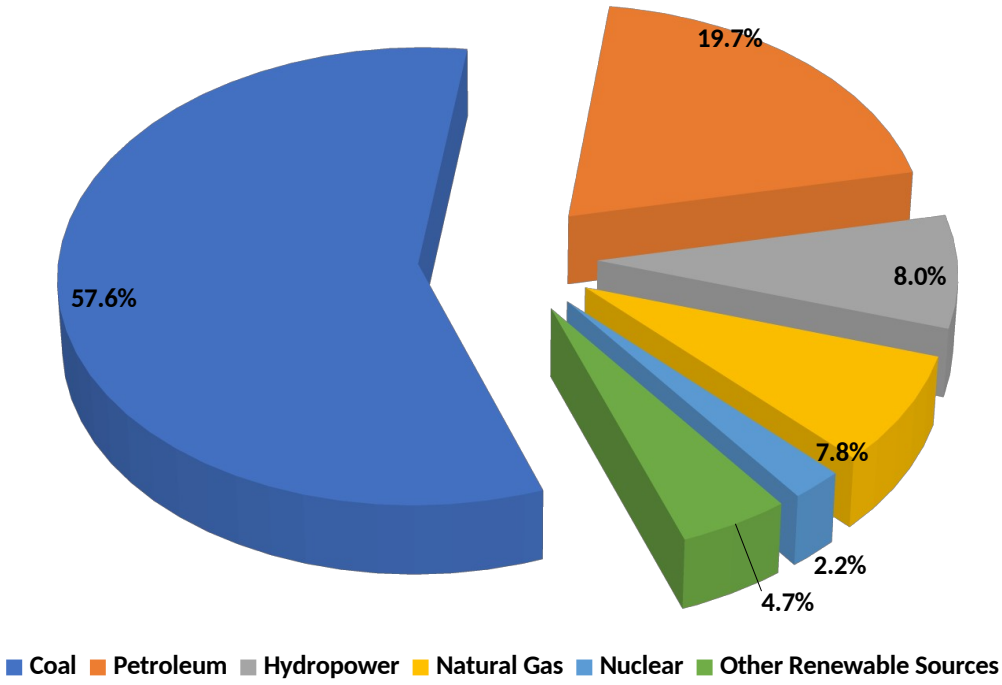
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II. Development Challenge

Development Challenge and Relevance to National Development Priorities, Global Environment and Sustainable Development Goals (SDGs)

China is one of the largest countries in the world, and the most populated with a population of approximately 1.4 billion people[2]. The gross domestic product (GDP) has grown from US$ 1.2 trillion in 2000 to US$ 14.3 trillion in 2019, measured in current US$[3]. This strong economic growth has been accompanied by an equally strong growth in energy consumption. According to the Energy Information Administration (EIA)[4] energy consumption (expressed in tons of coal equivalent: tce[5]) grew from 1,556 million tce in 2000 to 5,313 million tce in 2018, making China the largest energy consumer in the world. To meet this increasing energy demand, China turned from a nearly zero net importer in 2000, when 1,486 million tce were produced nationally, to a net importer in 2018, with only 4,240 million tce produced locally, posing some concerns on energy security. Figure 1 below gives a breakdown of total energy consumption by energy source.

Figure 1. Total Energy Consumption by Energy Source (www.eia.gov)



As shown in Figure 1, the large majority of energy demand is met with coal and other fossil fuels, which translates into substantial greenhouse gas (GHG) emissions. In fact, while in 2000 China ranked #2 in the world among the largest emitters with 3.5 billion tCO_2, trailing the USA with 5.9 billion tCO_2, in 2018 GHG emissions in China more than tripled reaching 10.8 billion tCO_2, more than double the emissions of the USA ranked #2 with 5.3 billion tCO_2.

[2] https://data.stats.gov.cn/english/tablequery.htm?code=AC03

[3] https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=CN

[4] https://www.eia.gov/international/overview/country/CHN

[5] 1 toe = 1.4286 tce https://www.iea.org/reports/unit-converter-and-glossary

According to the International Monetary Fund (IMF), although at a lower rate, China's GDP growth is expected to continue in the near future, outpacing the rest of the world[6]. Similarly, energy consumption and GHG emissions will continue to increase peaking in 2035 and 2030, respectively, according to a recent announcement by the China National Petroleum Corp[7], which will pose significant pressure on the environment and calls for significantly increasing the portion of energy demand met with renewable energy sources.

These growth trends trickle down to all energy end-user sectors as well as all geographic regions in China, including rural areas. China's rural population is currently about 580 million, which is over 40% of the total population. Although the per capita energy consumption in rural areas is considerably lower than in urban areas, owing it to the very large rural population, the total 2017 energy consumption in rural areas stood at approximately 590 million tce (or ~12% of the total energy consumption in that same year). Improving living standards and increasing energy consumption in agricultural production of the rural population translates into a yearly energy demand growth rate larger than 4%, which is above the country average.

The energy sector in rural China faces several challenges:

- Since rural villages and towns are very small and scattered throughout the country, delivery of energy is very expensive and at times unreliable. Electric transmission lines have losses larger than 7%, which is above the accepted European and American standards. On the other hand, coal is very expensive at over US$ 150/ton, which affects greatly rural economy and its development potential.
- Most households use a combination of coal, straws, and firewood for cooking and domestic heating. Combustion is often done inefficiently and in an uncontrolled way, causing severe health consequences because of release of air pollutants. In addition, the excess waste biomass generated from agriculture and livestock is just disposed of in landfills, which threatens both soil and water basins.
- Many rural villages and towns in an effort of improving their living standards are moving away from the use of straws and firewood and switching towards coal, despite its high price, which would contribute to significantly increasing GHG emissions.
- Renewable Energy (RE) sources and Energy Efficiency (EE) technologies and measures are barely utilized because of the inadequate local capacity and awareness about these technologies.

These challenges could be overcome by increasing the utilization of indigenous RE resources, which in rural China include solar energy, both thermal and photovoltaic, wind energy, geothermal energy, and biomass. Increasing the use of indigenous RE resources would not only lead to considerable environmental and economic benefits, but it would also reduce the raising concerns about energy security, since China is increasingly becoming a net energy importer. However, only biomass is utilized to a significant extent, although very inefficiently, and there are few and small applications of solar thermal water heaters and solar photovoltaic panels for public illumination. China's central government has made great efforts to promote RE and EE technologies and measures in rural villages and towns, but the limited educational level and insufficient financial resources available in rural areas have so far prevented the achievement of the desired results.

[6] https://www.china-briefing.com/news/china-2021-economic-outlook-foreign-investor-expectations/

[7] https://www.reuters.com/article/us-china-energy-cnpc-forecast-idUSKBN28R0SE

In this context, low/zero carbon development of rural towns and villages is based on the improved development and utilization of renewable energy resources in rural areas, adopting renewable energy technologies and integrated energy management and energy efficiency technologies/techniques to meet the increasing energy demand in rural areas, and accelerate the rural energy transformation. Low/zero carbon development will contribute to improved "ecological livability" in China's rural revitalization strategy, promotion of circular economy, and contribute to the achievement of China's National Determined Contributions (NDCs) targets under the Paris Agreement.

The implementation of EZCERTV project will contribute to the achievement of several Sustainable Development Goals (SDGs) as set by the United Nations General Assembly. EZCERTV is relevant primarily to Goal 7: “Ensure access to affordable, reliable, sustainable and modern energy for all”, but also to Goal 8: “Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all”, Goal 11: “Make cities and human settlements inclusive, safe, resilient and sustainable”, Goal 13: “Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy”, and to a lesser extent to Goal 2: “End hunger, achieve food security and improved nutrition and promote sustainable agriculture”.

Barrier Analysis

The approach taken for the design of the EZCERTV project is barrier removal. The barriers hindering the transition towards zero-carbon villages and towns in China rural areas have been identified in the EZCERTV Project Information Form (PIF). During the Logical Framework Analysis (LFA) Workshop, which was held virtually, because of the travel restrictions due to the outspread of the Covid-19 pandemic, on August 5-7, 2020, the participants stakeholders have reassessed the validity of the identified barriers, which were further confirmed by the team of national consultants during the Fall 2020 field missions to the demo locations. The major identified barriers include:

Policy Barriers - Rural areas lack clear policies for the application and promotion of low-carbon and low GHG emitting energy technologies. Policies can mobilize investment and financing resources to promote the application of rural renewable energy technology and low-carbon development. The central government has not yet established and improved laws and regulations and long-term mechanism for zero carbon rural development, let alone low-carbon rural development, especially in energy supply and consumption. Currently rural energy policies can be changed to facilitate the increased utilization of available renewable energy resources in these areas. The potentials for distributed energy generation, distribution and utilization, micro-grid distribution system, biomass gasification, biogas power generation, solar and wind hybrid power generation, etc. can make good use of supporting policies that would allow the widespread applications of these technologies that can bring about zero carbon town and village development. The current rural energy policies do not include decentralized RE-based power generation, distribution, and utilization, including energy storage. However, the policies lack detailed implementing rules and regulations making it difficult to adapt to the rapid development of rural areas in China. Although the revised Renewable Energy Law and Cleaner Production Promotion Law have laid the foundation for the development of rural renewable energy, they do not include encouraging economic incentive mechanism and impeccable financing mechanism.

Technical Barriers. Currently, the adoption of RE technologies in rural areas often do not have the benefit of proper screening and integration of other key aspects such as local natural, environmental, and ecological conditions, including economic conditions. China's rural renewable energy technology is still relatively weak, characterized by inadequate systematic and properly set processes, standards, and

normative systems. The development of decentralized RE-based energy generation projects has not yet led to significant rural energy system transformation model that suitable for different regional economic, climatic conditions and resources distribution. For example, there are no major breakthroughs in scientific research and development of energy efficient solid fuel fired stove for many years. Presently, many rural people use straw and corncob as solid fuels, but there are no suitable stoves for the combustion of these biomass energy forms. The straw gasification technology is basically still at the level of the 1980s. The process of building new rural communities is also weak in integrating the planning of each sector and the supply and utilization of energy (especially indigenous RE resources) to implement all sectoral activities. Rural areas have not yet formed a network of energy micro-grid and integrated management services, and the operation of micro-grid still lacks corresponding technology, equipment, and standards. Typically, in rural areas any plans to promote and implement RE, technology application promotion cannot be adequately be done because of the lack of baseline data on RE resources present and available in rural areas, and a dearth of data relating to the socio-economic conditions, and the degrees of adaptation between technology route and energy development is not high.

Awareness and Capacity Barriers. The concept of low-carbon development among rural and provincial government personnel is weak, and there is a lack of knowledge in the application of RE-based energy generation in rural areas, especially for electric power generation. At present, the phenomenon of hollow villages in China is widespread. The proportion of women left-behind in villages and towns is increasing, the trend of aging is going rapidly upwards, and the level of cultural awareness of women and the elderly is low. It is impossible to recognize the importance and necessity of rural green development and the application of renewable energy, and it cannot take the initiative to reduce energy consumption in production and living. Due to the low level of education and the entrenched old ideas of rural people, they dare not easily try new ways of life and production methods. Therefore, most of the rural people choose to use old, traditional, high-carbon energy consumption. The farmer's lack of awareness of green development, coupled with the lack of propaganda, guidance, and incentives, has made rural environmental pollution worse and worse, creating a vicious circle. As to the rural government’s capacity on rural energy issues, they typically have insufficient technical management capabilities, and their knowledge and understanding of the zero carbon, let alone the low carbon development concept needs to be strengthened. Managers have insufficient knowledge of the appropriate policies, regulations and technical standards that will enable them to come up with integrated policies and plans that are supportive of low carbon development. Many of the rural governments have limited information system and management system.

Financial Barriers. At present, the promotion and application of low carbon energy technologies in rural areas are still dominated by national fiscal funds, and there is only one source of funds. It is difficult to meet the demand for large-scale technology promotion and application by relying solely on government funds. The income of farmers is generally low, and for rural communities, it is difficult to raise funds for the construction of renewable energy facilities like for decentralized RE-based power generation. The enthusiasm of farmers to collectively finance and build such kind of facilities themselves is not high. This limits development and establishment of low carbon villages and towns. Coupled with the lack of successful demonstration and promotion of relevant technology application models, the effectiveness of the development of low carbon villages and towns is not significant. The concept of the low carbon development of rural towns and villages is not attractive enough to get the relevant stakeholders to participate in such initiative. In that case, it is difficult to fund and implement decentralized RE-based power generation initiatives. It will be difficult to establish effective coordination of sustainable commercialization and large-scale capital operation mechanism among government, farmers,

enterprises, and social funds to support rural clean energy infrastructure constructions.

The abovementioned barriers, if not properly and adequately addressed/removed will continue to prevent the transition towards zero/low-carbon rural villages and towns.

Business-As-Usual (BAU) Scenario

The 2030 climate change mitigation targets set by the Government of China (GOC) in its Nationally Determined Contributions (NDC) are as follows:

- Peaking CO2 emissions by no later than 2030.
- Lower CO2 emissions per unit of GDP by 60-65% compared to the 2005 levels.
- Increase the share of non-fossil fuels in primary energy consumption to approximately 20% (in 2018, the share of non-fossil fuel energy consumption was about 14.9%, including nuclear energy; see Figure 1).
- Improve the CO2 sink potential by increasing the forest stock volume by around 4.5 billion m3 compared to the 2005 levels.

In 2017, during the 19th Communist Party of China (CPC) National Congress, president Xi Jinping has ranked "Rural Revitalization Strategy" third among China's "New Vision of Development and Developing a Modernization" priorities[8]. In fact, there is a significant ongoing rural-to-urban migration, in the attempt of pursuing higher earnings and better quality of life. It is usually the younger male population who is migrating towards cities and urban areas, leaving behind an aging population with a larger percentage of women. Among other issues, this migration from rural areas is posing serious risks of food security, since if this trend continues, farming would not be sufficient to satisfy people's demand for agricultural and livestock products. President Xi has therefore proposed a major agriculture and land reform that inevitably relies, besides other sectors, also on investments in energy. However, the efforts undertaken to date have not been sufficient to achieve the set goals and more work and additional investments are needed.

Baseline Initiatives

The GOC as well as the private sector have embarked in several projects and initiatives aligned with the NDC energy targets. Table 1 summarizes the projects and programs that will be the baseline of the activities designed under this proposed EZCERTV project. These baseline projects and programs not only have been already approved and budgeted, but their execution is either ongoing or planned and they will be completed over the EZCERTV implementation period.

Table 1. Ongoing and Planned Baseline Projects and Initiatives

Baseline Project/Program	Description	Available Budget, US$
MARA and Local Governments: Comprehensive Utilization of Crop	Since 2019, the Chinese government has allocated 7.35 billion yuan (1.95 billion yuan in 2019, 2.7 billion yuan in 2020, 2.1 billion yuan in 2021), adopting the method of "reward for subsidy" to carry out the comprehensive utilization of straw throughout the country, and carry	18,000,000

[8] http://www.china.org.cn/opinion/2018-03/15/content_50711502.htm

Baseline Project/Program	Description	Available Budget, US$
Straw	out the system construction of straw utilization (treated as fertilizer, feed, base material, fuel and raw material), collection, storage and transportation. The Chinese government will continue the implementation of this program by arranging funds to support the comprehensive utilization of straw, which is estimated around 2 billion yuan per year. The activities of the national program on straw utilization focusing on the utilization of straw as fuel will be part of the EZCERTV Project. Such activities include supporting the construction and operation of direct straw fired central heating plants and the construction and operation of straw fuel pellet production facilities. The national program also includes the provision of subsidies for farmers to purchase straw pellet fuels and the cooking stoves that will use this fuel, etc. The EZCERTV Project will include the enhancement of the activities of the national program on the construction and operation of centralized straw collection, storage and transportation systems in the project areas were enhanced utilization of straw for energy. ***Timeline of Implementation: 2021-2026***	
MARA and Local Governments: Resource Utilization of Livestock Manure	During the "14th Five-Year Plan" period, the Chinese government plans to invest 14.504 billion yuan (5 years) to continue to promote and support the utilization of livestock and poultry manure resources, and improvement project of resource utilization capacity of livestock and poultry manure. This national program includes activities that are focused on improving the treatment, handling and utilization of livestock and poultry manure resources as fuel. Such activities will be part of the EZCERTV Project. These activities include construction and operation of biogas production facilities, including storage facilities, and distribution systems. The EZCERTV Project will include the enhancement of the activities of the national program on the construction and operation of biogas production, storage, and distribution facilities, including technical assistance to biogas users like rural households and rural industries. ***Timeline of Implementation: 2021-2026***	16,000,000
Ministry of Finance (MOF) and Local Governments: Pilot Demonstration of Winter Clean Heating in Northern Areas	Since 2017, the Chinese government has been carrying out demonstrations of the application of technologies for clean winter heating in northern areas, arranging special funds to support the promotion of clean energy, as well as coordinating utilization of various clean energy sources such as geothermal, biomass, solar energy, industrial waste heat, etc. for winter heating in the northern regions of the country. Under such national program, the government also carries out energy-saving renovations of existing buildings, encouraging local governments to innovate systems and mechanisms, improve policies and measures, and guide enterprises and society to increase capital investments on clean winter heating in northern areas. This is to realize the replacement of coal-fired heating in pilot areas. At the same time,	16,000,000

Baseline Project/Program	Description	Available Budget, US$
	subsidies for clean heating operations (such as "replacing coal with biomass") in rural areas are provided to subsidize additional operating expenses incurred by rural households, especially for the poor people to ensure the affordability and efficiency. The activities of this national program include to on the application of biomass energy and solar energy technologies and rural building energy-saving renovation. These activities will be part of the EZCERTV Project. The EZCERTV Project will include the enhancement of the activities of the national program on the construction and operation of facilities for the production, distribution and utilization of biomass, solar energy, and other available clean energy resources. This include enhancements in the design and specification of the installed equipment and systems, and subsidies for building energy-saving renovation, as well as operational subsidies for rural clean heating in rural areas. ***Timeline of Implementation: 2021-2026***	
Private Sector: Development of Renewable Energy Projects	Private sector participates in the project implementation of bulked straw direct heating, solar photovoltaic, centralized biogas supply in the project area. It also supports individual training for technical/design personnel; the demonstration and promotion of renewable energy/energy efficiency technologies and products; individual training programs for technical/design personnel; promotion of the marketization of renewable energy technologies and products, etc. ***Timeline of Implementation: 2021-2026***	38,200,000

The successful completion of all the baseline projects and programs listed in Table 1 will bring China closer to transitioning towards zero/low-carbon towns and villages and to achieving its NDC energy targets, but still they will not be enough to fully realize these goals. Improvements in the baseline efforts of the country to achieve zero/low carbon rural town and village development are necessary. Such improvements call for the removal of the identified existing barriers. The removal of such barriers will not only achieve the desired transformation of the current rural development pathway into something that leads to low/zero carbon footprint but also contribute to the realization of the climate change mitigation NDC targets of the country.

III. Strategy

Applied Project Strategy and Incremental Activities

The EZCERTV Project is designed based on a barrier removal approach. The proposed strategy calls for the design of *ad-hoc* incremental activities that by building on, and complementing, supplementing, or augmenting, the aforementioned baseline projects aim to remove all identified barriers. A barrier removal approach is the best available strategy because of its sustainability long after the implementation of the EZCERTV project has been completed. The strategy is illustrated in the Theory of Change (ToC) diagram illustrated in Figure 2 below. The ToC visually show how the identified barriers lead to the main problem for China's rural areas (yellow section of the figure), while the implementation of the EZCERTV incremental activities will allow the removal of the barriers and the achievement of the main Project Objective: ***"Acceleration of zero-carbon transformation in China's rural area to contribute to global climate change mitigation and the achievement of the United Nations sustainable development goals (SDG)" (purple section of the figure)***. The following are the expected outcomes of the project and the proposed ways in which the barrier removal strategy will be carried out to realize them:

Effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China: This first Outcome will be delivered by implementing four different sets of activities: 1) the first set of activities will collectively produce, approve and enforce national and local policies that will support and incentivize the development of zero-carbon rural communities and simultaneously contribute to the achievement of China's carbon neutrality goal by 2060, as pledged in 2020 by President Xi while addressing the United Nations General Assembly[9]; 2) the second set of activities aims to establish and operationalize an institutional framework for the promotion of zero-carbon towns and villages development, including a training program to strengthen the capacity of local governments for the integration of RE resources in their development plans; 3) the thirds set of activities will lead to the formulation, approval and enforcement of technical and operational quality standards for the implementation and management of RE, EE, and Energy Conservation (EC) technologies and measures; and 4) lastly, assessment studies will be conducted concerning RE resources, energy use trends, and economic and social development levels in selected rural areas. These assessments, together with a capacity building program for local government officials, will be the basis for the formulation of energy development plans of pilot villages and towns in the eight selected demo provinces.

Low/zero carbon technology applications are widely implemented in rural towns and villages in rural areas of China: A three-part plan will be executed for the realization of the second Outcome, which includes: 1) activities on the design and implementation of RE technologies, based on indigenous resources, in rural areas to facilitate the transition towards zero-carbon villages and towns and showcase the cost-effectiveness of such projects. The activities will be supplemented with monitoring and evaluation tools, capacity building programs for relevant stakeholders, and a promotional program for the replication and scale-up of the demonstrated RE technology applications; 2) activities similar to the previous one, except the focus will be on EE and EC technologies and measures. In addition, there will be activities targeting the widespread deployment of high EE appliances and equipment, and the development of supply-side energy optimization methods and demand-side energy management systems; and 3) initiatives aiming at drafting and disseminating guidelines for the replication and scale-

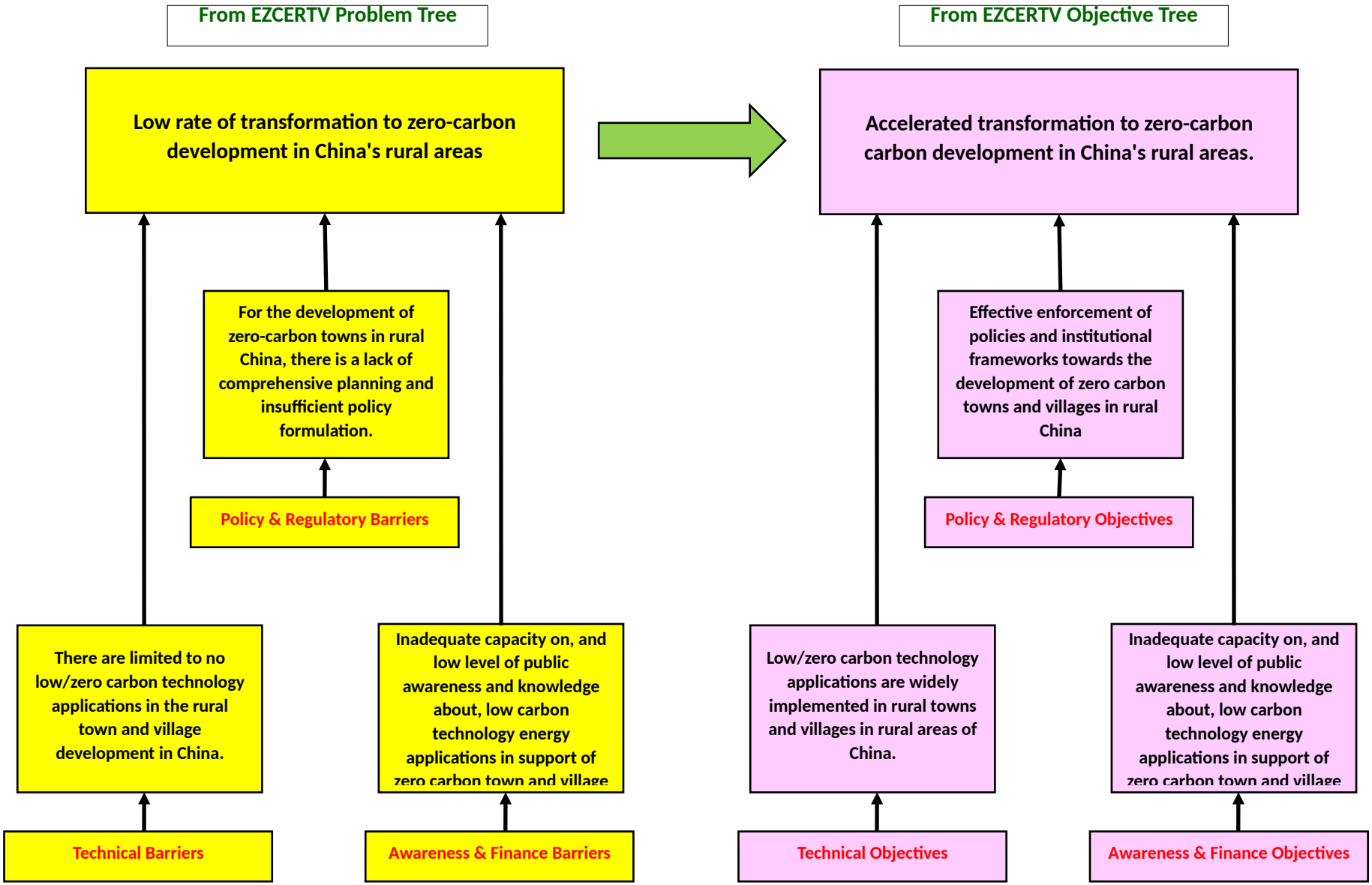
[9] https://www.cnbc.com/2020/09/23/china-claims-it-will-be-carbon-neutral-by-the-year-2060.html

up of zero carbon towns and villages development. The activities will include documentation of case studies and lessons learned as well as the demo implementation plans. The RE/EE/EC technology applications that will demonstrated and replicated in Parts 1 and 2 (for a detailed description see the supplemental annex on Description of Demonstrations) include:

- Bundled Straws-Fired Centralized Heating System
- Biomass Fuel Pellets Production and Use for Household Heating & Cooking
- Biomethane Use for Clean Heating and Cooking in Rural Households
- Piped Biogas for Household Cooking and Tobacco Flue-Curing
- Distributed Photovoltaic Power Generation and Application of a SMART Energy Management Service Platform
- Rural Residential Buildings Exterior Renovation
- Distributed Biogas Generation, Supply and Utilization System
- Integrated Solar PV-Storage-Direct-Flexible (PSDF) Technology coupled with Rural Houses Retrofits

Enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China: The activities that will bring about this last Outcome are organized into five groups, which will lead to the: 1) design, development and operationalization of an information, communication and education platform to be used as a media to hold and share data useful for the development of zero-carbon villages and towns in rural China. To maximize the use of this tool, capacity of local stakeholders will be assessed and training programs on RE/EE/EC technologies and measures for rural development will be designed, organized, and administered. Trainees will include local government officials, industry management teams, private sector players, rural communities, and Non-Governmental Organizations (NGOs); 2) establishment of a program that will strengthen international cooperation on clean energy development with countries adhering to the "Belt & Road" initiative. Capacity building programs will be organized and conducted in the "Belt & Road" countries assisted by China, and information concerning lessons learned from the implementation of the EZCERTV project will be prepared and shared to guide these countries in the preparation of rural development business models; 3) establishment of a mechanism meant to increase the participation of all rural social entities, including women and marginalized groups, in the efforts to shift towards zero carbon villages and towns. The mechanism will include training programs, promotion of RE/EE technologies and measures, and awareness raising initiatives concerning the sustainability of such technologies; 4) establishment of a private sector led service industry that will support and eventually overtake the installation, operation, maintenance and repair of the RE/EE technologies and measures adopted. Besides training programs on RE/EE technologies and measures, technical assistance activities to train rural communities on how to source local, national, and international funds to finance their projects will be also undertaken; and 5) establishment and operationalization of ad-hoc designed financial instruments and mechanisms, including financial and fiscal incentives. The design of such instruments will be based on a thorough assessment of similar initiatives in other countries adapted to China's local requirements, and training programs will be organized and administered to strengthen the financial skills of key rural stakeholders.

Figure 2. Theory of Change



Once successfully completed, EZCERTV will allow China's rural areas to significantly shift towards zero-carbon development, which will provide benefits to the entire rural community in four different aspects: 1) environmentally, since burning less fossil fuels will allow to reducing the emissions of GHG and particulate matter; 2) health benefits for the rural households occupants, since direct combustion of fossil-fuels for domestic heating and cooking systems is currently done in a very inefficient and uncontrolled way, with dangerous solid and gaseous emissions in closed spaces; 3) financially, because EZCERTV also includes activities devoted to the development of a local RE/EE technologies implementation, operation and maintenance market; and 4) security, with more indigenous energy generation, the scattered rural villages and towns will be less exposed to reliability of energy transmission and delivery.

The biggest innovation that the EZCERTV Project will introduce in China's rural areas, and potentially to other countries along the "Belt & Road", is the approach taken to shift the entire energy generation in rural villages and towns towards clean technologies virtually reaching zero carbon emissions. Furthermore, although China is a very large country with many different climate zones and different indigenous renewable energy resources, the demos proposed have been designed for villages selected in several different regions, covering a large spectrum of conditions. This choice will facilitate replication and scale-up of the demo activities in most, if not all, other rural villages, and towns. Finally, all the demos can be used to draw lessons that can be exported to, and used by, other countries that share similar issues and needs.

IV. Results and Partnerships

Expected Results

Goal: Reduced annual growth rate of GHG emissions in the energy and energy end-use sectors in the rural areas of the country

Project Objective: Acceleration of zero-carbon transformation in China's rural area to contribute to global climate change mitigation and the achievement of the United Nations sustainable development goals (SDG)

Component 1: Policy Formulation and Institutional Mechanisms on Zero Carbon Town and Village Development. *Please refer to Table 3 of Annex 9 for the project stakeholders that will be involved in the delivery of each output under Component 1.*

Outcome 1: Effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China

Output 1.1: Formulated, approved, and enforced policies on the promotion and support of zero carbon town and village development in line with China's rural revitalization strategy and goal of achieving carbon neutrality before 2060.

Activity	Description
Activity 1.1.1: Conduct of policy research to update existing rural energy policies and implementing rules and regulations (IRRs) on the various aspects of energy supply, demand, and utilization in rural development	Analyze and evaluate the effectiveness of the relevant rural energy policies and IRR implemented in China and provide recommendations to update and enforce existing policies and regulations currently in place to promote the realization of China's rural energy supply, demand, and utilization of zero carbon development goals.
Activity 1.1.2: Conduct of research studies on existing energy policies and implementing rules and regulations in other countries that can be used as reference for the rapid development of zero carbon in rural areas in China	Conduct research on, and analyze the existing energy strategies, plans, laws and regulations, and policy incentive devised and enforced in other countries for achieving carbon neutral development. Summarize relevant experiences and propose policy and regulatory frameworks to promote China's rural energy transition and accelerate the development of zero carbon in rural China.
Activity 1.1.3: Formulation, promotion, approval and enforcement of new policies and regulations on zero carbon development in rural area, and low GHG emitting energy technologies implementation	In accordance to the announcement of the ambitious goal to peak CO_2 emissions and achieve carbon neutrality in the upgraded Nationally Determined Contributions of China, and based on the research studies conducted under Activity 1.1.2, new policies and regulations will be formulated, approved and enforced, which will include the development and application of new technologies, as well as new models and mechanisms that are conducive to the promotion of zero carbon development in rural areas, and reduction of GHG emissions.
Activity 1.1.4: Formulation, promotion, approval and enforcement of new national and local policies and regulations on incentives and financial support for promotion of RE/EE/EC in rural areas, including increased role of	Conduct in-depth study on the development of RE/EE/EC in rural areas, and accordingly draft, promote, approve, and enforce new policies and regulations on incentive and financial support mechanisms to promote the adoption of RE/EE/EC technologies and measures in rural areas, with emphasis on increasing private sector involvement.

private sector	
Activity 1.1.5: Provision of recommendations to revise the laws related to RE/EE/EC and include incentives and financing mechanisms	According to the changing circumstances of low carbon development, study and analyze laws concerning zero-carbon development, both at the national and the local level, including but not limited to the Renewable Energy Law, the Cleaner Production Promotion Law, the Energy Conservation Law, and their relevant contents. Propose amendments to the articles that are not applicable, supplement and improve the missing content, particularly for zero-carbon rural development, propose feasible incentives and financing mechanisms, and lastly campaign and lobby for the approval of the proposed recommendations.

GEF support is required for the incremental technical assistance in the conduct of the review of existing national and international policies and implementing rules and regulations, in the formulation of energy policies, and in the provision of recommendations to revise current laws.

Output 1.2: Established and operational institutional framework to promote and support zero-carbon town and village development.

Activity	Description
Activity 1.2.1: Establishment of clearly defined and operational institutional arrangements for promoting zero-carbon energy development in rural areas, such as policy-making, planning and management	Devise and establish a working mechanism for national and local government officials operating in the project areas to plan, promote, monitor, and manage energy development in rural villages and towns. Concomitantly, establish a working mechanism for stakeholders to cooperate on promoting the utilization of zero carbon energy technologies and the development of zero carbon villages and towns.
Activity 1.2.2: Design and implementation of a capacity building program for local governments on integrated RE resource planning and analysis, including a post-training program evaluation	Organize relevant expert teams to study and prepare capacity building plans and programs for integrated planning and analysis of renewable energy resources. Conduct professional training for relevant personnel in the demonstration and promotion areas to improve comprehensively their capacity in energy planning, analysis, and management.

GEF support is required for the incremental technical assistance for the establishment of institutional mechanisms and for the design and administration of training programs.

Output 1.3: Formulated, approved, and enforced technical, management and operational standards in rural renewable energy (RE) development and utilization, energy storage, energy conservation and energy efficiency (EC&EE) and comprehensive energy management.

Activity	Description
Activity 1.3.1: Development and enforcement of guidelines and standards on RE/EE/EC technologies and measures used in rural areas	Based on an accurate assessment of existing clean technologies and their implementation; formulate, promote, and enforce relevant and stricter technical, management and operational standards and guidelines for the implementation of RE/EE/EC technologies and measures in rural villages and towns in order to improve their overall quality and simplify their adoption.
Activity 1.3.2: Standardization of RE/EE/EC technology specs and MEPS[10], testing and certification	Overall quality and performance of low/zero rural development is improved by formulating, promoting, approving, and implementing relevant rural energy standards throughout the entire value chain.

procedures for local and imported technologies, and design and operation of RE/EE/EC projects.	This includes: introduction of specs and MEPS for RE/EE/EC technologies and products; standardization of certification and testing procedures for of local and imported RE/EE/EC technologies and products; and best practice for EE/RE/EC project design and operation.

GEF support is required for the incremental technical assistance in the development of guidelines and standards on RE/EE/EC as well as testing and certification procedures.

Output 1.4: Completed assessment of renewable energy resource endowment, economic and social development level, and energy use trends and patterns in rural areas of selected provinces; and formulated energy-integrated development plans of pilot towns in selected 8 provinces.

Activity	Description
Activity 1.4.1: Conduct of a comprehensive assessment of the current status of rural energy in the selected project rural areas.	Evaluate the available renewable energy resources and potential socio-economic development levels and forecast local rural energy generation and utilization patterns and trends in the selected project rural areas. Design, establish and operate a RE/EE/EC data platform in the project rural areas. The platform will also contain relevant data on local socio-economic development.
Activity 1.4.2: Design and conduct of a capacity building program for rural governments on monitoring, recording, analysis and management of energy generation and utilization data, including a post-training program evaluation	A training program will be designed and administered for all relevant rural government officials responsible for zero/low carbon development. The training will enable the participants to acquire the required skills on how to monitor, record, analyze and manage data concerning both energy generation and consumption. A post-training program evaluation of the trainees' skills will be undertaken to assess the success of the capacity building program.
Activity 1.4.3: Preparation of comprehensive energy-integrated development plans for the selected project areas	Based on Activity 1.4.1 and building on the capacity building program conducted under Activity 1.4.2, comprehensive energy-integrated development plans for zero/low carbon development in demonstration villages and towns will be prepared.
Activity 1.4.4: Development of guidelines applicable to the promotion of zero carbon villages and towns	Based on the summary of the demonstration of integrated application of zero carbon energy technologies in the project areas, as well as on learned lessons from domestic and international experience, develop and formulate guidelines for the promotion of zero carbon villages and towns and zero carbon energy technologies, which will be used to carry out publicity and promotion of zero carbon village developments.

GEF support is required for the incremental technical assistance in the assessment of the current status of rural energy, design and administration of training programs, preparation of development plans, and formulation of guidelines to promote zero-carbon development.

Component 2: Integrated Renewable Energy and Energy Efficiency Technologies Applications for Zero Carbon Town and Village Development. *Please refer to Tables 3 and 4 of Annex 9 for the project stakeholders that will be involved in the delivery of each output under Component 2.*

Outcome 2: Low/zero carbon technology applications are widely implemented in rural towns and villages in rural areas of China.

[10] Minimum Energy Performance Standards

Output 2.1: Completed and operational zero carbon technology demonstration involving the installation of about 167.2 MW of RE-based power generation capacity in 120 rural villages and 43 towns for energy production and utilization showcasing the cost-effective application of new RE delivery mechanisms decentralized RE-based energy generation and distribution (through mini/micro-grids) and energy storage.

Activity	Description
Activity 2.1.1: Evaluation and improvement of existing initiatives and applications of renewable energy to facilitate the shift towards zero carbon villages and towns development.	Evaluate existing major renewable energy technologies and applications in China's rural villages and towns in terms of energy resource security, conversion efficiency, technical benefits over conventional technologies, environmental protection, and economic viability. On this basis, the proposed technology path of the demonstration projects is optimized by combining Activity 1.4.1 on RE resource endowment, climate characteristics, and economic and social levels of the various selected rural regions of China.
Activity 2.1.2: Development of engineering designs and implementation plans, and facilitation of their approval process for the selected demonstration areas.	Based on Activity 2.1.1, develop zero carbon energy technology implementation plans and work plans for demonstration villages and towns, including site selection of the project , technology route to be adopted, investment estimate, implementation schedule and expected economic and social benefits. The implementation and work plans will be illustrated to local energy authorities for approval, while wide publicity will be conducted targeting residents of demonstration villages and towns.
Activity 2.1.3: Implementation of the demos of new feasible and cost-effective RE technology application models designed in Activity 2.1.2	In accordance with the implementation plans and work plans formulated in Activity 2.1.2, carry out technology application demonstrations, which will include: a) rooftop and ground-mounted solar Photo Voltaic (PV) mini-grids with battery energy storage systems; b) anaerobic digesters; c) biogas-fired power generators; d) biogas upgraders for bio-methane production; e) biomass pelletizers; and ancillary equipment such as: f) biogas transmission pipelines; and g) skid-mounted biogas compressors to fill pressurized biogas cylinders.
Activity 2.1.4: Design and implementation of methods for monitoring and evaluation of the operational, energy and economic performances of RE technology application demonstrations.	Design and utilize methods for monitoring and evaluation of all RE demo technology applications, and evaluate their operational, energy and economic performances. Conduct in-depth analysis of the demo performance evaluation results. The results of the analysis will be utilized in the selection and optimization of renewable energy technologies that will be applied to enable zero carbon developments in rural villages and towns.
Activity 2.1.5: Design and conduct of a promotional program for local governments and rural communities on the technical and environmental advantages of adopting new and cost-effective RE technology applications	Based on the results of the demonstration activities of the project, identify feasible potential renewable energy technologies that can be applied to enable and support zero carbon development in rural villages and towns. Formulate and implement promotion plans for the replication and scale-up of the demonstrated RE technology applications.
Activity 2.1.6: Design and implementation of a capacity building program to facilitate the establishment of a rural renewable energy technical service industry.	This activity will entail the design and conduct of a capacity building program for the installation, repair, and maintenance services for renewable energy applications. The training initiatives for relevant personnel will include renewable energy technology principles, engineering design, construction, operation and maintenance, training

	on zero carbon development, and ensuring stable operation of rural zero carbon energy systems. A post-training program evaluation will be also carried out.

GEF support is required for the incremental technical assistance in the development of engineering design and implementation plans, installation, and operation of the RE demos, procurement of equipment and measurement instruments, development of M&E systems, design and conduct of promotional programs, and design and administration of training programs.

Output 2.2: Completed and operational demonstrations of the cost-effective applications of EC&EE technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development. Estimated annual energy savings of about 6,487.0 TJ/year in 4 towns and 120 villages.

Activity	Description
Activity 2.2.1: Development and approval of engineering designs and implementation plans of demos of new and cost-effective EE/EC technologies	Assess the current situation of energy utilization in selected rural areas and develop implementation programs and plans for the EE/EC demonstration projects. This includes the proposed technical route, investment estimates, implementation schedule and expected economic and social benefits.
Activity 2.2.2: Design and demonstration of new feasible and cost-effective EE/EC technology application models	Carry out EE/EC technology application demonstrations in accordance with the implementation plans for EE/EC technologies identified in Activity 2.2.1. These will include the following technologies: a) high energy efficiency cookers (e.g., induction, biogas stoves, and biomass stoves); b) high energy efficiency heaters (e.g., biogas-fired heaters, and biomass-fired heaters); c) heat recovery for biomass pre-heating; d) building insulation (e.g., for walls, windows and roofs); e) high energy efficiency kilns; and f) Smart Energy Management Systems.
Activity 2.2.3: Design and implementation of EE/EC technology monitoring and evaluation methods, and evaluate their operability, energy saving and economic efficiency in demonstration projects.	Design and implement EE/EC technology monitoring and evaluation methods, monitor EE/EC technologies and measures. Evaluate operability, energy saving and economic efficiency in demonstration projects. Conduct in-depth analysis of the results to support the screening and optimization of EE / EC technology applications that will be used to enable zero carbon development of rural villages and towns.
Activity 2.2.4: Design of EE/EC technology implementation promotion plans for local governments and rural communities to promote the application of relevant applicable and feasible EE/EC technologies.	Based on the successful demonstration of the EC/EE technology applications and identified promising EE/EC technologies, formulate, and implement relevant promotion plans, expand the application scope and sustainable impact of EE/EC technologies, and promote the development of zero carbon villages and towns in rural areas.
Activity 2.2.5: Design and implementation of a capacity building program to facilitate the establishment of a rural EE/EC technical service industry.	Provide capacity building for EE/EC installation repair and maintenance services, including training of relevant personnel on EE/EC technical principles, engineering design, construction, operation, and maintenance, and zero carbon knowledge. A post-training program evaluation will be also carried out.
Activity 2.2.6: Design and implementation of a promotional program for the improved sales and use of high-performance EE	Based on the demonstrations under Activity 2.2.2, combined with Activities 2.2.4 and 2.2.5, and jointly with NGOs, carry out various initiative for the promotion of EE/EC technology and products, raise public awareness on the importance of EE/EC, and promote the

equipment and devices in rural areas	development of rural energy-saving communities in response to the problems of insufficient awareness of EE products, low active participation and uneven quality of EE products among residents in rural areas in China.
Activity 2.2.7: Development and design of an integrated rural energy management system based on energy supply-side monitoring and optimization, and demand-side management	Based on the RE/EE/EC demonstration, develop and design an integrated rural energy management system. The system includes various management measures such as supply-side energy optimization, energy consumption statistics, management of key energy-consuming equipment and energy metering and auditing to achieve the purpose of reducing energy consumption, renewable energy substitution and improving energy utilization efficiency, and apply this system to project performance evaluation. The developed energy management system will be piloted in selected rural towns and villages and compared to demo locations.

GEF support is required for the incremental technical assistance in the development of engineering design and implementation plans, installation and operation of the EE & EC demos, procurement of equipment and measurement instruments, development of M&E systems, design and conduction of promotional programs, design and administration of training programs, and development of an integrated energy management system.

Output 2.3: Developed, published, and disseminated zero carbon town and village development guidelines to replicate and scale-up the impact of zero-carbon villages and towns.

Activity	Description
Activity 2.3.1: Preparation, publication and dissemination of case studies and lessons learned, based on the implemented demos, for the promotion of zero-carbon village and town energy technology applications	Prepare a profile/report for each demonstration, and compile all of the profiles as case studies that also include lessons learned. The compiled information shall be used for developing guidelines for the application and promotion of low/zero carbon technologies in rural towns and villages.
Activity 2.3.2: Development of implementation plans and action plans for demos replication and scale-up	Based on the reports developed in Activity 2.3.1 in combination with the national and regional zero-carbon/low-carbon development strategies, formulate sustainable development and implementation plans for replication and scale-up of zero/low-carbon projects in other regions of the country. The formulated plans will be adapted to local conditions, technical and economic characteristics, and in line with sustainable development of each region. Facilitate and secure approval and funding from national or local governments for the implementation of the plans.
Activity 2.3.3: Application of the development and implementation plans formulated under Activity 2.3.2 and promotion of the application of RE and EE technologies in different regions.	Based on the formulated development and implementation plans the successfully demonstrated RE/EE technologies and measures will be replicated and scaled-up, funded with government resources, under different climatic and socio-economic conditions. The plans also take into consideration local natural and human/technical resources as well as energy consumption patterns and local policies. Initiatives to promote the benefits of the proposed environmentally sustainable technologies to rural communities will also be undertaken.

GEF support is required for the incremental technical assistance for the preparation of case studies, development and application of implementation and action plans for demos scale-up and replication.

Component 3: Facilitating Rural Zero Carbon Development and Lifestyle Popularization and Knowledge Dissemination. *Please refer to Table 3 of Annex 9 for the project stakeholders that will be involved in the delivery of each output under Component 3.*

Outcome 3: Enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China.

Output 3.1: Completed information, communication, and education (ICE) program on zero carbon development in rural towns and villages in China for local authorities, service companies, rural energy cooperatives, public institutions, private entrepreneurs, farmers, and the public.

Activity	Description
Activity 3.1.1: Assessment of the ability and awareness level of relevant stakeholders about zero-carbon rural development	The focus of this activity is the evaluation of the capacities and level of awareness concerning RE/EE/EC and zero-carbon village and town development of relevant stakeholders from the selected project areas (including government departments, energy production and service companies, NGOs, scientific and technical personnel, farmers, etc.). To conduct this assessment, ad-hoc questionnaires will be prepared and administered, and the results will be used to recommend capacity building programs.
Activity 3.1.2: Design and implementation of a capacity building and information program for relevant local government officials and industry management teams.	Based on the results of the assessment Activity 3.1.1, a capacity building program will be designed and conducted for government industry managers at all levels, including the determination of trainers, the preparation of training materials concerning the development of RE/EE/EC and zero-carbon village and town related concepts, policies, regulations, technologies, and application prospects. The topics covered will effectively help improve the awareness and execution capabilities of government officials and industry managers. A post-training program evaluation will be also carried out.
Activity 3.1.3: Design and implementation of a capacity building and information program for the private sector, NGOs, and rural communities.	Based on the results of the assessment Activity 3.1.1, a capacity building program will be designed and conducted for the private sector, NGOs, and business associations[11], and rural communities on the concept of zero-carbon energy technologies and development, end-use energy guidance, and awareness raising concerning energy conservation and emission reduction. A post-training program evaluation will be also carried out.
Activity 3.1.4: Development and operationalization of a project management and information communication platform for zero-carbon villages and towns.	Design, establish, operate, and maintain a project information management platform and establish a multimedia-combined project information and knowledge sharing and media integration mechanism to strengthen project management capacity, facilitate information exchanges among managers, and improve the informatization level of project management.
Activity 3.1.5: Design, approval, and operationalization of an energy monitoring and reporting system and an energy benchmarking system for	This activity will involve the design, implementation and operationalization of an energy monitoring and reporting system and energy benchmarking system for the rural sector in China. Such systems will involve the reporting of energy consumption and energy

[11] China Association of Rural Energy Industry; China Energy Conservation Association; China Renewable Energy Society, China Household Electrical Appliances Association; etc.

rural towns and villages.	use-related GHG emissions, and the analysis of the specific energy consumption in various energy consuming activities in the rural sector. The recorded information will be used in rural energy planning in the country. These will also be used in tracking the progress and performance of, RE-based energy generation (power and non-power applications) system installations as part of the low/zero carbon development efforts in rural towns and villages, including the application of EE and energy conservation technologies/techniques in China's rural sector.

GEF support is required for the incremental technical assistance in the assessment of capacity gaps and needs, design and administration of training programs, design and implementation of awareness raising programs, and development and operationalization of an information sharing platform.

Output 3.2: Established international cooperation and knowledge sharing program on zero-carbon towns and villages development and green energy transformation with countries along the "Belt and Road", to support these countries in achieving their nationally determined contributions (NDC) commitments to the "Paris Agreement".

Activity	Description
Activity 3.2.1: Development of business models for zero-carbon project implementations for countries assisted by China.	The activity will summarize the good practices of China in carrying out the development of zero-carbon villages and towns based on the results of the project demonstration, as well as replication and scale-up, including the specific practices of renewable energy resources development and utilization technology, energy management mechanism and capacity building programs, and provide guidance and support for other countries participating in China's "Belt and Road Initiative".
Activity 3.2.2: Organization and conduct of international capacity building, communication and information sharing programs on the development of zero-carbon villages and towns with "Belt and Road" countries.	The main purpose of this activity is to organize and conduct capacity building initiatives in "Belt and Road" countries on zero/low carbon rural development based on training programs designed under EXCERTV. Furthermore communications between managers and experts from the "Belt and Road" countries will be established to promote EZCERTV project results to a broader audience, and to assist and further build capacity of relevant personnel from the participant "Belt and Road" countries. A post-training program evaluation of the trainees' skills will be undertaken to assess the success of the capacity building program.
Activity 3.2.3: Organization and conduct of international bilateral and multilateral cooperation programs on zero-carbon rural development.	Organize and conduct international bilateral and multilateral cooperation programs on zero-carbon energy technology and development of zero-carbon villages and towns. Organize international conferences on zero-carbon towns and villages development to communicate and share project experiences and results and support the achievement of international climate change targets as pledged in the participant countries' NDCs.

GEF support is required for the incremental technical assistance of "Belt & Road" countries in the organization and conduct of bilateral and multilateral cooperation program, design and administration of training programs, and development of business models.

Output 3.3: Established social mechanism for the promotion of zero-carbon town and village development and the sustainability of renewable energy industries.

Activity	Description
Activity 3.3.1: Establishment of a social participation mechanism conducive to strengthening stakeholders' capacity on zero-carbon towns and villages development.	NGOs, business associations, research institutions, private enterprises, farmers, women, and youth groups will be involved in the project activities through training programs on sustainable development of zero-carbon villages and towns, and RE and EE technologies and measures. As part of the training tools, it will be established a zero-carbon technology demonstration platform. A post-training program assessment will also be carried out.
Activity 3.3.2: Development of plans to publicize EZCERTV project and promote rural communities' participation.	This activity focuses on assessing the current status of awareness levels about RE/EE/EC technologies and products among relevant stakeholders in the selected project areas (e.g., government, researchers, technicians, rural residents, etc.) to develop an overall communication plan to publicize the benefits and achievement of the EZCERTV project, and to devise a promotional strategy to increase rural community participation in zero-carbon rural development initiatives.
Activity 3.3.3: Implementation of the promotional strategy to promote zero-carbon towns and villages development to rural communities.	Based on the project promotion strategy designed in Activity 3.3.2, implement the designed communication initiatives to promote zero/low-carbon rural development. A combination of traditional and modern media (i.e., television, radio, newspapers, internet, and social media) will be used to reach out the rural communities. As part of the strategy, knowledge products will also be prepared and disseminated (i.e., promotional videos, brochures, leaflets, billboards, etc.). The success of the promotional strategy will be assesses by preparing and administering a survey to evaluate the increased level of awareness of the rural communities compared to the initial assessment.
Activity 3.3.4: Design and conduct of promotional initiatives to increase the involvement of women, youth, and marginalized groups in zero-carbon development in rural areas.	A focused action plans will be developed to improve women, youth and marginalized groups knowledge of clean energy technologies, energy savings and GHG emission reduction. Initiatives will be designed to foster women empowerment, strengthen youth education, and improve marginalized groups benefits. The purpose of this activity is to ultimately increase the participation of these groups in zero-carbon rural development. Evaluation of the impacts of the initiatives organized will be conducted during the project implementation stage.
Activity 3.3.5: Establishment of a renewable energy industry group to sustain efforts of the rural villages and towns to replicate the low carbon energy technology application models.	This activity involves the development and implementation of a plan to establish a renewable energy industry association that will provide technical services for the design, manufacturing or fabrication and sales of RET systems, as well as the design, engineering, installation, operation, repair, and maintenance of RET system applications in China's rural sector. This can be the China Rural Distributed Renewable Energy Construction and Development Industry Association. Such industry and industry association will sustain efforts of the rural villages and towns to replicate the low carbon energy technology application models. Such industry association can also serve as a national RE/EE Accelerator that will also help ensure the sustainability of rural zero/low-carbon development efforts for the promotion and servicing of efforts towards EE rural buildings and industries in China.

GEF support is required for the incremental technical assistance in the assessment of needs and gaps, development and implementation of promotional programs, and design and administration of training programs, as well as in the setting up and operationalization of a renewable energy industry serving rural towns and villages.

Output 3.4: Established and operational market-oriented mechanism for the enhanced development and utilization of rural RE resources, EC&EE improvement, and comprehensive energy management systems for supporting sustainable socio-economic development, and creation of zero carbon towns and villages in rural China.

Activity	Description
Activity 3.4.1: Establishment of a private sector-led service industry in rural areas for RE and EE technologies	This activity is interconnected to the training programs implemented under Activities 2.1.6 and 2.2.5 to build local capacity in rural areas to implement, repair and maintain RE and EE technologies and measures. Trained specialists and private enterprises will establish in rural areas a service industry for sustainably operate RE and EE technologies and measures and accelerate the transition towards zero-carbon rural development.
Activity 3.4.2: Design and implementation of a program to assist rural municipalities/towns in sourcing funding for socio-economic development initiatives from available alternative sources of funds	Design and operationalize a technical assistance program to assist rural communities in obtaining financial support. Potential funding sources that will be considered include national and local governments, private enterprises, state-owned enterprises, financial institutions, bi-lateral and multi-lateral international donors, regional and national development banks, and NGOs. The funding obtained will be used to support for rural community development and promote the achievement of the community's zero carbon and low carbon goals.
Activity 3.4.3: Design and conduct of a capacity building and technical assistance program for the private sector to enable them to procure and supply new and cost-effective RE/EE technologies for rural development, including a post-training program evaluation	Design and implement a capacity-building training program for the private sector and relevant public and private stakeholders interested in developing rural energy businesses. The training content will include rural sustainable energy resources, RE and EE technologies and measures, fundamentals of energy management systems for both generation and consumption, etc., related to zero and low-carbon rural development. Technical support and guidance will be provided to private investors. A post-training program evaluation will be also included.

GEF support is required for the incremental technical assistance in the establishment of a RE & EE service industry, support rural stakeholders in sourcing funds, and design and administration of training programs.

Output 3.5: Established and operational investment and financing mechanisms for supporting the commercial viability and operation of zero-carbon towns and villages and the development of rural renewable energy industries.

Activity	Description
Activity 3.5.1: Evaluation of the existing financing mechanisms and incentives, and their adaptation to	Conduct research and analysis on the existing fiscal/financial incentives and financing mechanisms for promoting zero-carbon energy development in rural areas. Use the results of this assessment

zero-carbon rural development in China.	to recommend similar instruments for the development of zero and low carbon towns and villages in rural areas in China. Align the recommendations provided for facilitating the participations of the private sector with the national policy guidelines for the development of green finance and carbon neutral strategies.
Activity 3.5.2: Promotion of green finance and implementation of business practices to support the development of zero-carbon villages and towns	Use the recommendations developed under Activity 3.5.1 to design and implement sustainable and viable financial/fiscal incentives, financial instruments, and mechanisms to promote green finance for zero-carbon development in rural villages and towns. Support will be provided to participating financial institutions and private sector to facilitate their access to the existing and the newly established fiscal/financial incentives as well as financing instruments and mechanisms.
Activity 3.5.3: Design and conduction of a promotional, assistance, and capacity building program to enhance the involvement of private sector and financial practitioners, as well as national and local governments, and collectively finance and build community-based RE/EE facilities	According to the green financing mechanism developed in Activity 3.5.2, training programs will be organized and conducted on green finance, investment and financing methods and benefits. Participants will include national and local government officials, practitioners from financial institutions and the private sector to enhance their ability to use financial instruments to carry out zero carbon town and village development. Post-training program evaluations will be also carried out

GEF support is required for the incremental technical assistance in the assessment of existing financing mechanisms and incentives, promotion of green financing, and design and administration of training programs and promotional activities.

Partnerships:

The EZCERTV Project will be implemented in partnership with several stakeholders that also contribute to addressing the development challenges and support the transition towards zero-carbon rural villages and towns.

Table 2: EZCERTV Project Partners

Partners	Relevance to EZCERTV and Description of Initiatives
Ministry of Agriculture and Rural Affairs (MARA) of the People's Republic of China	The Ministry of Agriculture and Rural Affairs, which replaced the Ministry of Agriculture in 2018, manages several recurring government budgets devoted to the socio-economic development of rural areas. In particular, there is a current effort to utilize the huge amount of straws produced in agriculture with the double benefit of using it as feedstock for many application which removing the need for its disposal. Among the several uses supported by MARA, a significant portion of the annual budget is allocated to finance the use of straws as an energy resource for rural communities, particularly for replacing fossil fuels, such as coal and liquefied petroleum gas. The program will explore the use of pelletized, bundled and direct-combustion of straws in rural communities for heating, cooking, and power generation. Additionally, MARA also promotes the utilization of livestock and poultry manure as an energy resource. Similarly to what is proposed for the use of straws, there is a recurring annual government budget allocated to deploying biogas production

Partners	Relevance to EZCERTV and Description of Initiatives
	facilities in rural areas to replace fossil fuels. Both MARA led programs will facilitate the transition of China's rural villages and towns towards zero/low-carbon development.
Ministry of Finance (MOF)	The Ministry of Finance is responsible for handling economic regulations and fiscal policies, and for administering government public expenditures, including the national annual budget. Since 2017, as part of its mandate, the MOF is overseeing a recurring annual government budget allocated for the demonstration and deployment of clean technologies, based on indigenous renewable energy sources, to provide sustainable and efficient heating systems for the cold Northern rural regions in order to replace the use of fossil fuels, particularly coal. As part of this effort, the program also includes interventions on renovation of existing buildings to improve their thermal insulation and increase the efficiency of the energy use. Similarly to the two MARA-led programs described above, also this program managed by the MOF will contribute to the development of zero-carbon communities in rural areas.
Local Governments	Local provincial-level and country-level governments, especially from the eight provinces selected for the installation of the EZCERTV demos (Liaoning, Gansu, Shanxi, Anhui, Hubei, Hebei, Yunnan, and Beijing), will be engaged for the implementation of this proposed GEF-funded project. In particular, these local governments, and more specifically their Departments of Agriculture, will partner with MARA, MOF and UNDP for the advancements of the three government programs described above as well the eight demos designed under EZCERTV project. Therefore, the local governments not only will accomplish their mandate to foster and support socio-economic development of rural areas but will also contribute to their shifts towards zero-carbon development.
Private Sector Enterprises	Private sector investors and enterprises are very active in the implementation of RE and EE technologies and measures as well as the required ancillary equipment, including in rural areas. Several partners from the private sector will participate in the implementation of EZCERTV baseline projects, such as rooftop and ground-mounted solar PV installations completed with battery energy storage systems, pelletized, and bundled straws, biogas and bio-methane generation and distribution systems. Private investors will be extremely instrumental not only for the completion of the designed demos, but especially because they will be the backbone for the replication and scale-up of the demos in other rural villages and towns.

Risks:

As per standard UNDP requirements, the Project Manager will monitor risks quarterly and report on the status of risks to the UNDP Country Office. The UNDP Country Office will record progress in the UNDP ATLAS risk log. Risks will be reported as critical when the impact and probability are high (i.e. when impact is rated as 5, and when impact is rated as 4 and probability is rated at 3 or higher). Management responses to critical risks will also be reported to the GEF in the annual Project Implementation Review (PIR).

The risks that might prevent the project objectives from being achieved, during the project implementation, are listed as follows:

Table 3: EXCERTV Project Risks

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
The Project could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population in the rural areas where the on-the-ground project activities will be carried out.	Social Political Financial	I =3 P=1 Low	The designed demos are based on ongoing and planned rural revitalization projects of the national and provincial governments. In this case, there are generally no big issues that will prevent the demo implementation, or just minimal issues that are expected to be easily addressed. The project has been designed with extensive consultation with local people as well as the involvement of the relevant national and provincial government agency responsible for the formulation of relevant legislation on the installation of a decentralized RE-based power generation system or community mini-grids. A comprehensive Stakeholder Engagement Plan has been prepared during the PPG stage, with site-specific Plans prepared for each selected demo site.	PMU MARA	No Change
Potential occurrence of discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits.	Social Political Organizational	I =3 P =1 Low	The project could potentially offer a wide range of opportunities, for individuals and groups for support in the RE & EE technology applications demos and in the demo replications, participate and get trained in capacity development program activities, to be hired as a consultant or contractor for the project implementation. If care is not taken, any existing discrimination in the country towards woman could continue to come to play through the project. Special measures to ensure that any potential discrimination against women in the implementation of the project is countered have been identified and included in the project design. The results of such measures will be observed during project implementation, and if any corrective action is necessary, this will be addressed. Beyond that, the PDT will make special efforts to explore and facilitate the inclusion of interventions to enhance the role of women. Special activities have been designed to involve women, e.g., in the productive use of RE activities, to have strong representation of women in training courses and seminar-workshops, and to ensure a significant proportion of project consultants are women. The special measures are incorporated in the Gender Analysis and Action Plan that has been prepared for the EZCERTV Project.	PMU	No Change
Potential adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or	Environmental	I =3 P =3 Moderate	The demo RE-based power generation, distribution and storage systems may be sited on areas of habitat that could be adversely affected. The potential environmental impacts during the installation and operation of the demo RE-based power generation, distribution, and storage systems	PMU MARA	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
ecosystems and ecosystem services in some possible changes to the use of lands and resources in the implementation of the demonstrations.			depend on the state or condition of the selected sites. The potential environmental impacts during operation are mostly continuous, while those associated with construction activities are expected to be temporary and mostly reversible. The detailed feasibility studies (inclusive of SES screening) to be carried out during full project implementation will clearly articulate mitigation measures for any alterations of the general topographical and environmental nature of the demo sites. Following world class engineering and construction design, specific mitigation measures that are regarded as best practice with respect to environmental management of the systems that will be installed, ensuring compliance with the requirements of the applicable environmental regulations/legislation. Any other potential impacts and risks including mitigation measures will be elaborated in the limited environmental and social assessments to be conducted during project implementation for each demo in selected towns and villages and to be completed prior to any physical work beginning on the installation of the required hardware.		
The associated construction, operation, or decommissioning of the demo installations may have potential health and safety risks to local communities due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel, and other chemicals) that may be used during construction and operation.	Operational Environmental	I =2 P =4 Moderate	Construction aspects and operation of demo RE-based power generation, mini-grid systems, energy storage systems and the EE retrofits can pose safety risk. For example Li batteries for the demo solar PV power systems, energy storage systems can produce potentially dangerous waste products. The design of all demos involved site-specific environmental and social assessments and recommend measures to mitigate any identified health and safety risks. Appropriate training will be provided to the target beneficiaries of the demos to ensure that they will be able to operate the installed systems properly and safely in such a way that the release or handling of waste products are properly controlled and managed. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the minimization and management of waste generated from these demos (e.g., spent lead-acid batteries, spent Li batteries). *Please refer also to Annex G of the EZCERTV CEO Endorsement Request Document regarding the mitigation actions concerning the Covid-19 pandemic as this may impact on the project partners in the*	PMU MARA	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			project sites.		
The project would potentially result in the generation of waste (both hazardous and non-hazardous).	Environmental Technical	I =3 P =5 Moderate	The construction and operation of the RE-based energy generation (power and non-power applications) demos will generate wastes. PV panels and batteries will require disposal at end of life. EE retrofit projects (e.g., Use of LED lamps to replace CFLs/FLs) will potential, generate hazardous waste (Hg). The demos are designed considering the potential waste generation, and ensuring proper disposal of wastes from the construction, operation, and disposal stages. Disposal plans will be one of the requirements of the site-specific environmental and social impact assessment that will be conducted for each of demo. Those plans, for example, will include a disposal plan for the PV panels and batteries, which contain hazardous wastes, once their useful life is expired. For the EE retrofits, refrigerators are one example of an area that will require special plans for waste disposal in the E/S assessments. The refrigerators require proper disposal and may involve hazardous substances, particularly the refrigerant. Proper handling of the disposed refrigerants, as well as spent batteries, and Hg in spent CFLs/FLs will be incorporated in the operating manuals of these demo units.	PMU MARA	No Change
Some project activities / demo sites might be located in lands of ethnic minorities and might require FPIC.	Social Political Organizational	I =3 P =1 Low	The project activity or demo sites may be located in lands of ethnic minority and thus require a change in land use, such as a change in ethnic minority settlement patterns (scattered to centralized housing). The demos have been designed considering the potential impacts on land use in particular for ethnic minority lands. The project design documents need to be written in inclusive language so that minority groups can receive adequate knowledge sharing and training so that they fully understand the purpose and design logic of the project. If changes to minority land use are involved, the feasibility and cost-benefit should be carefully evaluated, and audience awareness and acceptance should be well researched. So far, all the demos have not involved changes in lads of ethnic minority.	PMU MARA	No Change
Some project activities/ demo sites might lead to physical displacement or forced evictions.	Social Political	I =4 P =1 Low	The construction of some large factories may require additional land that may need to be vacated by the relocation of residents (e.g. scattered dwellings to centralized residential buildings). The demos will be designed considering the potential land demand and take the local development planning as the references. So far, all the demos have	PMU MARA	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			not referred to the issue of forced evictions.		
Some project activities / demo sites might impact cultural heritage.	Social Political	I =2 P =2 Low	The substitution of different household energy resources may affect the transmission of certain cultures, for example, the food prepared by traditional wood stoves may lose its original taste due to the replacement of the cooking fuel; another example is the clay bed ("KANG") with wood stoves, which is widely used in the north, may be replaced in the process of energy substitution. The design and construction process of demonstration projects should consider the possible impact on traditional culture as much as possible, and some traditional applications (facilities) involving cultural heritage should be preserved and protected while introducing new low-carbon (RE or EE) technologies. For some application methods involving traditional culture, the comprehensive impact of all aspects, such as environmental and health effects, needs to be comprehensively assessed, and if the combined environmental health and social benefits are poor, its replacement can be considered.	PMU MARA	No Change
Risks to project workers during construction.	Organizational Operational Technical	I =3 P =2 Low	Project workers may face risks such as accidents and injuries during the construction process. The risks of workers' work during constructions that are involved in the designed demos will be further evaluated during the demo engineering design and measures will be identified and taken to avoid them, and insurance should be purchased for project workers.	PMU	No Change
Supply security risks (continuity, affordability) of renewable energy (or EE) alternatives to conventional energy for winter heating	Organizational Operational Technical	I =3 P =2 Moderate	Some renewable energy production has time discontinuity and uncontrollability due to the dependence on renewable resources, and against the background of limited development of energy storage technology, the replacement of traditional energy by renewable energy is likely to cause security risks in energy supply. This risk also exists for alternatives of energy efficient technologies, for example, after the replacement of coal to natural gas for winter heating in Hebei, there was a problem of insufficient natural gas supply in winter. The designed demos will be further evaluated during their engineering design. At that point, the RE resource supply security risks will be accurately and comprehensively assessed, including the continuity and economy of the RE energy supply (affordability for residents), and ensure a stable supply of the RE resource.	PMU	No Change
Risk of increased labor input by	Social Political	I =3 P =2	Some forms of energy substitution may require more labor input, such as traditional domestic biogas, and	PMU	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
women (e.g., will projects such as biomass fuels, biogas, etc., increase women's labor input for household energy access)	Financial	Low	households may need to invest more energy and time in collecting the raw materials for energy production, and these tasks are also likely to require more involvement of women. The design of the demos has considered resource, financial and labor inputs for energy supply substitution, and has assessed the feasibility of these inputs and their potential impact, and reduce end-user labor inputs, including through scaling up and centralizing production, to the extent possible, especially to avoid increasing the burden on women and children.		
Some project activities may require significant consumption of raw materials (e.g. biomass) and natural resources (energy and/or water)	Financial Technical	I =3 P =3 Moderate	Some renewable energy projects may present potential resource efficiency issues, such as whether biomass resources are being used more efficiently in biomass utilization projects; some renewable energy or energy efficiency project production may increase water demand, such as the high dependence of biomass power generation on water from a whole life cycle perspective. The design of the demos projects has fully considered the adequacy, availability and utilization efficiency of resources; for example, in biomass power projects, the utilization efficiency of biomass is much higher than that of traditional biomass (many straws are returned to the field); and for the energy and water consumption involved in RE/EE projects, the projects will conduct coupled analysis and accounting for the whole life cycle and water-energy nexus to avoid additional environmental pressure from new projects.	PMU MARA	No Change
The project management unit may not be able to effectively organize and coordinate the participation of relevant project stakeholders (e.g., relevant departments of local governments, specific project execution units, domestic and foreign experts, etc.).	Organizational, Operational	I = 4 P = 1 Low	The implementing agency of the project is the MARA, which is one of the main departments of the State Council's new rural construction management for rural revitalization. MARA has rich experience in GEF project design, implementation, and management. With this asset, the proposed project will be designed to minimize the chances of this risk from happening. If the PMU indeed become unable to prevent this risk, subject to the approval of the GEFSec, the UNDP China can assist in the procurement of internal/external technical experts who will work on the affected project activities. If need be, the affected activities may have to be modified to allow expeditious implementation and completion.	PMU MARA	No change
Inadequate and possibly	Organizational Operational	I=3 P=2	The MARA, NDRC and the project management unit shall maintain close working relations with relevant	PMU MARA	No change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
constrained coordination with the implementers of energy projects in local governments in the implementation of the project's barrier removal activities.	Regulatory	Moderate	departments of the State Council and can make necessary adjustments to the work in a timely manner according to the adjustment of national policies, which can ensure the implementation of the project and achieve optimal results. In case this happens, MARA/NDRC will facilitate discussions with project stakeholders and relevant government authorities through the project steering committee (PSC) to come up with decisions on the coordinated implementation of the agreed barrier removal activities.	NDRC	
The co-financing for the project activities may not be timely and enough for supporting project implementation.	Financial Political Strategic Operational	I=4 P=2 Medium	MARA attaches great importance to the implementation of the project and provides relevant technical, resource, capital, and institutional support. The relevant project provinces will also promise the project supporting funds in the early stage of project implementation. During the project design and implementation process, MARA will continuously promote the establishment of a diversified financing model to ensure that the project supporting funds are in place. In case this problem will occur, the reallocation of budget will be done to support the implementation of affected activities. This may entail the delivery of alternative outputs that are still contributing to the achievement of the relevant project outcome. Constant follow-up with the pertinent co-financers will be conducted either to secure the committed co-financing or renegotiate the amount of co-financing.	PMU MARA	Reducing
Rural communities may not support the project implementation promptly and sufficiently.	Organizational Operational	I=2 P=3 Low	Project includes strong promotional campaigns to target rural communities/towns, as well as outreach to relevant local governments to ensure their support. Liaison by capable project team personnel will further ensure support. Integration of productive use income generation opportunities will increase community interest in systems. If the project partners in the rural communities become remiss in their obligations and commitments to the project implementation, follow-up discussions between MARA/NDRC, and relevant local government leaders will be carried out to determine and resolve any issue.	PMU MARA NDRC	No Change
The private sector in the rural areas will not be interested in investing in green development technologies.	Financial Operational Strategic	I=4 P=1 Low	The project design, particularly the investment activities, will be coordinated with the relevant private sector entities in the target rural communities to involve them right away in the project planning, and get their inputs and comments in the design of the project activities. If this happens, the PMU, with the guidance of NDRC/UNDP will carry out direct consultations with	PMU MARA NDRC MOF	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
			target private sector entities to determine their concerns and explore ways to fulfill their initial commitments and ensure end-users better understanding about the use and benefits of EE/RE technology applications.		
Recommended policies may not be approved by the relevant authorities or may be approved but not effectively enforced.	Political	I=3 P=3 Moderate	The project will include the piloting of the application of the support policies for green town and village development to gauge the effectiveness of said policies. This will help guide the relevant government authorities in the finalization, approval, and effective enforcement of such policies. In case this happens, MARA/NDRC will facilitate discussions with project stakeholders and relevant government authorities through the project steering committee (PSC) to come up with decisions on expediting the approval, or reformulation, of the recommended policies/regulations.	PMU MARA NDRC MOF	No Change
Sustaining the outcomes and benefits of GEF investment on the activities implemented will not be fully sustained.	Technical Operational	I = 4 P = 2 Moderate	Sustainable follow-up plan for the replication of the demonstrated applicable and feasible EE/RE technologies associated with green town and village development in other provinces. The plan must be supported (financed) by the local governments. In case, despite the measures, sustainability of project outcomes and benefits is seen to be in jeopardy, MARA/NDRC, the project team, and the project steering committee will meet to come up with alternative measures to ensure sustainability.	PMU MARA NDRC	Reducing
Risks of non-availability of: (a) technical expertise and consequential changes in the project's schedule of activities; and (b) expected financing from the government and co-financing from all stakeholders during times of pandemic. *(Please refer to Annex G of the EZCERTV CEO Endorsement Request*	Environmental	I = 4 P = 1 Moderate	MARA will make use of its wide network of both external and internal expertise (individuals or groups of individuals) who they work with on project design, implementation, and management. A clear anticipation and understanding of the type and quality expertise that are needed, their locations and when these will be required will help potential alternatives in case these will be necessary. With such an asset, the chances of this risk from happening is minimized, and impacts such as delays in the implementation of scheduled/planned project activities can be prevented. MARA (REEA) and the PMO will: (a) closely monitor the public health situation in the project sites; and also (b) regularly update its list of technical experts that can be tapped to worked on the project implementation promptly and quickly or with minimal delays. If this risk cannot be prevented, there is also the option, subject to the approval of the GEFSec for UNDP China to assist in the procurement of internal or external experts who will work on the affected project activities. If need be, the affected activities may also be modified to allow expeditious implementation and completion.	MARA PMO	No Change

Description	Type	Impact & Probability	Mitigation Measures	Owner	Status
Document regarding addressing Covid-19 issues)			Regarding the potential risks of the non-availability in a timely manner of the committed financing and co-financing during pandemic situations, During the implementation process, MARA will continuously promote the establishment of a diversified financing model to ensure that the project supporting funds are in place. In case this problem will occur, the reallocation of budgets will be done to support the implementation of affected activities. This may also entail the delivery of alternative outputs that are still contributing to the achievement of the relevant project outcome. Constant follow-up with the pertinent co-financers will be conducted either to secure the committed co-financing or renegotiate the amount of co-financing.		
RE-based energy generation (power and non-power purposes) and EE system installations can be seriously affected by adverse climate-related events[12]. *(Please refer to Annex H of the EZCERTV CEO Endorsement Request Document regarding the project's Climate Risk Assessment)*	Environmental	I = 4 P = 1 Moderate	It is already common in international design and engineering practices, as well as in the construction/installation of RE-based energy generation units to follow proper engineering and construction design and construction that ensure not only structural integrity but also climate resilience. This applies also in the procurement, design/engineering, installation, and operation of the pertinent installations[13]. Depending on the extent of the impacts of the adverse climate –related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.	MARA PMO	No Change

A preliminary Social and Environmental Safeguard screening was conducted during the PIF preparation and it has been reviewed and updated during this project design process and included here as Annex 4. The overall social and environmental risk rating of the project is ***Substantial***. Because of this rating, all EXCERTV activities will be preceded by the preparation of an Environmental and Social Management Plan (ESMP). Since the completion of the ESMP is propaedeutic to any activities, it is important to

[12] Please refer to Annex E of the EZCERTV CEO Endorsement Request Document.

[13] The design and construction/installation of the physical infrastructures that will be installed will be based on the technical and structural specifications that major bilateral and multi-lateral donors require for the infrastructure projects that they are funding in China.

complete the plan during the first year of EZCERTV implementation. The activities that will have the strongest impact will be the demo projects. For example:

1. Environmental:
   a. All special waste generated by implementing RE and EE technologies will have to be properly disposed (i.e., battery banks for RE electricity storage, solar PV panels, incandescent lightbulbs, old households cooking and heating equipment, etc.).
   b. Locations for the biogas and bio-methane reticulation systems as well as for the ground-mounted PV solar field do not have a negative impact on the environment.

2. Social:
   a. Waste disposal facilities must be chosen without affecting the health and safety of local population.
   b. Health[14] and socio-economic benefits must be redistributed as uniformly as possible among the rural community members and fully integrate women, marginalized and youth groups.

Environmental and social grievances (if any) will be reported to the GEF during the annual Project Implementation Review (PIR) reporting exercise.

Stakeholder Engagement Plan:

Based on the stakeholder analysis, the project's key players include MARA, REEA, MOF, NDRC, MEE, local governments, and the private sector. They will take active part in the implementation of the EZCERTV project activities while the others will assume either supporting or beneficiary roles.

Table 4: Stakeholder Roles & Responsibilities in Project Implementation

Stakeholder	Roles and Responsibilities in Project Implementation
Ministry of Agriculture and Rural Affairs (MARA) of the People's Republic of China	MARA has been designated as the implementing partner for EZCERTV and will chair the Project Steering Committee. Within MARA, the Department of Science, Technology and Education (DSTE) will be the leading agency responsible for guiding the project implementation phase, working with the PMU to resolve key issues in the project design, provide inputs and support for all activities, particularly those related to enhancing national and provincial policy and institutional frameworks that are supportive of low carbon development in rural areas, and assure the appropriate flow of co-financing. It is also responsible for effective coordination and communication with UNDP, the Ministry of Finance and other relevant government departments.
United Nations Development Programme (UNDP)	As the implementing agency of GEF, UNDP will be a member of the Project Steering Committee and will be responsible for project coordination, implementation, monitoring and management. Furthermore, UNDP will actively participate in the design, organization, and administration of capacity building programs and in the implementation of awareness raising and project promotion activities.
Rural Energy & Environment Agency (REEA)	REEA is an institution operating directly under MARA. REEA is entrusted by the DSTE as the fund manager for international projects and will undertake the day to day implementation and management responsibilities of EZCERTV project. A

[14] Please refer also to Annex G of the EZCERTV CEO Endorsement Request Document regarding the mitigation actions concerning the Covid-19 pandemic as this may impact on the project partners in the project sites.

Stakeholder	Roles and Responsibilities in Project Implementation
	Project Management Office (PMO) will be established under REEA to provide technical support and coordination of the PMO and assure the timely preparation of reports on project achievements in line with UNDP rules, procedures, and quality standards.
Ministry of Finance (MOF) of the People's Republic of China	The Ministry of Finance is the GEF Operational Focal Point in China and is one of the GOC signatories to the EZCERTV project document. During the project implementation phase, MOF will be a member of the project steering committee and will be the recipient of the GEF grant on behalf of the GOC. MOF will play a pivotal role in: providing high-level strategic guidance to the PMU; coordinating and supervising all disbursements and expenses of the GEF grant; evaluating the project performance; and finally in summarizing and promoting the project results.
National Development and Reform Commission (NDRC)	NDRC is a government agency responsible for the formulation and implementation of national economic and social development strategies, as well as annual, medium, and long-term development plans. The agency will be a development partner of the EZCERTV project and a member of the Project Steering Committee. Its main contributions will be the promotion of zero/low-carbon technologies to the rural communities to stimulate their active participation, and provision of advice to the government on energy resources conservation, comprehensive energy utilization planning, and policy measures.
Ministry of Ecology and Environment (MEE) of the People's Republic of China	The Ministry of Ecology and Environment will be involved in promotion activities to stimulate rural communities and other key players' participation in the EZCERTV project. MEE experience and expertise will guarantee the environmental sustainability of the project activities and will assure that they comply with the UN Framework Convention on Climate Change requirements.
All China Women's Federation (ACWF) and provincial women's federations	As one of the most prominent and historical non-governmental women's right organizations in China, ACWF will be engaged in EZCERTV activities to assure that women will actively participate in the project implementation. The NGO will be a member of the Project Steering Committee and will assure women's needs are properly addressed, which will lead to an improvement of working and living conditions for everybody. Local provincial chapters of ACWF will also be involved in EZCERTV activities to maximize the impact of training programs and awareness raising initiatives.
Local governments	Local governments at different levels will participate in the project, particularly from the eight selected provinces: Anhui, Beijing, Gansu, Hebei, Heilongjiang, Hubei, Qinghai, Sichuan, and Yunnan. Provincial-level agricultural departments and county-level governments will be actively engaged for the implementation of baseline and demonstration projects and subsequent monitoring and evaluation activities. Furthermore, local government officials will participate into capacity building programs and awareness raising initiatives.
Relevant Scientific Research Institutions	Several prestigious scientific institutions will be involved in the project implementation, including: the Chinese Academy of Agricultural Engineering Planning & Design; the Chinese Academy of Building Research; the Agricultural Science Institute; the Beijing Haoying Environmental Science and Technology Development Center; the Chinese Academy of Sciences; the Tsinghua University; and the Renmin University of China. These stakeholders will be primarily engaged for the design, organization, and administration of capacity building initiatives, and for the implementation, replication and scale-up of demos.
Private sector	Stakeholders from the private sector will be involved in several project activities, such as the implementation of baseline and demonstration projects as well as their replication and scale-up, and the establishment of a technologies supply and service market. The private sector will help to create and fulfill market demand

Stakeholder	Roles and Responsibilities in Project Implementation
	concerning RE & EE technologies and measures and to provide alternative sources of financing. Key players include: China Association of Rural Energy Industry, Ltd; Hebei Jing'an Bioenergy Technology Co., Ltd.; Shanxi Golden Cooperate DC Power Distribution Engineering Technology Co., Ltd.; Chaoyang Jiajiadian Farm Co., Ltd.; Hailun Limin Energy-saving Boiler Manufacturing Co., Ltd.; Songzi Senchun Agricultural Development Co., Ltd.; Yunan Fu Hui Technology Co., Ltd.; and Ningxia Runwell Energy & Environment Engineering Co, Ltd.
NGOs and rural towns and villages social communities and groups	NGOs and rural communities will participate in all activities related to RE & EE technology promotions and awareness raising initiatives. Assistance will be requested by groups such as: the China Association of Rural Energy Industry; the China Energy Conservation Association; the China Association of Agricultural Science Societies; the China Biogas Society; the Beijing Energy Association; and the China Bricks & Tiles Industrial Association.

Annex 9 presents the EZCERTV Project's detailed Stakeholder Engagement Plan.

South-south and triangular cooperation to present opportunities for replication in other countries:

In addition, to bring the voice of China to global and regional fora, the EZCERTV project will explore opportunities for meaningful participation in specific events where UNDP could support engagement with the global development discourse on zero-carbon rural villages and towns development. The project will furthermore provide opportunities for regional and international cooperation with countries that are implementing similar initiatives on zero-carbon rural villages and towns development in geopolitical, social, and environmental contexts relevant to the proposed project in China. This international cooperation aspect is particularly important for the EZCERTV project because of the leading role played by China towards countries along the Belt & Road. In fact, several project activities have been designed and will be implemented in these Belt & Road countries and foundations will be laid to foster future cooperation and further developments.

Gender equality and Women's Empowerment:

In rural area of China, due to natural conditions, historical culture and traditional customs, many families have relatively obvious social division of labor, i.e., the family division mode of "male outside, female inside", and adult male family members are responsible for relative heavy physical labor and the majority of the family's economic income sources, while adult female family members are responsible for the entire family's dietary life. To put it another way, in a typical rural Chinese family, women's main work is done under indoor environmental conditions, including cooking and heating. Thus, through addressing the barriers in clean energy supply in the construction of villages, on one hand, it can ease the labor intensity of rural women in access and using energy; one the other hand, project implementation can also improve indoor air quality, and strengthen women's health.

The project also enables improvement of the working and living conditions of rural women who are typically responsible for household chores through clean energy supply that improve indoor air quality and improve women's health. The envisioned project activities will significantly improve living environment conditions, reduce threats and damages to rural residents caused by energy-related indoor pollution (carbon monoxide, haze, and dust) cause by using traditional energy sources. The project also facilitates rural household solid and liquid waste management through their processing and utilization

as energy resources. In addition, this project will have a positive impact on rural women. Through targeted training, this project will improve women's energy conservation and environmental protection skills at the project site and promote them to apply low energy consumption technology and clean-living practices, thereby reducing the use of traditional rural energy sources and reducing rural living resources to increase their economic income.

The proposed GEF project presents opportunities for the involvement of women in the deployment of low carbon technologies and mitigation options and come up with gender-sensitive policies in the energy consuming operations and processes in China is an essential feature of the design and preparation of this proposed project. This also includes putting the potential contributions of women in the management and implementation of measures to mitigate climate change. Adequate consideration has been done during the design and preparation of this project to enhance chances for the engagement of the technical services of women professional and indigenous people if there are opportunities to involve them.

Innovativeness, Sustainability and Potential for Scaling Up:

Innovativeness: The innovativeness of the proposed project resides in two main aspects: 1) this project promotes and demonstrates the application of zero carbon villages and towns development in China rural areas. Such development is expected to lead to green development as advocated by the country's rural revitalization strategy. This will involve the enhanced use of available indigenous renewable and zero GHG emitting energy sources to meet the energy demand of rural towns and villages. The proposed project introduces a model for rural villages and towns development that involves the application of appropriate technological, institutional and policy-oriented options that increase the community's climatic resilience, productivity and income of its citizens, while significantly reducing emissions of greenhouse gases (GHGs); and 2) the project demonstrates another novel feature, which is the application of decentralized commercial RE-based power generation systems in rural villages and towns. As part of low carbon development of rural areas, model low carbon villages will be established as demos where appropriate low carbon energy technologies applications will be showcased to other rural towns to replicate and emulate. This will also include the demonstration of the application of: a) energy storage as supplemental component of the decentralized RE-based power generation system; b) smart mini/micro-grid distribution system; c) high energy efficient appliances and space heaters; and c) clean energy distribution systems. In addition, energy-integrated rural development planning will be part and parcel of the zero-carbon rural development process.

Sustainability: The proposed GEF project will develop and work towards the adoption of supportive policies/regulations and institutional mechanisms focused on low carbon development planning, incentive mechanisms for decentralized commercial RE-based energy generation, technical standards, and specifications for EE and RE technologies, that contributes to sustainable rural development in China. The objective for ecological environmental protection and renewable energy development in the countryside is a significant component of the country's sustainable agricultural and rural development and is one of the major drivers for the sustained efforts towards green rural villages and towns development. To ensure avoidance of the recurrence of the barriers and the continuance of the enabling environments that will be created and/or facilitated by the project, appropriate sustainable follow-up actions will be planned as part of the project activities. The establishment of China Rural Distributed Renewable Energy Construction and Development Industry Association will also sustain efforts of the rural villages and towns to replicate the low carbon energy technology application models.

Under this project, the development of this association into something like a national RE/EE Accelerator, will also help ensure the sustainability of rural zero/low-carbon development efforts for the promotion and servicing of efforts towards EE rural buildings and industries in China.

Scale-up Potential: The proposed GEF project will lay a solid basis for scaling up with its exploration of the effective models and mechanics for decentralized RE-based energy generation, energy efficiency technologies and measures, as well as RE and EE financing models that will support low carbon development of rural towns and villages. These provide the possibility for scaling-up the application of these models/schemes in other regions of China. The proposed project can be scaled up in the following directions: (1) application of low carbon development policies, standards, IRRs and best practices in rural towns and villages in other provinces; 2) application of the successful demo decentralized RE-based power generation and distribution and EE technology applications in other rural towns/villages; and (3) enhancement of the low carbon village models to low carbon town models, The scale-up will also be made possible through extensive information sharing, technical assistance, ICE programs, and promotional campaigns. In addition, all these efforts have high potential to be replicated and scaled-up also in countries along the “Belt and Road”. In fact several activities of EZCERTV have been devised to lay the ground to replicate the proposed model for rural zero-carbon development in these countries assisted by China.

V. Project Results Framework

This project will contribute to the following Sustainable Development Goal (s): Goal 7: "Ensure access to affordable, reliable, sustainable and modern energy for all"; Goal 8: "Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all"; Goal 11: "Make cities and human settlements inclusive, safe, resilient and sustainable"; Goal 13: "Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy"; and to a lesser extent to Goal 2: "End hunger, achieve food security and improved nutrition and promote sustainable agriculture".				
This project will contribute to the following country outcome (UNDAF/CPD, RPD, GPD): UN Development Assistance Framework China (2016-2020): More people enjoy a cleaner, healthier, and safer environment as a result of improved environmental protection and sustainable green growth (Outcome 2).				
	Objective and Outcome Indicators	**Baseline**	**Mid-term Target**	**End of Project Target**
Project Objective: Acceleration of zero-carbon transformation in China's rural area to contribute to global climate change mitigation and the achievement of the United Nations sustainable development goals (SDG)[15]	**Mandatory** Indicator 1: # direct project beneficiaries disaggregated by gender (individual people)	0	250,000 (women 128,000) (men122,000	918,500 (women 460,000) (men 458,500)
	Mandatory GEF Core Indicators: Cumulative GHG emission reduction from the rural sector of China, tCO_2	0	100,000	4,079,000
	Cumulative reduction in fossil fuel consumption due to the implemented zero/low carbon energy technologies and techniques in rural areas, tce	0	11,297.8	446,223.6
	% RE generation in rural areas in China	0	75%	85%
Project Component 1. Policy Formulation and Institutional Mechanisms on Zero Carbon Town and Village Development				
Project Outcome 1: Effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China	Number of approved and enforced new and updated zero/low carbon policies, at both national and local levels	0	5	10
	Cumulative number of rural towns/villages that adopt and enforce developed guidelines and standards on RE/EE/EC technologies and measures	0	8	50
Outputs to achieve Outcome 1	Output 1.1: Formulated, approved, and enforced policies on the promotion and support of zero carbon town and village development in line with China's rural revitalization strategy and goal of achieving carbon neutrality before 2060. Output 1.2: Established and operational institutional framework to promote and support zero-carbon town and village			

[15] The EZCERTV Project is in line with the GEF-7 climate change mitigation (CCM) objective of promoting innovation and technology transfer to achieve sustainable energy breakthroughs, and specifically to the following CCM objectives: (1) Decentralized renewable power with energy storage; and (2) Accelerating energy efficiency adoption. While it will be contributing to the achievement of SDGs 7, 8, 11, 13 and 2, the main SDG targeted is Goal 7: "Ensure access to affordable, reliable, sustainable and modern energy for all"; and to some extent Goal 13: "Take urgent action to combat climate change and its impacts by regulating emissions and promoting developments in renewable energy". Hence, apart from the general gender equity core indicator, the indicators of the Project Objective are related to GHG emission reduction and the extent of application of renewable energy interventions.

	development. Output 1.3: Formulated, approved, and enforced technical, management and operational standards in rural renewable energy (RE) development and utilization, energy storage, energy conservation and energy efficiency (EC&EE) and comprehensive energy management. Output 1.4: Completed assessment of renewable energy resource endowment, economic and social development level, and energy use trends and patterns in rural areas of selected provinces; and formulated energy-integrated development plans of pilot towns in selected 8 provinces.			
Project Component 2. Integrated Renewable Energy and Energy Efficiency Technologies Applications for Zero Carbon Town and Village Development				
Project Outcome 2: Low/zero carbon technology applications are widely implemented in rural towns and villages in rural areas of China	Cumulative number of rural towns/villages that replicate and scale-up the implemented demo projects	0	20	200
	Cumulative number of private sector entities investing in zero/low carbon energy in rural towns/villages	0	50	200
Outputs to achieve Outcome 2	Output 2.1: Completed and operational zero carbon technology demonstration involving the installation of about 167.2 MW of RE-based power generation capacity in 120 rural villages and 4 towns for energy production and utilization showcasing the cost-effective application of new RE delivery mechanisms decentralized RE-based energy generation and distribution (through mini/micro-grids) and energy storage. Output 2.2: Completed and operational demonstrations of the cost-effective applications of EC&EE technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development. Estimated annual energy savings of about 6,487.0 TJ/year in 4 towns and 120 villages. Output 2.3: Developed, published, and disseminated zero carbon town and village development guidelines to replicate and scale-up the impact of zero-carbon villages and towns.			
Project Component 3. Facilitating Rural Zero Carbon Development and Lifestyle Popularization and Knowledge Dissemination				
Project Outcome 3: Enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China	Cumulative number of local firms in rural areas that can capably provide technical, engineering and maintenance services for zero/low carbon technology application projects	0	20	100
	Number of stakeholders from BRI countries trained on zero-carbon/low-carbon energy technologies and zero-carbon villages and towns	0	100	500
Outputs to achieve Outcome 3	Output 3.1: Completed information, communication, and education (ICE) program on zero carbon development in rural towns and villages in China for local authorities, service companies, rural energy cooperatives, public institutions, private entrepreneurs, farmers, and the public. Output 3.2: Established international cooperation and knowledge sharing program in zero-carbon towns and villages development and green energy transformation with countries along the "Belt and Road", to support these countries in achieving their nationally determined contributions (NDC) commitments to the "Paris Agreement". Output 3.3: Established social mechanism for the promotion of zero-carbon town and village development and the			

	sustainability of renewable energy industries. Output 3.4: Established and operational market-oriented mechanism for the enhanced development and utilization of rural RE resources, EC&EE improvement, and comprehensive energy management systems for supporting sustainable socio-economic development, and creation of zero carbon towns and villages in rural China. Output 3.5: Established and operational investment and financing mechanisms for supporting the commercial viability and operation of zero-carbon towns and villages and the development of rural renewable energy industries.

VI. Monitoring and Evaluation (M&E) Plan

The project results, corresponding indicators and mid-term and end-of-project targets in the project results framework will be monitored annually and evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. The Monitoring Plan included in Annex details the roles, responsibilities, and frequency of monitoring project results.

Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP and UNDP Evaluation Policy. The UNDP Country Office is responsible for ensuring full compliance with all UNDP project monitoring, quality assurance, risk management, and evaluation requirements.

Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[16]. The costed M&E plan included below, and the Monitoring plan in Annex 3, will guide the GEF-specific M&E activities to be undertaken by this project.

In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed during the Project Inception Workshop and will be detailed in the Inception Report.

Additional GEF monitoring and reporting requirements:

Inception Workshop and Report: A project inception workshop will be held within 60 days of project CEO endorsement, with the aim to:

a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
c. Review the results framework and monitoring plan.
d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
g. Plan and schedule Project Board meetings and finalize the first-year annual work plan.
h. Formally launch the Project.

GEF Project Implementation Report (PIR):

[16] See https://www.thegef.org/gef/policies_guidelines

The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. Any environmental and social risks and related management plans will be monitored regularly, and progress will be reported in the PIR. The PIR submitted to the GEF will be shared with the Project Board. The quality rating of the previous year's PIR will be used to inform the preparation of the subsequent PIR.

GEF Core Indicators:

The GEF Core indicators shown in Annex 16 will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the project team is responsible for updating the indicator status. The updated monitoring data should be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent ground truthing. The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

Independent Mid-term Review (MTR):
The terms of reference, the review process and the final MTR report will follow the standard templates and guidance for GEF-financed projects available on the UNDP Evaluation Resource Center (ERC).

The evaluation will be 'independent, impartial and rigorous'. The evaluators that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing, or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project under review.

The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the evaluation process. Additional quality assurance support is available from the BPPS/GEF Directorate.

The final MTR report and MTR TOR will be publicly available in English and will be posted on the UNDP ERC by the 2nd quarter of 2024. A management response to MTR recommendations will be posted in the ERC within six weeks of the MTR report's completion.

Terminal Evaluation (TE):
An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance for GEF-financed projects available on the UNDP Evaluation Resource Center.

The evaluation will be 'independent, impartial and rigorous'. The evaluators that will be hired to undertake the assignment will be independent from organizations that were involved in designing, executing, or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated.

The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/GEF Directorate.

The final TE report and TE TOR will be publicly available in English and posted on the UNDP ERC by the 3rd Quarter of 2026. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

Final Report:
The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information**:** To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgement to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[17] and the GEF policy on public involvement[18].

Table 5: Monitoring and Evaluation Plan and Budget

GEF M&E requirements	Indicative costs (US$)	Time frame
Inception Workshop	20,000	Within 60 days of CEO endorsement of this project.
M&E of GEF core indicators and project results framework	5,000 per year	Annually and at mid-point and closure.
GEF Project Implementation Report (PIR)	None	Annually typically between June-August
Monitoring all risks	3,000 per year	On-going.
Monitoring of social and environmental safeguard screening	3,000 per year	On-going.
Monitoring of stakeholder engagement plan	3,000 per year	On-going.
Monitoring of gender action plan	3,000 per year	On-going.
Project Board Meetings	3,000 per year	Annually
Reports of Project Board Meetings	None	Annually
Lessons learned and knowledge generation	5,000 per year	Annually
Supervision missions	None	Annually
Oversight missions	None	Troubleshooting as needed
Mid-term GEF and/or LDCF/SCCF Core indicators and METT or other required Tracking Tools	15,000	Before mid-term review
Independent Mid-term Review (MTR)	35,000	2Q 2024
Terminal GEF and/or LDCF/SCCF Core indicators and METT or other required	15,000	Before terminal evaluation

[17] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[18] See https://www.thegef.org/gef/policies_guidelines

Table 5: Monitoring and Evaluation Plan and Budget		
GEF M&E requirements	**Indicative costs (US$)**	**Time frame**
Tracking Tools		
Independent Terminal Evaluation (TE)	35,000	3Q 2026
Translation of MTR and TE reports into English	None	As needed
TOTAL indicative COST	**245,000**	

VII. Governance and Management Arrangements

Roles and responsibilities of the project's governance mechanism:

The EZCERTV Project will be implemented in accordance with the UNDP's national implementation modality (NIM). In this implementation modality, the project is implemented by the government. UNDP identifies a suitable national entity as Implementing Partner (IP) to implement the project. Where necessary other national/local government entity/entities can be designated as responsible partner(s) to assist the IP. The IP directly assumes the responsibility for the delivery of the project outputs and carries out all activities towards the achievement of these outputs. UNDP has no role in the execution of the project activities. Its role is limited to overseeing the project implementation on behalf of the GEF. In NIM, the government regulations, rules and procedures shall apply to project implementation to the extent that they do not contravene the principles of the financial regulations and rules of UNDP and GEF, and unless requested by the IP, in which case UNDP rules shall apply.

UNDP (as the GEF Agency) oversees the implementation of the project. It will not be executing any project activity apart from the required GEF Agency project cycle management services. It will ensure compliance with the specific instructions that must be followed regarding the financial management requirements to be complied with the IP, as well as ensure that the GEF Minimum Fiduciary Standards Requirements are met by the IP and other designated responsible party/parties at all stages of project implementation.

Implementing Partner: The Implementing Partner for this project is the Ministry of Agriculture and Rural Affairs (MARA).

The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

The Implementing Partner is responsible for executing this project. Specific tasks include:

- Project planning, coordination, management, monitoring, evaluation, and reporting. This includes providing all required information and data necessary for timely, comprehensive, and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
- Risk management as outlined in this Project Document;
- Procurement of goods and services, including human resources;
- Financial management, including overseeing financial expenditures against project budgets;
- Approving and signing the multiyear workplan;
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

Responsible Parties: During the 5-year implementation period of the EZCERTV Project, the MARA, as the implementing partner will require the cooperation and services of several responsible parties to implement or support the implementation of the project activities.

MOF will receive the GEF grant for the implementation of the EZCERTV project on behalf of the Chinese Government. MOF will be a member of the Project Steering Committee, and it will have a key role in providing guidance, coordination and supervision on the disbursement and expense of GEF grant, performance evaluation, summary, and promotion of project results.

REEA will, as the day-to-day project implementer and fund manager of the Implementing partner MARA, establish a PMO which will provide technical support and coordination to the project team. REEA will be engaged in as the project coordinator in the majority of the designed EZCERTV activities.

MEE will be involved in promotion activities to stimulate rural communities and other key players' participation in the EZCERTV project. MEE experience and expertise will guarantee the environmental sustainability of the project activities and will assure that they comply with the UN Framework Convention on Climate Change requirements.

NDRC's main contributions to the EZCERTV project will be the promotion of zero/low-carbon technologies to the rural communities to stimulate their active participation, and provision of advice to the government on energy resources conservation, comprehensive energy utilization planning, and policy measures.

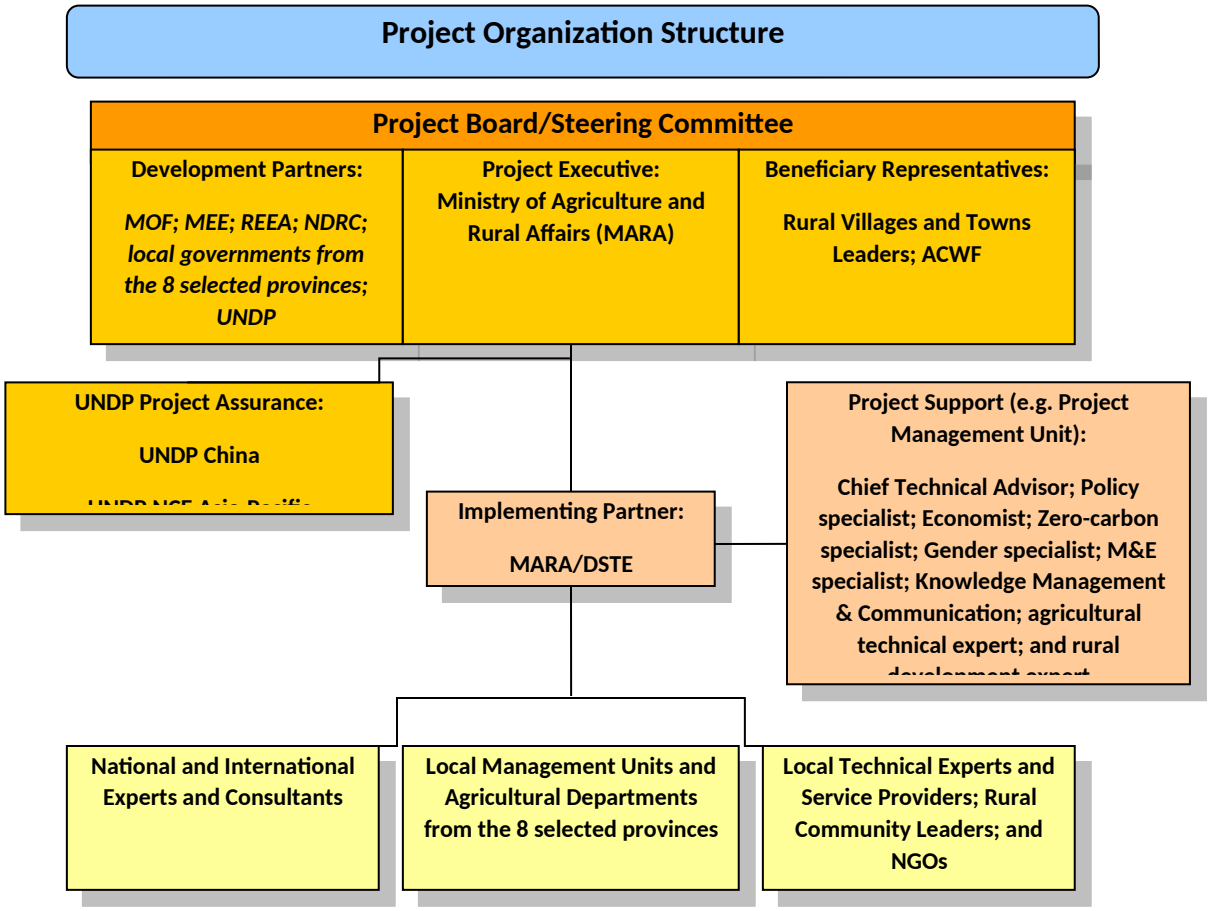
ACWF and provincial women's federations will be engaged in EZCERTV activities to assure that women will actively participate in the project implementation and that women's needs are properly addressed, which will lead to an improvement of working and living conditions for everybody. ACWF and the provincial chapters will also participate in capacity building programs as well as promotion and awareness raising initiatives to promote zero-carbon rural development.

Local governments at provincial and county levels, and particularly their agricultural departments, will be actively engaged for the implementation of baseline and demonstration projects and subsequent monitoring and evaluation activities. Furthermore, local government officials will participate into capacity building programs and awareness raising initiatives.

Project stakeholders and target groups: For project stakeholders and target groups, please refer to the Stakeholder Engagement Plan in section IV and to Annex 9.

UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes oversight of project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Board and attends Project Board meetings as a non-voting member.

Project organization structure:



Second line of defense

- Regional Bureau oversees RR and Country Office compliance at portfolio level.
- BPPS NCE RTA oversees technical quality assurance and GEF compliance. BBPS NCE RTA oversees RTA function.
- UNDP GEF Executive Coordinator and Regional Bureau Deputy Director can revoke DOA/cancel/suspend project or provide enhanced oversight.

The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

Project Board: All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to

ensure quality delivery of results. The Project Board (also called the Project Steering Committee) is the most senior, dedicated oversight body for a project.

The two main (mandatory) roles of the project board are as follows:

1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure the project achieves the desired results.
2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

Requirements to serve on the Project Board: *to be included in the TOR of the Project Board*

- ✓ Agree to the Terms of Reference of the Board and the rules on protocols, quorum and minuting.
- ✓ Meet annually; at least once.
- ✓ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✓ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✓ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

Responsibilities of the Project Board: *to be included in the TOR of the Project Board*

- ✓ Consensus decision making:
  - The project board provides overall overall guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.
  - Review project performance based on monitoring, evaluation and reporting, including progress reports, risk logs and the combined delivery report;
  - The project board is responsible for making management decisions by consensus.
  - In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.
  - In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.
- ✓ Oversee project execution:

- Agree on project manager's tolerances as required, within the parameters outlined in the project document, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded.
- Appraise annual work plans prepared by the Implementing Partner for the Project; review combined delivery reports prior to certification by the implementing partner.
- Address any high-level project issues as raised by the project manager and project assurance;
- Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies);
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
- Track and monitor co-financed activities and realisation of co-financing amounts of this project.
- Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.

✓ Risk Management:

- Provide guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks.
- Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
- Address project-level grievances.

✓ Coordination:

- Ensure coordination between various donor and government-funded projects and programmes.
- Ensure coordination with various government agencies and their participation in project activities.

The composition of the Project Board must include the following roles:

a. Project Executive: Is an individual who represents ownership of the project and chairs the Project Board. The Executive is normally the national counterpart for nationally implemented projects. MARA, as the implementing partner, represents the Project Executive, and chairs the Project Board. The Executive is ultimately responsible for the project, supported by the Beneficiary Representatives and the Development Partners. The Executive's role is to ensure that the project is focused throughout its life cycle on achieving its objectives and delivering outputs that will contribute to higher level outcomes. The Executive must ensure that the project gives value for money, ensuring cost-conscious

approach to the project, balancing the demands of Beneficiary Representatives and Development Partners.

Specific Responsibilities: (as part of the above responsibilities for the Project Board)

- Ensure that there is a coherent project organization structure and logical set of plans.
- Set tolerances in the AWP and other plans as required for the Project Manager.
- Monitor and control the progress of the project at a strategic level.
- Ensure that risks are being tracked and mitigated as effectively as possible.
- Brief relevant stakeholders about project progress.
- Organize and chair Project Board meetings.

b. Beneficiary Representatives: Individuals or groups representing the interests of those who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. The Beneficiary representatives are: Rural Villages and Towns Leaders

The Beneficiary Representatives are responsible for validating the needs and for monitoring that the solution will meet those needs within the constraints of the project. The Beneficiary Representative role monitors progress against targets and quality criteria. This role may require more than one person to cover all the beneficiary interests. For the sake of effectiveness, the role should not be split between too many people.

Specific Responsibilities (as part of the above responsibilities for the Project Board)

- Prioritize and contribute beneficiaries' opinions on Project Board decisions on whether to implement recommendations on proposed changes.
- Specification of the Beneficiary's needs is accurate, complete, and unambiguous.
- Implementation of activities at all stages is monitored to ensure that they will meet the beneficiary's needs and are progressing towards that target.
- Impact of potential changes is evaluated from the beneficiary point of view.
- Risks to the beneficiaries are frequently monitored.

c. Development Partners: Individuals or groups representing the interests of the parties concerned that provide funding and/or technical expertise to the project. The Development Partners are: MOF, MEE, REEA, NDRC, ACWF, local governments, private sector, rural communities, and UNDP CO.

The Development Partners' primary function within the Board is to provide guidance regarding the technical feasibility of the project.

Specific Responsibilities (as part of the above responsibilities for the Project Board)

- Make sure that progress towards the outputs remains consistent from the Development Partners perspective.

- Promote and maintain focus on the expected project output(s) from the point of view of project development management.
- Ensure that the technical and financial resources required for the project are made available.
- Contribute technical and financial opinions on Project Board decisions on whether to implement recommendations on proposed changes.
- Arbitrate on, and ensure resolution of, any technical and financial priority or resource conflicts.

As a development partner the specific responsibilities of UNDP are also in line with its oversight services as the GEF Agency for this project. This oversight role involves ensuring that actions carried out by the implementing partner in regards its management and execution of the project are as per the signed project document, aligned to the country program, and do not contravene the principles of the financial regulations and rules of UNDP and GEF. To emphasize, UNDP has no role in the execution of any project activity nor any project management task. Its role is limited to overseeing the project implementation on behalf of the GEF.

d. Project Assurance: Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including compliance with the risk management and social and environmental standards of UNDP. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. Project assurance is totally independent of project execution.

Governance role for project target groups: The project target groups will be engaged in decision making for the project through the Project Board consultation and meetings to be coordinated by the Project Management Office. The broader governance environment may include various stakeholder groups. The engagements will also be done though the activities under each project component relevant to policy making, data gathering, participation and procedural documentation. In this way, the project's governance will be integrated within the wider governance arena considering the capacity building of project management capability in the project organization as well as the target groups.

Project Management Office (PMO): MARA will establish a Project Management Office (PMO) to be based in MARA and recruit its personnel for the day-to-day management of project activities. The PMO to be headed by a Project Manager (PM) will provide administrative, technical, management and coordination roles in collaboration with MARA, and other project partners (e.g., for demonstrations), provincial officers and provincial government institutions, and UNDP. The UNDP will provide induction and regular training to PMO staff on UNDP procedures and policies so that they may effectively execute their roles and responsibilities.

VIII. Financial Planning and Management

The total cost of the project is *USD 98,532,420.* This is financed through a GEF grant of *USD* 8,932,420 administered by UNDP, and US$ 89,600,000 co-financing (*USD 88,350,000* cash, and USD 1,250,000 in-kind). UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

Confirmed Co-financing: The actual realization of project co-financing will be monitored during the *mid-term review* and terminal evaluation process and will be reported to the GEF. Note that all project activities included in the project results framework that will be delivered by co-financing partners (even if the funds do not pass through UNDP accounts) must comply with UNDP's social and environmental standards. Co-financing will be used for the following project activities/outputs:

Table 6: List of Project Co-Financing

Co-financing source	Co-financing type	Co-financing amount (US$)	Planned Co-financing Activities/Outputs	Risks	Risk Mitigation Measures
Ministry of Agriculture and Rural Affairs (MARA) and Local Governments	Grant	17,000,000	The GOC supports and incentivize the utilization of crop straws as renewable energy resources for both direct combustion and as pellets to be used for domestic and commercial purposes	The government divert funds to other uses (this risk is extremely low, since all three programs are part of a multi-year recurring budget that is renewed every 5 years) The activity of the programs are delayed and not completed as scheduled	Facilitate through government procurement system as national priority. Provide assistance to complete the programs on schedule
	In-kind	1,050,000			
	Grant	17,200,000	The GOC promotes and supports the utilization of livestock and poultry manure as feedstock to generate biogas to be used for domestic and commercial purposes		
MOF and Local Governments	Grant	15,950,000	The GOC is supporting the demonstration of indigenous renewable energy sources (i.e., solar energy, wind energy, geothermal, and biomass) to replace fossil fuels for space heating. The program also supports building retrofits to improve energy efficiency		
Private Investors[19]:	Grant	33,200,000	Private investors and CSO will finance several baseline projects, such	Investors change their priorities or	Support MARA in liaising with the private

			as biogas and bio-methane generation and distribution, solar PV systems, bulk and pelletized biomass for heating and cooking, and deployment of EE equipment and products	delay the disbursement of the funds	investors to secure the budget
CSO[20]	Grant	5,000,000			
UNDP	In-kind	200,000	Project management and M&E	None	N/A
Total		**89,600,000**			
Grant		**88,350,000**			
In-Kind		**1,250,000**			

Budget Revision and Tolerance: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
c) Project Management Cost (PMC): budget under PMC component is capped and cannot be increased.

Any over expenditure incurred beyond the available GEF grant amount must be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

[19] This include: China Association of Rural Energy Industry; Hebei Jing'an Bioenergy Technology Co., Ltd.; Shanxi Golden Cooperate DC Power Distribution Engineering Technology Co., Ltd.; Chaoyang Jiajiadian Farm Co., Ltd.; Hailun Limin Energy-saving Boiler Manufacturing Co., Ltd.; Songzi Senchun Agricultural Development Co., Ltd.; Yunan Fu Hui Technology Co., Ltd.; and Ningxia Runwell Energy & Environment Engineering Co, Ltd.

[20] China Association of Rural Energy Industry

Project extensions: The UNDP Resident Representative and the UNDP-GEF Executive Coordinator must approve all project extension requests. Note that all extensions incur costs and the GEF project budget cannot be increased. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP; the project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs will be covered by non-GEF resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's guidance set out in UNDP POPP.

Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies. The costs for audit cannot be included under M&E component and budget must be included under PMC.

Project Closure: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

Operational completion: The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-of-project review Project Board meeting. **Operational closure must happen at the end date calculated by the approved duration after the Project Document signature or at the revised operational closure date as approved in the project extension. Any expected activity after the operational date requires project extension approval.** The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. At this time, the project should have completed the transfer or the disposal of any equipment that is still the property of UNDP.

Transfer or disposal of assets: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project (it is strongly encouraged to be done before the operational closure date). In all cases of transfer, a transfer document must be prepared and kept on file[21]. The transfer should be done before Project Management Unit complete their assignments.

Financial completion (closure): The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

[21] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

The project will be financially completed **within 6 months of operational closure or after the date of cancellation**. If Operational Closure is delayed for any justified and approved reason, the Country Office should do all efforts to Financially Close the project within 9 months after TE is completed. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the BPPS/NCE-VF Unit for confirmation before the project will be financially closed in Atlas by the UNDP Country Office.

Refund to GEF: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE-VF Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF Trustee.

IX. Total Budget and Work Plan

Total Budget and Work Plan			
Atlas Award ID:	00129191	**Atlas Output Project ID:**	00122943
Atlas Proposal or Award Title:	EZCERTV		
Atlas Business Unit	CHN10		
Atlas Primary Output Project Title	EZCERTV		
UNDP-GEF PIMS No.	6431		
Implementing Partner	Ministry of Agriculture and Rural Affairs (MARA)		

Atlas Activity (GEF Component)	Atlas Implementing Agent (Responsible Party , IP, or UNDP)	Atlas Fund ID	Donor Name	Atlas Budgetary Account Code	ATLAS Budget Account Description	Amount Year 1 (USD)	Amount Year 2 (USD)	Amount Year 3 (USD)	Amount Year 4 (USD)	Amount Year 5 (USD)	Total (USD)	*See Budget Note:*
COMPONENT 1: Energy Policy & Regulatory Framework Strengthening												
Outcome 1: Effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China	MARA	62000	GEF	71200	International Consultants	31,500	31,500	24,500	35,000	24,500	147,000	1
				71300	Local Consultants	47,000	41,750	52,250	45,250	48,750	235,000	2
				71600	Travel	10,000	10,000	10,000	10,000	10,000	50,000	3
				72100	Contractual Services -company	150,000	215,000	65,000	75,000	45,000	550,000	4
				75700	Training, Workshops and Conferences	53,000	43,000	23,000	33,000	24,000	176,000	5
					Total Component 1	**291,500**	**341,250**	**174,750**	**198,250**	**152,250**	**1,158,000**	
Component 2: Integrated Renewable Energy and Energy Efficiency Technologies Applications for Zero Carbon Town and Village Development												
Outcome 2: Low/zero carbon technology applications are widely implemented in rural towns and villages in rural areas of China.	MARA	62000	GEF	71200	International Consultants	41,000	52,500	88,500	42,000	77,000	301,000	6
				71300	Local Consultants	7,000	22,750	34,250	24,500	36,000	124,500	7
				71600	Travel	6,000	6,000	6,000	6,000	6,000	30,000	8
				72100	Contractual Services -company	156,000	222,400	590,050	863,800	866,600	2,698,850	9
				72200	Equipment and Furniture	-	2,100,000	900,000	87,500	-	3,087,500	10
				72800	Information Tech. Equipment	30,000	1,500	1,700	1,700	1,717	36,617	11

				75700	Training, Workshops and Conferences	-	30,000	30,000	-	-	60,000	12
					Total Component 2	**240,000**	**2,435,150**	**1,650,500**	**1,025,500**	**987,317**	**6,338,467**	
Component 3: Facilitating Rural Zero Carbon Development and Lifestyle Popularization and Knowledge Dissemination												
Outcome 3: Enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China.	MARA	62000	GEF	71200	International Consultants	10,500	32,900	28,000	23,800	16,800	112,000	13
				71300	Local Consultants	24,000	19,750	19,750	18,000	21,500	103,000	14
				71600	Travel	10,000	10,000	10,000	10,000	10,000	50,000	15
				72100	Contractual Services -company	66,000	133,000	125,000	66,000	40,000	430,000	16
				75700	Training, Workshops and Conferences	-	75,600	90,000	80,000	70,000	315,600	17
					Total Component 3	**110,500**	**271,250**	**272,750**	**197,800**	**158,300**	**1,010,600**	
Project management costs												
	MARA	62000	GEF	71800	Contractual services – Implementing Partner	71,000	71,000	71,000	71,000	71,000	355,000	18
				74100	Professional services	5,000	5,000	5,000	5,000	5,000	25,000	19
				71300	Local Consultants	5,000	5,000	5,000	5,000	5,000	25,000	20
				71600	Travel	4,000	4,000	4,000	4,000	-	16,000	21
				72500	Office Supplies	-	1,000	1,000	1,353	1,000	4,353	22
					Total PMC	85,000	86,000	86,000	86,353	82,000	425,353	
					Total	**727,000**	**3,133,650**	**2,184,000**	**1,507,903**	**1,379,867**	**8,932,420**	

Summary of Funds:

	Amount Year1	Amount Year 2	Amount Year 3	Amount Year 4	Amount Year 5	Total
GEF	727,000	3,133,650	2,184,000	1,507,903	1,379,867	8,932,420
UNDP	40,000	40,000	40,000	40,000	40,000	200,000
Government of China	10,480,000	10,240,000	10,240,000	10,240,000	10,000,000	51,200,000
Private Sector	6,640,000	6,640,000	6,640,000	6,640,000	6,640,000	33,200,000

CSO	1,000,000	1,000,000	1,000,000	1,000,000	1,000,000	5,000,000
TOTAL	18,887,000	21,053,650	20,104,000	19,427,903	19,059,867	98,532,420

Budget Notes:

No.	Explanation
1	**<u>International consultants</u>: US$147,000** (including professional fees, travel expenses and DSA) for 210 working days at US$700 per working day (multiple activities could be conducted by one consultant):
	• 180 working days in total for activities under Output 1.1, and specifically: 30 working days for Activity 1.1.1; 30 working days for Activity 1.1.2; 40 working days for Activity 1.1.3; 40 working days for Activity 1.1.4; 40 working days for Activity 1.1.5; (**US$126,000**)
	• 30 working days for Activity 1.2.1; (**US$21,000**)
2	**<u>National consultants:</u> US$235,000** (including professional fees, travel expenses and DSA) for 500 working days at US$350 per working day (multiple activities could be conducted by one consultant), and local consultants for monitoring activities (multiple activities could be conducted by one consultant):
	• 130 working days for activities under Output 1.1, and specifically: 20 working days for Activity 1.1.1; 20 working days for Activity 1.1.2; 30 working days for Activity 1.1.3; 30 working days for Activity 1.1.4; 30 working days for Activity 1.1.5; (**US$45,500**)
	• 20 working days for activities under Output 1.2, and specifically: 20 working days for Activity 1.2.1; (**US$7,000**)
	• 250 days for a local RE/EE/EC technology procurement specialist to support all stakeholders, consultants, both international and local, and service companies, for all EZCERTV activities to facilitate and guarantee uniformity of RE/EE/EC technologies, measures, guidelines and standards for all selected locations as well as replication sites; (**US$87,500**)
	• 100 days for a national expert to liaise with national and local government to facilitate the approval and enforcement of all formulated policies and regulations in all EZCERTV activities; (**US$35,000**)
	• Monitoring of project risks (**US$15,000**); Gender action plan monitoring (**US$15,000**), Social and environmental safeguard monitoring (**US$15,000**), and stakeholder engagement plan monitoring (**US$15,000**). (M&E activities)
3	**<u>Travel:</u> US$50,000**: the travel and DSA expenses for 2 PMO officers participating to each of the 3 training sessions (2 sessions for Activity 1.2.2 and 1 session for Activity 1.4.2) and to the 2 workshops (Activities 1.1.3 and 1.1.4), and for 8 stakeholders from the selected rural locations participating to the same 3 training sessions, and 2 workshops (50 travels at US$1,000 each), (The other participants to the 3 training sessions and 2 workshops will be connected via videoconference)
4	**<u>Contractual Services-Companies: US$550,000:</u>** All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):
	• **Sub-contract 1 – US$40,000**: Design and evaluate the 2 training sessions in Activity 1.2.2 (administration of training programs are under a separate budget, see note 6 below)
	• **Sub-contract 2 – US$70,000**: Develop guidelines and standards on the application of RE/EE/EC technologies and measures in rural areas
	• **Sub-contract 3 – US$70,000**: Standardization of RE/EE/EC technology specs and MEPS, testing and certification procedures for local and imported technologies, and design and operation of RE/EE/EC projects
	• **Sub-contract 4 – US$120,000**: Assess rural energy status in the selected project rural areas (US$15,000 x 8 = US$120,000)
	• **Sub-contract 5 – US$30,000**: Design and evaluate the training sessions in Activity 1.4.2 (administration of training programs are under a separate budget, see note 6 below)
	• **Sub-contract 6 – US$160,000**: Prepare energy-integrated development plans in the selected project rural areas (US$20,000 x 8 = US$160,000)
	• **Sub-contract 7 – US$60,000**: Develop guidelines for the promotion of zero-carbon villages and towns development

5	**<u>Training, Workshops and Conferences</u>: US$176,000**
	• Workshops for Formulation, recommendation of new and improved policies and regulations on zero carbon development in rural areas (Activities 1.1.3 and 1.1.4), 2 sessions: US$20,000 x 2 = **US$40,000**
	• Capacity building program for local governments on integrated RE resource planning and analysis, including a post-completion evaluation of the training program (Activity 1.2.2). 2 sessions US$30000 x 2 = **US$60,000**
	• Capacity building program for rural governments on monitoring, recording, analysis and management of energy generation and utilization data, including a post-completion evaluation of the training program (Activity 1.4.2). **US$41,000**
	• Project inception workshop: **US$20,000** (part of M&E); and Annual PSC meeting US$3,000 x 5 = **US$15,000 (M&E)**
6	**<u>International consultants</u>**: **US$301,000** (including professional fees, travel expenses and DSA) for 330 working days at US$700 per working day (including the Chief Technical Advisor), and both Mid-Term Review and Terminal Evaluation:
	• 50 working days for activities under Output 2.1, and specifically: 50 working days for Activity 2.1.1; (**US$35,000**)
	• 280 working days for the CTA, and specifically: 40 working days in year 1, and 60 working days for year 2 through year 5; (**US$196,000**)
	• **US$35,000** for the MTR and **US$35,000** for the TE (part of M&E)
7	**<u>National consultants:</u> US$124,500** (including professional fees, travel expenses and DSA) for 270 working days at US$350 per working day, and both Mid-Term Review and Terminal Evaluation:
	• 30 working days for activities Activity 2.1.1; (**US$10,500**)
	• 240 working days for 8 local consultants to support for demos replication and scale-up in each provinces (Activity 2.3.2). US$350 x 30 x 8 = **US$84,000**
	• **US$15,000** for the MTR and **US$15,000** for the TE (part of M&E)
8	**<u>Travel:</u> US$30,000**: the travel and DSA expenses for 2 PMO officers participating to each of the two training programs (Activities 2.1.6 and 2.2.5) and for 8 stakeholders from the selected rural locations participating to the same 2 training programs (a total of 20 travels at US$1,500 each). (The other participants to both training programs will be connected via videoconference)
9	**<u>Contractual Services-Companies: US$2,698,850:</u>** All sub-contract budgets include: national and international experts, **Equipment for Demos Replication**, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):
	• **Sub-contract 1 – US$160,000**: Develop engineering designs and implementation plans for the RE and EE demos in selected rural areas (Activity 2.1.2 and Activity 2.2.1). US$20,000 x 8 = US$160,000
	• **Sub-contract 2 – US$240,000**: Implementation of the RE and EE demos in selected rural areas (Activity 2.1.3 and Activity 2.2.2). US$30,000 x 8 = US$240,000
	• **Sub-contract 3 – US$120,000**: Design of M&E systems of the RE and EE demos in selected rural areas (Activity 2.1.4 and Activity 2.2.3). US$15,000 x 8 = US$120,000
	• **Sub-contract 4 – US$40,000**: Design and implement a promotional program for rural stakeholders on technical advantages of RE technologies measures (Activity 2.1.5)
	• **Sub-contract 5 – US$30,000**: Design and evaluate the training sessions in Activity 2.1.6 (administration of training programs are under a separate budget, see note 14 below)
	• **Sub-contract 6 – US$40,000**: Design and implement a promotional program for rural stakeholders on technical advantages of EE/EC technologies measures (Activity 2.2.4)

	• **Sub-contract 7 – US$30,000**: Design and evaluate the training sessions in Activity 2.2.5 (administration of training programs are under a separate budget, see note 14 below)
	• **Sub-contract 8 – US$30,000**: Design and implement a promotional program to improve use and sales of high EE appliances and equipment in rural areas (Activity 2.2.6)
	• **Sub-contract 9 – US$50,000**: Design an integrated energy management system for rural areas (Activity 2.2.7)
	• **Sub-contract 10 – US$105,000**: Prepare and disseminate case studies based on the demos (Activity 2.3.1)): US$10,000 x 8 = US$80,000; and prepare lessons learned: **US$25,000** (part of M&E)
	• **Sub-contract 11 – US$500,000**: Develop implementation plans for demos replication (Activity 3.3.2). US$10,000 x 50 = US$500,000
	• **Sub-contract 12 – US$1,353,850**: Procure **Equipment** for the replication of a demo project at a town level
10	**Equipment and Furniture: US$3,087,500**
	US$2,000,000 for activities under Output 2.1: provide the equipment for demos to facilities the application of RE technology in villages.
	US$1,000,000 for activities under output 2.2: provide the equipment for demos to facilities the application of EE technology in villages.
	US$87,500 for the activities under output 2.3: provide the equipment for replication sites
11	**Information Technology Equipment: US$36,617**
	US$15,000 for office equipment (computers, printers, copier, etc.); **US$15,000** for Communications and conferencing equipment (digital camera, overhead projector, and screen, etc.) and US$6,617 for supplies and maintenance.
12	**Training, Workshops and Conferences: US$60,000**
	• Capacity building program to facilitate the establishment of a rural **RE** service industry (Activity 2.1.6). **US$30,000**
	• Capacity building program to facilitate the establishment of a rural **EE/EC** service industry (Activity 2.2.5). **US$30,000**
13	**International consultants**: **US$112,000** (including professional fees, travel expenses and DSA) for 160 working days at US$700 per working day (multiple activities could be conducted by one consultant):
	• 80 working days in total for activities under Output 3.2, and specifically: 40 working days for Activity 3.2.1; 40 working days for Activity 3.2.2; (**US$56,000**)
	• 40 working days for Activity 3.4.2; (**US$28,000**)
	• 40 working days for Activity 3.5.1; (**US$28,000**)
14	**National consultants**: **US$103,000** (including professional fees, travel expenses and DSA) for 80 working days at US$350 per working day (multiple activities could be conducted by one consultant), a project knowledge management officer :
	• 40 working days in total for activities under Output 3.2, and specifically: 20 working days for Activity 3.2.1; 20 working days for Activity 3.2.2; (**US$14,000**)
	• 20 working days for activities 3.4.2; (**US$7,000**)
	• 20 working days for activities 3.5.1; (**US$7,000**)
	• 60 months for the project knowledge management officer, US$1,250 per month x 60 months = **US$75,000**: part–time staff responsible for communicating with all stakeholders, support the project dissemination and knowledge products development
15	**Travel: US$50,000**: the travel and DSA expenses for 2 PMO officers participating to each of the 4 training programs (Activities 3.1.2, Activity 3.1.3, Activity 3.4.3, and Activity 3.5.2), and for 8 stakeholders from the selected rural locations participating to the same 4 training programs (a total of 40 travels at US$1,250 each). (The other participants to the 4 training programs will be connected via videoconference)

16	**Contractual Services-Companies: US$430,000:** All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company):
	• **Sub-contract 1 – US$20,000**: Assess awareness levels of relevant stakeholders about zero-carbon rural development (Activity 3.1.1)
	• **Sub-contract 2 – US$30,000**: Design and evaluate the training sessions in Activity 3.1.2 (administration of training programs are under a separate budget, see note 20 below)
	• **Sub-contract 3 – US$30,000**: Design and evaluate the training sessions in Activity 3.1.3 (administration of training programs are under a separate budget, see note 20 below)
	• **Sub-contract 4 – US$50,000**: Develop a project management and information communication platform for zero-carbon rural development (Activity 3.1.4)
	• **Sub-contract 5 – US$30,000**: Establish a social participation mechanism to strengthen stakeholders' capacity on zero-carbon rural development (Activity 3.3.1)
	• **Sub-contract 6 – US$30,000**: Develop a plan to publicize EZCERTV project (Activity 3.3.2)
	• **Sub-contract 7 – US$30,000**: Implement the promotional strategy to promote zero-carbon development to rural communities (Activity 3.3.3)
	• **Sub-contract 8 – US$40,000**: Design and implement promotional initiatives to involve women and marginalized groups in zero-carbon rural development (Activity 3.3.4)
	• **Sub-contract 9 – US$40,000**: Facilitate the establishment of a private sector-led service industry for RE/EE technologies in rural areas (Activity 3.4.1)
	• **Sub-contract 10 – US$30,000**: Design and evaluate the training sessions in Activity 3.4.3 (administration of training programs are under a separate budget, see note 20 below)
	• **Sub-contract 11 – US$60,000**: Design and evaluate the training sessions and the promotional program in Activity 3.5.2 (administration of training programs are under a separate budget, see note 20 below)
	• **Sub-contract 12 – US$40,000**: Design and implement promotional initiatives to enhance green financing to support zero-carbon rural development (Activity 3.5.3)
17	**Training, Workshops and Conferences: US$315,600**
	• Capacity building program for local government and officials and industry management teams about zero-carbon rural development (Activity 3.1.2). **US$35,600**
	• Capacity building program for the private sector, NGOs and rural communities about zero-carbon rural development (Activity 3.1.3). **US$30,000**
	• **US$120,000** for knowledge exchange programs with a minimum of 4 "Belt and Road" countries on the development of zero-carbon villages and towns. Each exchange program will include training programs and information sharing initiatives (Activity 3.2.2)
	• **US$60,000** to conduct international bilateral and multilateral cooperation programs on zero-carbon rural development, training the people from" Belt and Road "countries (Activity 3.3.3). US$30,000 x 2 = US$60,000
	• Capacity building program for the private sector to enable them to procure RE/EE technologies for rural development (Activity 3.4.3). **US$30,000**
	• Capacity building and promotional program to enhance private sector participation (Activity 3.5.2). **US$40,000**
18	• **Contractual services – Implementing Partner: US$355,000 in total (**US$225,000 for a full time project manager and US$130,000 for the project finance officer for 50 months to support all components.)
19	• **Professional services: US$25,000** for project audit
20	• **US$25,000** for GEF core indicators monitoring before MTR and TE (part of M&E)

21	• **<u>Travel:</u> US$16,000**: for Project Management Office (PMO) travels cost and DSA for 2 officers to each of the 8 selected locations (US$1,000 per person per location).
22	• **US$ 4,353 Office supplies** such as cartridges, printing papers, courier, etc

X. Legal Context

This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of China and UNDP, signed on 29 June 1979. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

This project will be implemented by [name of entity] ("Implementing Partner") in accordance with its financial regulations, rules, practices, and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

XI. Risk Management

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999)[22].

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

   (a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").
   (b) Moreover, and without limitation to the application of other regulations, rules, policies, and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment, or creates an intimidating, hostile or offensive work environment.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan

[22] http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;
ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;
iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;
iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and
v. Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).
7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.
8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental

Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud, or corruption, by its officials, consultants, responsible parties, subcontractors, and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption, and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

10. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

11. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies, and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', sub-contractors', and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

12. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

13. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, sub-contractors, and sub-recipients.

14. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

15. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover, and return any recovered funds to UNDP.

16. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

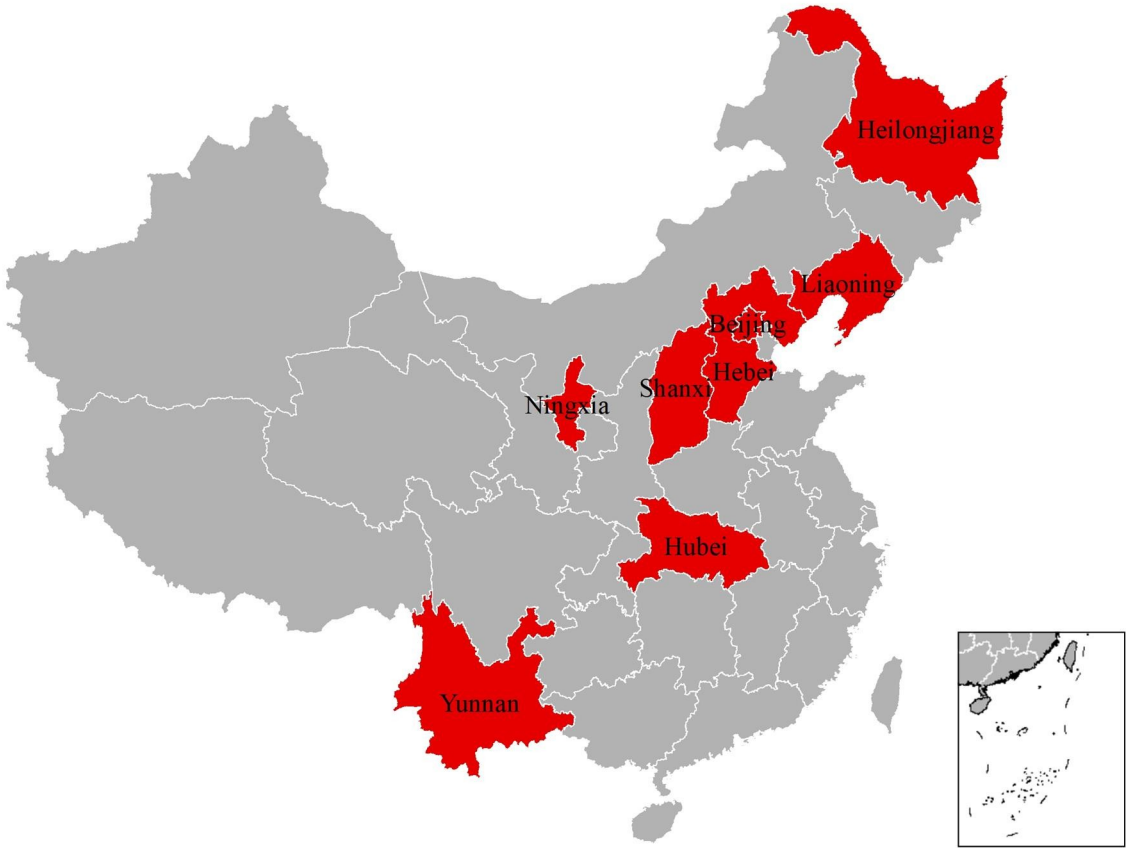
XII. Mandatory Annexes

1. Project Map and geospatial coordinates of the project area
2. Multiyear Workplan
3. Monitoring Plan
4. Social and Environmental Screening Procedure (SESP)
5. UNDP Atlas Risk Register
6. Overview of technical consultancies/subcontracts
7. Partners Capacity Assessment Tool and HACT assessment
8. GEF Budget Template
9. Stakeholder Engagement Plan
11. Gender Analysis and Gender Action Plan
12. Procurement Plan
13. A. Description of Demonstration
    B. GHG Emission Reduction Estimates
14. Co-Financing Letters
15. GEF Checklist for Audit Risk Review

Supplementary documents as part of the submission package to the GEF and are entered into the GEF Portal.

16. GEF and/or LDCF/SCCF Core indicators
17. GEF Taxonomy

Annex 1: Project map and Geospatial Coordinates of project sites



Geo-Coordinates of the Project Sites:

- Yanqing, Beijing (Lat: 40° 28' 0" North, Long: 115° 58' 0" East)
- Hengshui, Hebei (Lat: 37° 43' 55.99" North; Long: 115° 42' 4.00" East)
- Anping, Hebei (Lat: 37° 32' 59.99" North; Long: 115° 34' 59.99" East)
- Yuncheng, Shanxi (Lat: 35° 1' 35.09'' North; Long: 111° 0' 27.11'' East)
- Chaoyang, Liaoning (Lat: 41° 34' 59.99" North; Long: 120° 25' 59.99" East)
- Suihua, Heilongjiang (Lat: 46° 40' 32.98'' North; Long: 127° 1' 6.36'' East)
- Hailun, Heilongjiang (Lat: 47° 27' 23.40" North; Long: 126° 57' 13.25" East)
- Jingmen, Hubei (Lat: 31° 02' 1.00" North; Long: 112° 12' 16.99" East)
- Yuxi, Yunnan (Lat: 24° 20' 59.99" North; Long: 102° 31' 59.99" East)
- Tonghai, Yunnan (Lat: 24° 05' 21.60" North; Long: 102° 44' 59.99" East)
- Qintongxia, Wuzhong, Ningxia (Lat: 37° 59' 13.41'' North; Long: 106° 1' 49.56'' East)

NOTE: The planned project sites in the provinces of Heilongjiang, Liaoning and Yunnan are far from the geographical borders with China's neighboring countries, and that there are no known territorial disputes with the neighboring countries that border these provinces.

Annex 2: Multi Year Work Plan

Outcomes	Outputs	Activities	Year 1				Year 2				Year 3				Year 4				Year 5			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Component 1: Policy Formulation and Institutional Mechanisms on Zero Carbon Town and Village Development																						
Outcome 1: Effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China	**Output 1.1:** Formulated, approved, and enforced policies on the promotion and support of zero carbon town and village development in line with China's rural revitalization strategy and goal of achieving carbon neutrality before 2060	**Activity 1.1.1:** Conduct of policy research to update existing rural energy policies and implementing rules and regulations (IRRs) on the various aspects of energy supply, demand, and utilization in rural development																				
		Activity 1.1.2: Conduct of research studies on existing energy policies and implementing rules and regulations in other countries that can be used as reference for the rapid development of zero carbon in rural areas in China																				
		Activity 1.1.3 Formulation, promotion, approval and enforcement of new policies and regulations on zero carbon development in rural area, and low GHG emitting energy technologies implementation																				
		Activity 1.1.4: Formulation, promotion, approval and enforcement of new national and local policies and regulations on incentives and financial support for promotion of RE/EE/EC in rural areas, including increased role of private sector																				
		Activity 1.1.5: Provision of recommendations to revise the laws related to RE/EE/EC and include incentives and financing mechanisms																				
	Output 1.2: Established and operational institutional framework to	**Activity 1.2.1:** Establishment of clearly defined and operational institutional arrangements for promoting zero-carbon energy development in rural areas, such as policy-making, planning and																				

	promote and support zero-carbon town and village development	management																					
		Activity 1.2.2: Design and implementation of a capacity building program for local governments on integrated RE resource planning and analysis, including a post-training program evaluation																					
	Output 1.3: Formulated, approved, and enforced technical, management and operational standards in rural renewable energy (RE) development and utilization, energy storage, energy conservation and energy efficiency (EC&EE) and comprehensive energy management	**Activity 1.3.1**: Development and enforcement of guidelines and standards on RE/EE/EC technologies and measures used in rural areas																					
		Activity 1.3.2: Standardization of RE/EE/EC technology specs and MEPS[23], testing and certification procedures for local and imported technologies, and design and operation of RE/EE/EC projects																					
	Output 1.4: Completed assessment of renewable energy resource endowment, economic and social development level, and energy use trends and	**Activity 1.4.1:** Conduct of a comprehensive assessment of the current status of rural energy in the selected project rural areas																					
		Activity 1.4.2: Design and conduct of a capacity building program for rural governments on monitoring, recording, analysis and management of energy generation and utilization data, including a post-training program evaluation																					
		Activity 1.4.3: Preparation of																					

[23] Minimum Energy Performance Standards

	patterns in rural areas of selected provinces; and formulated energy-integrated development plans of pilot towns in selected 8 provinces	comprehensive energy-integrated development plans for the selected project areas																					
		Activity 1.4.4: Development of guidelines applicable to the promotion of zero carbon villages and towns																					
Component 2. Integrated Renewable Energy and Energy Efficiency Technologies Applications for Zero Carbon Town and Village Development																							
Outcome 2: Low/zero carbon technology applications are widely implemented in rural towns and villages in rural areas of China	**Output 2.1:** Completed and operational zero carbon technology demonstration involving the installation of about 167.2 MW of RE-based power generation capacity in 120 rural villages and 4 towns for energy production and utilization showcasing the cost-effective application of new RE delivery mechanisms decentralized RE-based energy generation and distribution (through mini/micro-grids)	**Activity 2.1.1:** Evaluation and improvement of existing initiatives and applications of renewable energy to facilitate the shift towards zero carbon villages and towns development																					
		Activity 2.1.2: Development of engineering designs and implementation plans, and facilitation of their approval process for the selected demonstration areas																					
		Activity 2.1.3: : Implementation of the demos of new feasible and cost-effective RE technology application models designed in Activity 2.1.2																					
		Activity 2.1.4: Design and implementation of methods for monitoring and evaluation of the operational, energy and economic performances of RE technology application demonstrations																					
		Activity 2.1.5: Design and conduct of a promotional program for local governments and rural communities on the technical and environmental advantages of adopting new and cost-effective RE technology applications																					
		Activity 2.1.6: Design and implementation of a capacity building program to facilitate the establishment of a rural renewable																					

	and energy	energy technical service industry																					
	Output 2.2: Completed and operational demonstrations of the cost-effective applications of EC&EE technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development. Estimated annual energy savings of about 6,487.0 TJ/year in 4 towns and 120 villages	**Activity 2.2.1:** Development and approval of engineering designs and implementation plans of demos of new and cost-effective EE/EC technologies																					
		Activity 2.2.2: Design and demonstration of new feasible and cost-effective EE/EC technology application models																					
		Activity 2.2.3: Design and implementation of EE/EC technology monitoring and evaluation methods, and evaluate their operability, energy saving and economic efficiency in demonstration projects																					
		Activity 2.2.4: Design of EE/EC technology implementation promotion plans for local governments and rural communities to promote the application of relevant applicable and feasible EE/EC technologies																					
		Activity 2.2.5: Design and implementation of a capacity building program to facilitate the establishment of a rural EE/EC technical service industry																					
		Activity 2.2.6: Design and implementation of a promotional program for the improved sales and use of high-performance EE equipment and devices in rural areas																					
		Activity 2.2.7: Development and design of an integrated rural energy management system based on energy supply-side monitoring and optimization, and demand-side management																					
	Output 2.3: Developed, published, and disseminated zero carbon town and	**Activity 2.3.1:** Preparation, publication and dissemination of case studies and lessons learned, based on the implemented demos, for the promotion of zero-carbon village and town energy																					

	village development guidelines to replicate and scale-up the impact of zero-carbon villages and towns	technology applications																					
		Activity 2.3.2: Development of implementation plans and action plans for demos replication and scale-up																					
		Activity 2.3.3: Application of the development and implementation plans formulated under Activity 2.3.2 and promotion of the application of RE and EE technologies in different regions																					
Component 3: Facilitating Rural Zero Carbon Development and Lifestyle Popularization and Knowledge Dissemination																							
Outcome 3: Enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China	**Output 3.1:** Completed information, communication, and education (ICE) program on zero carbon development in rural towns and villages in China for local authorities, service companies, rural energy cooperatives, public institutions, private entrepreneurs, farmers, and the public	**Activity 3.1.1:** Assessment of the ability and awareness level of relevant stakeholders about zero-carbon rural development																					
		Activity 3.1.2: Design and implementation of a capacity building and information program for relevant local government officials and industry management teams.																					
		Activity 3.1.3: Design and implementation of a capacity building and information program for the private sector, NGOs, and rural communities																					
		Activity 3.1.4: Development and operationalization of a project management and information communication platform for zero-carbon villages and towns																					
		Activity 3.1.5: Design, approval, and operationalization of an energy monitoring and reporting system and an energy benchmarking system for rural towns and villages																					
	Output 3.2: Established international cooperation and	**Activity 3.2.1:** Development of business models for zero-carbon project implementations for countries assisted by China																					

	Knowledge sharing program in zero-carbon towns and villages development and green energy transformation with countries along the “Belt and Road”, to support these countries in achieving their nationally determined contributions (NDC) commitments to the “Paris Agreement”	**Activity 3.2.2:** Organization and conduct of international capacity building, communication and information sharing programs on the development of zero-carbon villages and towns with "Belt and Road" countries																					
		Activity 3.2.3: Organization and conduct of international bilateral and multilateral cooperation programs on zero-carbon rural development																					
	Output 3.3: Established social mechanism for the promotion of zero-carbon town and village development and the sustainability of renewable energy industries	**Activity 3.3.1:** Establishment of a social participation mechanism conducive to strengthening stakeholders’ capacity on zero-carbon towns and villages development																					
		Activity 3.3.2: Development of plans to publicize EZCERTV project and promote rural communities’ participation																					
		Activity 3.3.3: Implementation of the promotional strategy to promote zero-carbon towns and villages development to rural communities																					
		Activity 3.3.4: Design and conduct of promotional initiatives to increase the involvement of women, youth, and marginalized groups in zero-carbon development in rural areas																					
	Output 3.4:	**Activity 3.4.1:** Establishment of a private																					

	Established and operational market-oriented mechanism for the enhanced development and utilization of rural RE resources, EC&EE improvement, and comprehensive energy management systems for supporting sustainable socio-economic development, and creation of zero carbon towns and villages in rural China	sector-led service industry in rural areas for RE and EE technologies																					
		Activity 3.4.2: Design and implementation of a program to assist rural municipalities/towns in sourcing funding for socio-economic development initiatives from available alternative sources of funds																					
		Activity 3.4.3: Design and conduct of a capacity building and technical assistance program for the private sector to enable them to procure and supply new and cost-effective RE/EE technologies for rural development, including a post-training program evaluation																					
	Output 3.5: Established and operational investment and financing mechanisms for supporting the commercial viability and operation of zero-carbon towns and villages and the development of rural renewable energy industries	**Activity 3.5.1:** Evaluation of the existing financing mechanisms and incentives, and their adaptation to zero-carbon rural development in China																					
		Activity 3.5.2: Promotion of green finance and implementation of business practices to support the development of zero-carbon villages and towns																					
		Activity 3.5.3: Design and conduct of a promotional, assistance, and capacity building program to enhance the involvement of private sector and financial practitioners, as well as national and local governments, and collectively finance and build community-based RE/EE																					

		facilities																					

Annex 3: Monitoring Plan

This Monitoring Plan and the M&E Plan and Budget in Section VI of this project document will both guide monitoring and evaluation at the project level for the duration of project implementation.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source/Collection Methods[24]	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		Mid-term	End of Project						
Project objective: Acceleration of zero-carbon transformation in China's rural area to contribute to global climate change mitigation and the achievement of the United Nations sustainable development goals (SDG)	**Indicator 1:** # direct project beneficiaries disaggregated by gender (individual people)	250,000 (women: 128,000 Men: 122,000)	918,500 (women: 460,000 Men: 458,500)	Cumulative number of people in the rural towns and villages where demos and their replication and scale-up will be implemented that will benefit from cleaner and healthier environment and socio-economic growth	List the source of the baseline data and targets and explain how this data was collected and how it should be collected during project implementation. Detail which methodology should be used (e.g. GEF GHG measurement methodology).	Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants	Project M&E and activity reports Annual reports submitted by relevant national and local GOC entities.	Continuous commitment, support & active participation of the national and local governments in the development of zero-carbon rural towns and villages Realization of committed co-financing from the national and local governments as well as the private sector in the implementation of project activities and monitoring
	Indicator 2: Cumulative GHG emission reduction from the	100,000	4,079,000	Cumulative reduction of GHG emissions, over the implementati		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants		

[24] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Monitoring	Indicators	Targets		Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		Mid-term	End of Project						
	rural sector of China, tCO_2			on period of the EZCERTV Project, attributable to the implementation of zero/low-carbon technologies proposed under the Alternative Scenario					systems
	Indicator 3: Cumulative reduction in fossil fuel consumption due to the implemented zero/low carbon energy technologies and techniques in rural areas, tce	11,297.8	446,223.6	Cumulative reduction of fossil fuel consumption, over the implementation period of the EZCERTV Project, due to the application of all EE measures and RE technology		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants		
	Indicator 4: % RE generation in rural areas	75%	85%	Annual amount of energy generated		Annually Reported in DO tab	Project Manager and Consultant		

Monitoring	Indicators	Targets		Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		Mid-term	End of Project						
	in China			from renewable energy resources (e.g., biomass, solar PV)		of the GEF PIR	s		
Project Outcome 1: Effective enforcement of policies and institutional frameworks towards the development of zero carbon towns and villages in rural China	***Indicator 5:*** Number of approved and enforced new and updated zero/low carbon policies, at both national and local levels.	5	10	Number of policy documents drafted and approved at the national level and at the provincial level in the selected provinces concerning the development of zero-carbon rural towns and villages		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants	Policies, regulations, and standards approved by the central and local governments. EZCERTV Project M&E and activity reports Surveys to track No. of villages and towns participating in the project activities	Full and continuous commitment and support of the national and local governments in the formulation, approval and enforcement of policies, regulations, and standards to support the development of zero-carbon rural areas
	Indicator 6: Cumulative number of rural towns/villages that adopt and enforce developed guidelines	8	50	Number of rural village and towns that will participate in the project activities by adopting the guidelines and standards that		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants		

Monitoring	Indicators	Targets		Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		Mid-term	End of Project						
	and standards on RE/EE/EC technologies and measures			will be developed to assure high quality of RE, EE and EC technologies and measures					
Project Outcome 2: Low/zero carbon technology applications are widely implemented in rural towns and villages in rural areas of China	***Indicator 7:*** Cumulative number of rural towns/villages that replicate and scale-up the implemented demo projects.	20	200	Number of rural village and towns that will participate in the project by replicating and scaling-up, during the project implementation phase, the demo projects that have been implemented in the eight selected locations		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants	EZCERTV Project M&E and activity reports. Engineering designs and implementation plans of replication and scale-up projects. Completed/ installed zero-carbon technologies and measures, and monitoring reports.	Relevant entities will be fully cooperating in the sharing of data and information. The private sector fully support and commit to the feasible replication of successful results of the EZCERTV project demonstrations. The planned demonstrations are implemented and completed as per schedule
	Indicator 8: Cumulative number of private sector entities investing in	50	200	Number of private investors that will be involved in the implementati		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants		

Monitoring	Indicators	Targets		Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		Mid-term	End of Project						
	zero/low carbon energy in rural towns/villages			on of baseline projects, and replication and scale-up of project demos					
Project Outcome 3: Enhanced and strengthened public awareness and knowledge about low carbon technology energy applications in support of zero carbon town and village development in rural China	***Indicator 9 :*** Cumulative number of local firms in rural areas that can capably provide technical, engineering and maintenance services for zero/low carbon technology application projects	20	200	Cumulative number of existing or newly established local firms that will provide technical, engineering and maintenance services for zero/low carbon technology application projects to guarantee the continuation and sustainability of the project activities past the project implementation phase		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants	EZCERTV Project M&E and activity reports Registration of new businesses to design, operate and maintain RE & EE technologies and measures in rural areas. Training program evaluation reports for stakeholders from the Belt and Road countries	Continuous commitment and support by the private sector and general public on the establishment of a local market for RE and EE technologies and measures Full commitment of governments of Belt and Road countries that share issues concerning zero-carbon rural development similar to China

Monitoring	Indicators	Targets		Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
		Mid-term	End of Project						
	Indicator 10: Number of stakeholders from BRI countries trained on zero-carbon/low-carbon energy technologies and zero-carbon villages and towns	50	200	Number of relevant stakeholders from the Belt and Road countries that will be trained, as part of knowledge sharing, on zero/low-carbon energy technologies and zero-carbon villages and towns development based on training programs and lessons learned drawn from the EZCERTV project		Annually Reported in DO tab of the GEF PIR	Project Manager and Consultants		

Annex 4: UNDP Social and Environmental Screening Procedure (SESP)

Project Information

Project Information	
1. Project Title	Enabling Zero Carbon Energy in Rural Towns and Villages in China (EZCERTV) Project
2. Project Number (i.e. Atlas project ID, PIMS+)	PIMS 6431
3. Location (Global/Region/Country)	China, People's Republic
4. Project stage (Design or Implementation)	Design
5. Date	10 June 2022

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
As in other UNDP-GEF projects, the EZCERTV Project is designed considering a human-rights based approach to programming. This project will particularly and for most part focus on the local communities in the rural areas of China. It is envisaged that like in the initial discussions with potential provincial governments partners, during the project design, consultations have been carried out with the local communities in selected rural towns and villages where the demonstrations that have been preliminarily designed to showcase the applications of RE and EE technologies to bring about zero carbon rural development will be implemented. The project design involved the conduct of additional, in-depth focus consultations. Such consultations will be a continuous process even during project implementation to ensure that the project activities, particularly the demonstrations (e.g., zero carbon village demos), constantly consider the will and requirements of indigenous peoples and all other residents of the communities. The project development team (PDT) met and discussed with communities any potential RE-based energy generation and EE technology applications that will be developed and implemented in their villages and assessed their interest in such initiatives and the potential productive applications in the use of RE resources. The preliminary design of each demo that will showcase the devised low/zero carbon energy technology applications is based on site inspections, as well as data gathering, and analyses conducted by the PDT during the design stage. This due diligence allowed the PDT to determine the technical specs, the investment costs and the environmental and financial benefits of the demos and pre-assess their feasibility. This work has been integrated with site specific environmental and social ("E/S") assessments of each demos on RE-based energy generation and EE system retrofit for implementation under the project. Based on the agreement expressed by the communities in full community hearings/meetings that were conducted during the project design period, additional in-depth social and environmental assessments will be carried out during project implementation. Such consultations will include households and where possible, separate men's and women's groups consultations within each partner town/village. The project design also underscored human rights by designing the project activities (e.g., village demos) in such a way to also promote livelihood improvement/ income generation via use of the new RE-based energy generation facilities that will be established under the demos as part of the initiative towards achieving zero carbon development. As part of the demos, activities for stimulating productive use of RE in specific towns/villages were also included in the project. These activities were also designed to specifically include women and children. To support this, the demos were designed so that their implementation will also provide villagers (where the demos will be implemented) access to a grievance redress mechanism. Such mechanism also emphasizes and ensures that access to opportunities for involvement in productive use initiatives and work opportunities are equitable.

Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment
The design of the proposed project has been guided by principles of gender equality and women's empowerment. The project has been designed with the intention to actively enable and involve qualified and capable men and women working in both management and technical departments of the CHINA Government agencies/institutions as well as those in the rural communities hosting the project demos. These are the entities who can play important roles in the design, development and implementation of the project. Since the focus is on RE and EE technology applications, the project activities have been designed to facilitate benefits from RE and EE to be enjoyed by both women and men. The design of the project activities considered the opportunity for the country to further enhance the role of women in deployment of low carbon technologies and climate change mitigation options. The project component addressing policy barriers has been designed to also come up with gender-sensitive policies in the energy sector and the energy end-use sectors of the country. This is in recognition of the potential contributions of women in the management and implementation of climate change mitigation measures. The contributions, impacts and benefits of productive applications from community based EE and RE technology applications, including for women, men, youth, and children were considered in the design and preparation of the project. The capacity building and training activities of the project are also designed to make possible high rates of participation from women. The activities on productive applications of RE have been designed to target at least half of the benefits to flow to women-led and women-operated businesses. Lastly, as in the project design, qualified and capable women consultants/experts will be targeted to work in the implementation of the designed project.
Briefly describe in the space below how the project mainstreams sustainability and resilience
Sustainable zero carbon rural development in China is the main context within which the proposed project is being developed. The zero-carbon development is mainly in line with the energy supply and utilization in the Chinese rural towns and villages, and the project is intended to facilitate the transformation of the rural sector to this growth pattern. This is based on the improved development and utilization of renewable energy resources in rural areas, adopting energy storage technology and integrated energy management and energy efficiency technologies/techniques to meet the increasing energy demand in rural areas, accelerate the rural energy transformation, and lead to the development of zero carbon towns and villages in rural China. Moreover, such transformation is expected to contribute to the achievement of China's Nationally Determined Contributions (NDC). Hence, a consequence of the achievement of the NDC targets will be, among others, the enabling of the enhanced sustainable development of the rural areas of the country. It is envisaged that this will not only bring about local benefits, mainly through contributions to the improvement of the living conditions of the rural people and in turn to the national economy but will also contribute to the protection of the natural environment. In terms of indigenous RE resources, this project focuses on the enhanced sustainable utilization of solar energy, as well as the abundant biomass waste resources that can be used as energy resource in the rural agricultural areas of the country as cost-effective substitutes to coal and coal-generated electricity. The global environmental benefits from the project will mainly come from GHG emission reductions from fossil fuel displacement by RE resources in electricity generation and for thermal energy uses, and in the improvement of the specific energy consumption of the energy end use sectors in the rural areas through improved energy utilization efficiency. These will be facilitated by the barrier removal approach that this project has been designed to employ. The synergistic aspect of the integrated way the key stakeholders work together, and the higher chances of scaling-up/replication of the low carbon development techniques/practices will be introduced, demonstrated, and promoted under this project contributing to the achievement of environmental sustainability. The RE-based energy generation demos under this project have been designed to showcase the adoption, and application of best practices in carrying out limited, site specific environmental/social assessments as a part of the demo project feasibility studies, prior to any construction or installation work. Where required by law and UNDP's SES (whichever standard is higher), site-specific environmental and social assessments will also be carried out prior to implementation of the EE retrofit demos.
Briefly describe in the space below how the project strengthens accountability to stakeholders
The proposed project has been designed based on several site studies, allowing for full participation of the local community in decision-making. Provision of timely, accessible and functional information regarding supported activities, including on potential environmental and social risks and impacts and management measures (SESP, ESMF) will be disclosed on UNDP China websites to ensure transparency of programming interventions. The Project Management Office (PMO) shall establish a project-level Grievance Redress Mechanism (GRM) during project inception. The full details of the GRM will be agreed upon during the project's inception phase. The Mechanism (GRM) will ensure that individuals and communities affected by the project's activities have access to appropriate grievance resolution procedures for hearing and jointly addressing complaints and disputes related to the social and/or environmental impacts of the proposed interventions. Interested stakeholders may raise a grievance at any time to the PMO, the Executing

Agency, the Implementing Agency (UNDP), or the GEF. A monitoring and evaluation procedure is also designed to ensure effective monitoring—and where appropriate, participatory monitoring with stakeholders—and reporting on implementation of social and environmental risk management measures.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
1. Without adequate consultation, awareness raising and engagement with local communities, the Project could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population in the rural areas where the on-the-ground project activities will be carried out. Inadequate consultation and engagement on policy formulation (i.e. under Component 1 of the project) could also lead to adverse impacts on the enjoyment of human rights. *Human Rights: P.3, P.4* *Accountability: P.13, P.14* *Standard 6, 6.4*	I=4 L=2	Moderate	The human rights of people in the project areas are protected under the "Law on the Promotion of Rural Revitalization" (Article 4, Section 2). The requirements for continued consultation have been factored into subsequent SES assessment tools that shall be applied to the project during implementation (and as specified in the ESMF).	The designed demos are based on ongoing and planned rural revitalization projects of the national and provincial governments. The design of the project is consistent with the strategic requirements of the country's rural revitalization program, and the implementation of the project activities will be carried out to ensure compliance with the requirements of the country's Law on the Promotion of Rural Revitalization. The project has been designed with extensive consultation with local people as well as the involvement of the relevant national and provincial government agency responsible for the formulation of relevant legislation on the installation of decentralized RE-based power generation system or community mini-grids. A comprehensive Stakeholder Engagement Plan has been prepared during the PPG stage, with site-specific Plans being required or each selected demo site. In addition, as prescribed in the ESMF, for upstream/policy formulation and support activities (under Component 1), the project will utilize a SESA approach

				that shall include robust stakeholder engagement and consultation.
2. Potential occurrence of discrimination against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits. *Gender Equality and Women's Empowerment: P.9, P.10*	I=4 L=2	Moderate	The Chinese government attaches great importance to gender equality. During the PPG stage, about 40% women staff were involved in the project design. The project could potentially offer a wide range of opportunities, for individuals and groups for support in the RE & EE technology applications demos and in the demo replications, participate and get trained in capacity development program activities, to be hired as a consultant or contractor for the project implementation.	Special measures to ensure that any potential discrimination against women in the implementation of the project is countered have been identified and included in the project design, and as part of the gender assessment conducted for the gender action plan.. The results of such measures will be observed during project implementation, and if any corrective action is necessary, this will be addressed. Beyond that, the PDT will make special efforts to explore and facilitate the inclusion of interventions to enhance the role of women. Special activities have been designed to involve women, e.g., in the productive use of RE activities, to have strong representation of women in training courses and seminar-workshops, and to ensure a significant proportion of project consultants are women. The special measures are incorporated in the Gender Analysis and Action Plan that has been prepared for the EZCERTV Project. As per UNDP SES requirements, gender issues will be assessed during the conduct of all SES studies during project inception/implementation, including: SESA, site specific ESIA's, Labour Assessment and management Plan and continued SESP screening.
3. Given the fact that the project will operate in rural areas of China, potential adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services may arise in some of the demonstration sites. Upstream policy interventions under Component 1 may also lead to unintended negative changes to ecosystems and the use of ecosystem services/natural resources. *Standard 1: 1.1, 1.2, 1.3, 1.7, 1.11, 1.14*	I=4 L=2	Moderate	Demos in Shanxi and Yunnan are close to the riverside, and these could have potential impacts to the riverine habitats. The potential environmental impacts during the installation and operation of the demo RE-based power generation, distribution, and storage RE/EE/EC systems depend on the state or condition of the selected sites. The potential environmental impacts during operation are mostly continuous, while	Prior to the construction/implementation of activities relating to the demo sites (i.e. Outputs 2.1 and 2.2), site specific environmental and social impact assessments will be required. Once a demo site has been chosen and, and the plans for it designed in full, the demo site and its associated activities will be screened against the UNDP SES. This will include assessment of impacts to habitats In line with UNDP SES 1 requirements. In addition, Chinese government regulations prohibit the establishment of manufacturing and large scale infrastructure in or around ecosystem function zones. The selection criteria of demo sites will consider to avoid the impact to habitats. The detailed feasibility studies (inclusive of SES screening) to be carried out during full project

			those associated with construction activities are expected to be temporary and mostly reversible.	implementation will clearly articulate mitigation measures for any alterations of the general topographical and environmental nature of the demo sites. Following world class engineering and construction design, specific mitigation measures that are regarded as best practice with respect to environmental management of the systems that will be installed, ensuring compliance with the requirements of the applicable environmental regulations/legislation. Any other potential impacts and risks including mitigation measures will be elaborated in the specific environmental and social assessments o be conducted during project implementation for each demo in selected towns and villages and to be completed prior to any physical work beginning on the installation of the required hardware.
4. The associated construction, operation, or decommissioning of the demo installations may have potential health and safety risks to local communities due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel, and other chemicals) that may be used during construction and operation. *Standard 3: 3.1, 3.2, 3.3, 3.5, 3.7, 3.8*	I=3 L=3	Moderate	Construction aspects and operation of demo RE-based power generation, mini-grid systems, energy storage systems and the EE retrofits can pose safety risk. For example Li-ion batteries for the demo solar PV power systems, energy storage systems can produce potentially dangerous waste products.	The preliminary design of all confirmed demos will involve initial site-specific environmental and social assessments and recommend measures to mitigate any identified health and safety risks. In the detailed design, appropriate training will be provided to the selected companies responsible for the safe and proper installation, operation, and maintenance of the demos, as well as targeted beneficiaries of the demos. This is to ensure that the installed systems will be properly and safely operated and maintained, in compliance with Government of China regulations and UNDP SES requirements in such a way that the release or handling of waste products are properly controlled and managed. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the minimization and management of waste generated from these demos (e.g., spent lead-acid batteries, spent Li-ion batteries). As specified in the ESMF, during the conduct of the site-specific ESIA's that are to be undertaken for the demo sites, issues pertaining to community health and safety 9most notably SES 3.1, 3.2, 3.3, 3.5, 3.7, 3.8) will be assessed within the scope of such studies. Any residual impacts will be managed/mitigated through appropriate site-specific ESMPs (or relevant targeted malmanagement plans).
5. The project would potentially result in the generation of waste (both hazardous and	I=3 L=4	Moderate	The project design is in line, and its implementation will	All the enterprises involved in the project will operate in accordance with the relevant regulations and standards

	non-hazardous). The construction and operation of the RE-based energy generation (power and non-power applications) demos will generate wastes. PV panels and batteries will require disposal at end of life. EE retrofit projects (e.g., Use of LED lamps to replace CFLs/FLs) will potentially generate hazardous waste (Hg). *Standard 8: 8.1, 8.2*			comply, with the pertinent requirements of the "Law on the Promotion of Rural Revitalization", Article 40., and UNDP SES 8 requirements.	of the government as they relate to waste managemmnt (both hazardous and non-hazardous).. During the project implementation , relevant training will also be conducted for enterprises. The detailed design of the confirmed demos will consider the potential waste generation and ensure proper disposal of wastes from the construction, operation, and maintenance stages. Site-specific ESIAs 9as required and outlined in the ESMF) will serve as the entry point for the development of site-specific disposal plans (which may be a sub-element of a larger site-specific ESMP). Those plans, for example, will include a disposal plan for the PV panels and batteries, which contain hazardous wastes, once their useful life is expired. While there are environmental laws that apply to this, such will be checked further during the detailed design of the demos and replications and appropriate decisions and/or measures will be carried out. For example, compliance with the regulations concerning waste management can be among the criteria in the approval of the applications for financial support for the demo replications.
6.	Some project activities / demo sites might be located in lands of ethnic minorities and might require FPIC. *Standard 6: 6.1, 6.3, 6.4, 6.6, 6.9*	I=4 L=2	Moderate		At present, the designed demonstration activities projects will be carried out on land that has been approved by the government and will not occupy the land of ethnic minorities. The detailed design of the confirmed demos will consider the potential impacts on land use in particular for ethnic minority lands. During the design of project demo sites, scoped ESIAs will be conducted. If, during the conduct of such scoped EISA's, impacts to ethnic minority communities are foreseen, the project will undertake FPIC and develop the appropriate S6 tool (i.e. IPP). The project publicity documents need to be written in inclusive language so that minority groups can receive adequate knowledge sharing and training so that they fully understand the purpose and design logic of the project. If changes to minority land use are involved, the feasibility and cost-benefit should be carefully evaluated, and audience awareness and acceptance should be well

					researched.
7.	Some project activities/ demo sites might lead to physical displacement or forced evictions. There is also the potential that economic displacement of local communities could occur as a result of the project's demo interventions. *Standard 5: 5.1, 5.2, 5.3*	I=4 L=2	Moderate	The project design is in line and its implementation will comply, with the pertinent requirements of the "Law on the Promotion of Rural Revitalization", (Article 14).	The project sites and the technologies to be demonstrated are yet to be fully confirmed, and preliminary design of each demo that will showcase agreed/confirmed low/zero carbon energy technology application has been done during the project development stage. The design of each demo is based on the findings of the project design team in each designated demo site. The initial demo design (please refer to Supplementary Annex on Description of Demonstrations) is based on the information from the site inspections and data gathering, processing and analyses that were done to define the demo scheme including the quantities and technical specifications of the system and equipment that have to installed. This also enable the project development team to determine the investment cost and come up with the pre-feasibility analyses. Information regarding the land requirements, social, economic, and cultural aspects of each village were collected, including topographical, geo-physical and environmental conditions. While general feasibility/contextual information has been ascertained during project preparation, the final siting and selection of exact technologies to be used has not been completed during project preparation. As such, this will need to be assessed (including SES 5 related impacts) during the conduct of site specific ESIA's that are to be undertaken during project implementation for each of the demo sites. Moreover, after consulting again with local authorities in the demo villages, they have confirmed that there is no displacement and resettlement of residential houses. The demos such as those on connecting houses to biogas pipeline, installation of solar home systems, and renovation of house walls and windows will not require people in these houses to temporarily relocate. For demos involving the installation of biomass-based power generation units, these will be in done in the premises and properties of the partner companies (i.e. will not affect local landowners). Where confirmed demos will be designed considering the potential land demand and take the local development planning as the reference.

				So far, the demos have been evaluated preliminarily but did not refer to the issue of forced evictions. This will be further verified, and this issue will be used as an exclusionary criterion for the selection of the demo replications that will be supported by the local governments.
8. Some project activities / demo activities may lead to changes in traditional way of life such as the customary use of energy in households. *Standard 4: 4.3*	I=3 L=3	Moderate	The substitution of different household energy resources may affect the transmission of certain cultures, for example, the food prepared by traditional wood stoves may lose its original taste due to the replacement of the cooking fuel; another example is the clay bed ("KANG") with wood stoves, which is widely used in the north, may be replaced in the process of energy substitution.	The promotion of the planned demo activities for improving energy use in households through the application of low carbon (RE/EE) technologies will take into consideration the concerns (if any) of rural households who may like to hold on to traditional or customary practices. With the combination of local traditional culture and farmers' living habits, the project will be promoted to foster a low and zero carbon lifestyle combined with the rural simple ecological culture. The project implementation will comply with the "Law on the promotion of rural vitalization", Articles 4, 7 and 32. Local communities who may encounter SES 4-related impacts will also be engaged in upstream/policy level decisisons. With their input being included through meaningful engagement during the conduct of the SESA (as prescribed in the ESMF). Cultural inheritance is also the concept that the project adheres to. In the promotion of the demos, the potential users will be adequately appraised of the benefits of, and the potential changes in current way of life and practices due, to the applied new technologies. For some application methods involving traditional culture, the comprehensive impact of all aspects, such as environmental and health effects, needs to be comprehensively assessed, and managed, per the ESMF.
9. There is a potential for workplace accidents and injuries to occur/impact project workers during construction and operation of demo facilities. *Standard 7: 7.1, 7.6*	I=4 L=2	Moderate	Project workers may face risks such as accidents and injuries during the construction process; Workers in biogas plants may face the risk of accidents such as gas leakage; workers in hog farms and biomass fuel production plants may face health risks caused by	The risks of workers' work during constructions that are involved in the designed demos will be further evaluated during the demo engineering design and measures will be identified and taken to avoid them. Construction companies whose services will be engaged in the engineering design and construction of the demo facilities will be required to provide adequate protection and insurance for their own workers who will be deployed.

			the working environment	The requirements of the Standard 7 (7.1 and 7.6) are to be applied in an appropriately-scaled manner based on the nature and scale of the project, its specific activities, the project associated social and environmental risks and impacts, and the type of contractual relationships with project workers. During Year 1 of project implementation, a project-wide Labour Assessment and Management Procedure will need to be developed (with particular attention to the Standard 7 related risks associated with the demos sites as outlined in Output 2). An indicative outline for this LAMP is outlined in Annex 1 of the ESMF. The LAMP must be in place before Outputs 2.1 and 2.2 are initiated. The LAMP will be prepared by the PMO and submitted to the PSC for endorsement. The LAMP will include key information on the various aspects of labour (particularly in the sourcing, production, installation, operation and maintenance) that have to be managed and monitored, etc. The partner companies that will be responsible for the operation of the demo facilities will be assisted in making their existing operational safety, health and environmental plans to be used for the demo facilities to avoid, and where avoidance is not possible, reducing, mitigating, and managing adverse impacts, via completion of stand-alone management plans, as planned in the ESMF.
10. Risk of increased labor input by women (e.g., projects such as biomass fuels, biogas, etc., will increase women's labor input for household energy access) *Gender Equality and Women's Empowerment: P.9, P.10*	I=3 L=2	Moderate	Some forms of energy substitution may require more labor input, but not in this case since the supply of RE resources (biogas) will be conveniently done through biogas piping. Biomass waste collection and use will be done on a commercial and centralized scale. Work in these facilities may limit the involvement of women.	The preliminary design of the demos fully considered resource, financial and labor inputs for energy supply substitution, and has assessed the feasibility of these inputs and their potential impact, and reduce end-user labor inputs, including through scaling up and centralizing production, to the extent possible, especially to avoid increasing the burden on women and children. Impacts to gender dynamics will also be assessed as part of the projects SES documentation, including the SSA (for upstream impacts associated with Component 1), site-specific ESIAs (for demo site construction and implementation), and the LAMP. Appropriate management plans will be prepared for

				avoiding, and where avoidance is not possible, reducing, mitigating, and managing adverse impacts, via completion of stand-alone management plans, as planned in ESMF.
11. Some project activities (most notably under Component 2) may require significant consumption of raw materials (e.g. biomass) and natural resources (energy and/or water) *Standard 8: 8.6*	I=3 L=3	Moderate	Some renewable energy projects may present potential resource efficiency issues, such as whether biomass resources are being used more efficiently in biomass utilization projects. Note that in this project, the main biomass resource that will be used are agricultural waste (e.g., straw and animal manure). There are no growing of trees for harvesting for fuel use, nor harvesting of natural forests nor reforestation activities. Some renewable energy production may increase water demand, such as the high dependence of biomass power generation on water from a whole life cycle perspective.	The preliminary design of the demos projects considered the adequacy, availability and utilization efficiency of resources; for example, in biomass power projects, the utilization efficiency of biomass is much higher than that of traditional biomass (many straws are returned to the field); and for the energy and water consumption involved in RE/EE projects, the detailed demo engineering design will involve coupled analysis and accounting for the whole life cycle and water-energy nexus to avoid additional environmental pressure from the demos and their replications. Appropriate management plans will be prepared for avoiding, and where avoidance is not possible, reducing, mitigating, and managing adverse impacts, via completion of stand-alone management plans, as planned in ESMF.
12. Risk of disrupting people's daily lives and of conflict/grievances from local communities if project implementation disruptions arise as a result of financial access issues for private sector project partners. *Human Rights: P.4, P.6, P.7*	I=3 L=3	Moderate	Energy service delivery private companies involved in project construction may face financing difficulties causing them to stop the implementation or the operation of the demos. In such events, disruptions in the daily lives of people in the demo villages will be affected and may result in them suffering from disruptions to their access of energy sources.	The demos are part of the corporate business plans of the partner companies, and they have also done their own due diligence and financial feasibility analyses of such investments While these partner companies have sufficient financial resources to implement the demos, the local governments are also in hand to continue supporting the implementation of the demos should there be some disruptions in the operations of these partners, since this is part of their responsibilities to support the country's rural revitalization program. For the replication of the demos, the project includes a study on green investment and financing, a series of project activities including: assessment of the investment and financing capacity of participating companies; design of incentives for financial companies and their introduction in rural areas; training of financial practitioners, etc.

				The project also includes green finance related activities that will help local banks in the project areas develop green finance measures to support the demo replications.
13. RE-based energy generation (power and non-power purposes) and EE system installations can be seriously affected by adverse climate-related events . *Standard 2: 2.2, 2.3*	I = 4 P = 2	Moderate	Some renewable energy projects (e.g., wind and solar) may be sensitive to climate change; the structural integrity of RE-based energy generation facilities can be sensitive to extreme events such as earthquakes and floods. Freezing damage can affect crop straw production, resulting in insufficient biogas production feedstock and affecting biogas production. Freezing damage can also affect livestock production efficiency and livestock manure production, resulting in insufficient biogas production feedstock. Low temperatures can affect biogas production rates, resulting in insufficient biogas production and unstable community energy supply.	It is already common in international design and engineering practices, as well as in the construction/installation of RE-based energy generation units to follow proper engineering and construction design and construction that ensure not only structural integrity but also climate resilience. This applies also in the procurement, design/engineering, installation, and operation of the pertinent installations. As specified in the project document , the design and construction/installation of the physical infrastructures that will be installed will be based on the technical and structural specifications that major bilateral and multi-lateral donors require for the infrastructure projects that they are funding in China. Climate and disaster related impacts will also be assed for the 8 demo sites as part of the site specific ESIAs that are to be conducted. Depending on the extent of the impacts of the adverse climate –related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes. Plan for addressing this potential risk (i.e. as it pertains to the thematic area of UNDP SES 2) is also included in the risk register (see Annex E of CEO Endorsement Request Document).

14. The project will support 8 demonstrations, of which 2 demonstrations involve solar PV, in Beijing and Shanxi Province (the '2 Solar Demonstrations'). These demonstrations will be implemented in years 2 and 3 of the project. In both of the 2 Solar Demonstrations: - Co-financing will be used to purchase the Solar PV equipment. - GEF INV will be used to purchase related equipment, such as smart meters and energy efficiency equipment. GEF INV will NOT be used to procure any Solar PV equipment. A particular risk area of potential concern for the co-financing aspect of the project is related to labour issues (i.e. UNDP Standard 7) *Human Rights: P.2* *Standard 7: 7.1, 7.6*	I=3 L=3	Moderate	UNDP's corporate policy on social and environmental safeguards applies to both GEF INV and any co-financing.	Amongst other national laws, the Labor Law of the People's Republic of China provides legal protection for Chinese workers and establishes a system of judicial labor arbitration. In order to comply with UNDP's corporate standard 7 on labor issues[25], UNDP will develop a labour assessment and management procedure for the project. This management plan will be finalized in year 1 of the project, prior to the implementation of the 8 demonstrations in year 2 onwards. The management plan will also be used for the replication of these demonstrations under the project. .
15. The project could contribute to cumulative environmental or social impacts in the area through unintended negative consequences from policy or legislative change as a result of activities under Component 1. *Standard 1: 1.14*	I=4 L-2	Moderate	There is a possibility that upstream policy or legislative changes supported by Component 1 of the project may negatively impact the biophysical receptor environment, as well as people's livelihoods if not properly managed.	Environmental and social impacts as a result of the project are expected to be overwhelmingly positive. However, upstream policy interventions/initiatives will need to be adequately assessed to ensure that any adverse social or environmental are adequately avoided, managed or mitigated during the development of such policy interventions. In order to assess and manage upstream policy risks, the project will undertake a SESA that shall inform activities supported as part of Component 1 of the project.

[25] https://info.undp.org/sites/bpps/SES_Toolkit/SES%20Document%20Library/Learning%20Materials/UNDP_S7_Labour%20Guidance%20Note_June2021.pdf

	QUESTION 4: What is the overall project risk categorization?				
	Low Risk	☐			
	Moderate Risk	☐			
	Substantial Risk	√		**The overall risk is Substantial. The project has been categorized as "Substantial" given the varied range of risks rated as "Moderate" that require more extensive assessment and management measures.** **An ESMF has been developed during the PPG Stage.** **Further studies/assessments will be required during project implementation, including site specific E&S assessments (and appropriate management plans) at demo sites, a SESA for Policy Formulation activities under Component 1, and the development of a labor management procedure that will be required in the first year of project implementation and before any construction activities can take place.**	
	High Risk	☐			
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)				
	Question only required for Moderate, Substantial and High Risk projects				
	Is assessment required? (check if "yes")	**X**			***Status? (completed, planned)***
	if yes, indicate overall type and status		**X**	Targeted assessment(s)	Completed: stakeholder analysis, gender analysis Planned: Labor Assessment

					and Management Procedures
			X	ESIA (Environmental and Social Impact Assessment)	Prior to the construction/implementation of activities relating to the demo sites (i.e. Outputs 2.1 and 2.2), site specific environmental and social impact assessments will be required
			X	SESA (Strategic Environmental and Social Assessment)	Planned (most specifically in relation to activities under Component 1)
	Are management plans required? (check if "yes)	**X**			
	If yes, indicate overall type		**X**	Targeted management plans (e.g. Gender Action Plan, Emergency Response Plan, Waste Management Plan, others)	Completed: Gender Action Plan, Stakeholder Engagement Plan has been completed Labor management plan to be completed by end of yr. 1.
			X	ESMP (Environmental and Social Management Plan which may include range of targeted plans)	Planned (as an output of the site-specific ESIAs that are required for Outputs 2.1

					and 2.2)
			X	ESMF (Environmental and Social Management Framework)	Completed.
	Based on identified <u>risks</u>, which Principles/Project-level Standards triggered?		**Comments (not required)**		
	Overarching Principle: Leave No One Behind				
	Human Rights	√			
	Gender Equality and Women's Empowerment	√	The site-specific environmental and social assessments of each demo site shall include the evaluation of impacts on gender equality and women's empowerments. The assessments shall identify mitigation measures to be carried out during demo implementation. A gender action plan is also prepared.		
	Accountability	√			
	1. Biodiversity Conservation and Sustainable Natural Resource Management	√	All demos are subject to site-specific environmental and social assessments during their engineering design. The assessments shall include the assessment of biodiversity conservation and natural resource management issues. The assessments will identify the required mitigation measures which will be carried out during demo implementation.		
	2. Climate Change and Disaster Risks	√			
	3. Community Health, Safety and Security	√	The site-specific environmental and social assessments that will be conducted for each demo site shall include impact assessments on community health, safety, and working condition issues. The assessments will identify mitigation measures, which will be carried out during demo implementation.		
	4. Cultural Heritage	√			
	5. Displacement and Resettlement	√			
	6. Indigenous Peoples	√			

	7. Labour and Working Conditions	√	All demos will be subject to a labor management plan, to be developed in year 1 of the project by UNDP, together with the implementing partner (MARA), PMO and project partners.
	8. Pollution Prevention and Resource Efficiency	√	Each demo are subject to site-specific environmental and social assessments during their engineering design. The assessments shall include the assessment of biodiversity conservation and natural resource management issues. The assessments will identify the required mitigation measures which will be carried out during demo implementation.

Final Sign Off

Final Screening at the design-stage is not complete until the following signatures are included

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks		
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.		
Overarching Principle: Leave No One Behind **Human Rights**		**Answer (Yes/No)**
P.1	Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	*No*
P.2	Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	*Yes*
P.3	Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	*Yes*
Would the project potentially involve or lead to:		
P.4	adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	Yes
P.5	inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities? [26]	No
P.6	restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	Yes
P.7	exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	Yes
Gender Equality and Women's Empowerment		
P.8	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:		
P.9	adverse impacts on gender equality and/or the situation of women and girls?	*Yes*
P.10	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
P.12	exacerbation of risks of gender-based violence? *For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.*	No

[26] Prohibited grounds of discrimination include race, ethnicity, sex, age, language, disability, sexual orientation, gender identity, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender and transsexual people.

Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below		
Accountability		
Would the project potentially involve or lead to:		
P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes
P.14	grievances or objections from potentially affected stakeholders?	Yes
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		
1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	Yes
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	Yes
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	Yes
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5	exacerbation of illegal wildlife trade?	No
1.6	introduction of invasive alien species?	No
1.7	adverse impacts on soils?	Yes
1.8	harvesting of natural forests, plantation development, or reforestation?	No[27]
1.9	significant agricultural production?	No
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	Yes
1.12	handling or utilization of genetically modified organisms/living modified organisms?[28]	No
1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[29]	No

[27] The main sources of raw materials for biogas and biomass energy production are: 1) agricultural waste such as straw; and livestock manure resources from animal husbandry. China's arable land resources are limited and cannot be used to produce dedicated energy crops, and the destruction of forest resources is not allowed in China.

[28] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.

[29] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic

1.14	adverse transboundary or global environmental concerns?	Yes
Standard 2: Climate Change and Disaster Risks		
Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	No
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? *For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes*	Yes
2.3	increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	Yes
2.4	increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	No
Standard 3: Community Health, Safety and Security		
Would the project potentially involve or lead to:		
3.1	construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	Yes
3.2	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	Yes
3.3	harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	Yes
3.4	risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	No
3.5	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	Yes
3.6	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	No
3.7	influx of project workers to project areas?	Yes
3.8	engagement of security personnel to protect facilities and property or to support project activities?	Yes
Standard 4: Cultural Heritage		
Would the project potentially involve or lead to:		
4.1	activities adjacent to or within a Cultural Heritage site?	No
4.2	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	No
4.3	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect, and conserve Cultural Heritage may also have inadvertent adverse impacts)	Yes
4.4	alterations to landscapes and natural features with cultural significance?	No

resources.

4.5	utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
Would the project potentially involve or lead to:		
5.1	temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	Yes
5.2	economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	Yes
5.3	risk of forced evictions?[30]	Yes
5.4	impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No
Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	Yes
6.2	activities located on lands and territories claimed by indigenous peoples?	No
6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk*	Yes
6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	Yes
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	Yes
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8	risks to the physical and cultural survival of indigenous peoples?	No
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	Yes
Standard 7: Labour and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		
7.1	working conditions that do not meet national labour laws and international commitments?	Yes

[30] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families, or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

7.2	working conditions that may deny freedom of association and collective bargaining?	No
7.3	use of child labour?	No
7.4	use of forced labour?	No
7.5	discriminatory working conditions and/or lack of equal opportunity?	No
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		
8.1	the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	Yes
8.2	the generation of waste (both hazardous and non-hazardous)?	Yes
8.3	the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	No
8.4	the use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention*	No
8.5	the application of pesticides that may have a negative effect on the environment or human health?	No
8.6	significant consumption of raw materials, energy, and/or water?	Yes

Annex 5: UNDP Risk Register

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
1	The Project could lead to adverse impacts on enjoyment of the human rights (civil, political, economic, social, or cultural) of the affected population in the rural areas where the on-the-ground project activities will be carried out.	Social Political Financial	Impact (I) =3 Likelihood (P) =1 Risk level: Low	The designed demos are based on ongoing and planned rural revitalization projects of the national and provincial governments. In this case, there are generally no big issues that will prevent the demo implementation, or just minimal issues that are expected to be easily addressed. The project has been designed with extensive consultation with local people as well as the involvement of the relevant national and provincial government agency responsible for the formulation of relevant legislation on the installation of a decentralized RE-based power generation system or community mini-grids. A comprehensive Stakeholder Engagement Plan has been prepared during the PPG stage, with site-specific Plans prepared for each selected demo site.	PMU MARA
2	Potential occurrence of discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits.	Social Political Organizational	I =3 P =1 Risk Level: Low	The project could potentially offer a wide range of opportunities, for individuals and groups for support in the RE & EE technology applications demos and in the demo replications, participate and get trained in capacity development program activities, to be hired as a consultant or contractor for the project implementation. If care is not taken, any existing discrimination in the country towards woman could continue to come to play through the project. Special measures to ensure that any potential discrimination against women in the implementation of the project is countered have been identified and included in the project design. The results of such measures will be observed during project implementation, and if any corrective action is necessary, this will be addressed. Beyond that, the PDT will make special efforts to explore and facilitate the inclusion of interventions to enhance the role of women. Special	PMU

				activities have been designed to involve women, e.g., in the productive use of RE activities, to have strong representation of women in training courses and seminar-workshops, and to ensure a significant proportion of project consultants are women. The special measures are incorporated in the Gender Analysis and Action Plan that has been prepared for the EZCERTV Project.	
3	Potential adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services in some possible changes to the use of lands and resources in the implementation of the demonstrations.	Environmental	I =3 P =3 Risk Level: Medium	The demo RE-based power generation, distribution and storage systems may be sited on areas of habitat that could be adversely affected. The potential environmental impacts during the installation and operation of the demo RE-based power generation, distribution, and storage systems depend on the state or condition of the selected sites. The potential environmental impacts during operation are mostly continuous, while those associated with construction activities are expected to be temporary and mostly reversible. The detailed feasibility studies (inclusive of SES screening) to be carried out during full project implementation will clearly articulate mitigation measures for any alterations of the general topographical and environmental nature of the demo sites. Following world class engineering and construction design, specific mitigation measures that are regarded as best practice with respect to environmental management of the systems that will be installed, ensuring compliance with the requirements of the applicable environmental regulations/legislation. Any other potential impacts and risks including mitigation measures will be elaborated in the limited environmental and social assessments to be conducted during project implementation for each demo in selected towns and villages and to be completed prior to any physical work beginning on the installation of the required hardware.	PMU MARA
4	The associated construction, operation, or	Operational Environmental	I =2 P =4	Construction aspects and operation of demo RE-based power generation, mini-grid systems, energy storage	PMU MARA

	decommissioning of the demo installations may have potential health and safety risks to local communities due to the transport, storage, and use and/or disposal of any hazardous or dangerous materials (e.g. explosives, fuel, and other chemicals) that may be used during construction and operation.		Risk Level: Medium	systems and the EE retrofits can pose safety risk. For example Li batteries for the demo solar PV power systems, energy storage systems can produce potentially dangerous waste products. The design of all demos involved site-specific environmental and social assessments and recommend measures to mitigate any identified health and safety risks. Appropriate training will be provided to the target beneficiaries of the demos to ensure that they will be able to operate the installed systems properly and safely in such a way that the release or handling of waste products are properly controlled and managed. This will minimize or avoid any community health risks and safety issues for the communities regarding construction work involved in the installation of the demos, and the minimization and management of waste generated from these demos (e.g., spent lead-acid batteries, spent Li batteries).	
5	The project would potentially result in the generation of waste (both hazardous and non-hazardous).	Environmental Technical	I =3 P =5 Risk Level: Medium	The construction and operation of the RE-based energy generation (power and non-power applications) demos will generate wastes. PV panels and batteries will require disposal at end of life. EE retrofit projects (e.g., Use of LED lamps to replace CFLs/FLs) will potential, generate hazardous waste (Hg). The demos are designed considering the potential waste generation, and ensuring proper disposal of wastes from the construction, operation, and disposal stages. Disposal plans will be one of the requirements of the site-specific environmental and social impact assessment that will be conducted for each of demo. Those plans, for example, will include a disposal plan for the PV panels and batteries, which contain hazardous wastes, once their useful life is expired. For the EE retrofits, refrigerators are one example of an area that will require special plans for waste disposal in the E/S assessments. The refrigerators require proper disposal and may involve hazardous substances, particularly the refrigerant. Proper handling of the disposed refrigerants, as well as spent batteries,	PMU MARA

				and Hg in spent CFLs/FLs will be incorporated in the operating manuals of these demo units.	
6	Some project activities / demo sites might be located in lands of ethnic minorities and might require FPIC.	Social Political Organizational	I =3 P =1 Risk Level: Low	The project activity or demo sites may be located in lands of ethnic minority and thus require a change in land use, such as a change in ethnic minority settlement patterns (scattered to centralized housing). The demos have been designed considering the potential impacts on land use in particular for ethnic minority lands. The project design documents need to be written in inclusive language so that minority groups can receive adequate knowledge sharing and training so that they fully understand the purpose and design logic of the project. If changes to minority land use are involved, the feasibility and cost-benefit should be carefully evaluated, and audience awareness and acceptance should be well researched. So far, all the demos have not involved changes in lads of ethnic minority.	PMU MARA
7	Some project activities/ demo sites might lead to physical displacement or forced evictions.	Social Political	I =4 P =1 Risk Level: Low	The construction of some large factories may require additional land that may need to be vacated by the relocation of residents (e.g. scattered dwellings to centralized residential buildings). The demos will be designed considering the potential land demand and take the local development planning as the references. So far, all the demos have not referred to the issue of forced evictions.	PMU MARA
8	Some project activities / demo sites might impact cultural heritage.	Social Political	I =2 P =2 Risk Level: Low	The substitution of different household energy resources may affect the transmission of certain cultures, for example, the food prepared by traditional wood stoves may lose its original taste due to the replacement of the cooking fuel; another example is the clay bed (“KANG”) with wood stoves, which is widely used in the north, may be replaced in the process of energy substitution. The design and construction process of demonstration projects should consider the possible impact on traditional culture as much as possible, and some traditional applications (facilities) involving cultural heritage should be preserved and protected while	PMU MARA

				introducing new low-carbon (RE or EE) technologies. For some application methods involving traditional culture, the comprehensive impact of all aspects, such as environmental and health effects, needs to be comprehensively assessed, and if the combined environmental health and social benefits are poor, its replacement can be considered.	
9	Risks to project workers during construction.	Organizational Operational Technical	I =3 P =2 Risk Level: Medium	Project workers may face risks such as accidents and injuries during the construction process. The risks of workers' work during constructions that are involved in the designed demos will be further evaluated during the demo engineering design and measures will be identified and taken to avoid them, and insurance should be purchased for project workers.	PMU
10	Supply security risks (continuity, affordability) of renewable energy (or EE) alternatives to conventional energy for winter heating	Organizational Operational Technical	I =3 P =2 Risk Level: Medium	Some renewable energy production has time discontinuity and uncontrollability due to the dependence on renewable resources, and against the background of limited development of energy storage technology, the replacement of traditional energy by renewable energy is likely to cause security risks in energy supply. This risk also exists for alternatives of energy efficient technologies, for example, after the replacement of coal to natural gas for winter heating in Hebei, there was a problem of insufficient natural gas supply in winter. The designed demos will be further evaluated during their engineering design. At that point, the RE resource supply security risks will be accurately and comprehensively assessed, including the continuity and economy of the RE energy supply (affordability for residents), and ensure a stable supply of the RE resource.	PMU
11	Risk of increased labor input by women (e.g., will projects such as biomass fuels, biogas, etc., increase women's labor input for household energy access)	Social Political Financial	I =3 P =1 Risk Level: Low	Some forms of energy substitution may require more labor input, such as traditional domestic biogas, and households may need to invest more energy and time in collecting the raw materials for energy production, and these tasks are also likely to require more involvement of women. The design of the demos has considered resource,	PMU

				financial and labor inputs for energy supply substitution, and has assessed the feasibility of these inputs and their potential impact, and reduce end-user labor inputs, including through scaling up and centralizing production, to the extent possible, especially to avoid increasing the burden on women and children.	
12	Some project activities may require significant consumption of raw materials (e.g. biomass) and natural resources (energy and/or water)	Financial Technical	I =3 P =3 Risk Level: Medium	Some renewable energy projects may present potential resource efficiency issues, such as whether biomass resources are being used more efficiently in biomass utilization projects; some renewable energy or energy efficiency project production may increase water demand, such as the high dependence of biomass power generation on water from a whole life cycle perspective. The design of the demos projects has fully considered the adequacy, availability and utilization efficiency of resources; for example, in biomass power projects, the utilization efficiency of biomass is much higher than that of traditional biomass (many straws are returned to the field); and for the energy and water consumption involved in RE/EE projects, the projects will conduct coupled analysis and accounting for the whole life cycle and water-energy nexus to avoid additional environmental pressure from new projects.	PMU MARA
13	The project management unit may not be able to effectively organize and coordinate the participation of relevant project stakeholders (e.g., relevant departments of local governments, specific project execution units, domestic and foreign experts, etc.).	Organizational, Operational	I = 4 P = 1 Risk Level: Low	The implementing agency of the project is the MARA, which is one of the main departments of the State Council's new rural construction management for rural revitalization. MARA has rich experience in GEF project design, implementation, and management. With this asset, the proposed project will be designed to minimize the chances of this risk from happening. If the PMU indeed become unable to prevent this risk, the UNDP China will manage and expedite the procurement of external personnel who will work on the affected project activities. If need be, the affected activities may have to be modified to allow expeditious implementation and completion.	PMU MARA
14	Inadequate and possibly	Organizational	I=3	The MARA, NDRC and the project management unit shall	PMU

	constrained coordination with the implementers of energy projects in local governments in the implementation of the project's barrier removal activities.	Operational Regulatory	P=2 Risk Level: Medium	maintain close working relations with relevant departments of the State Council and can make necessary adjustments to the work in a timely manner according to the adjustment of national policies, which can ensure the implementation of the project and achieve optimal results. In case this happens, MARA/NDRC will facilitate discussions with project stakeholders and relevant government authorities through the project steering committee (PSC) to come up with decisions on the coordinated implementation of the agreed barrier removal activities.	MARA NDRC
15	The co-financing for the project activities may not be timely and enough for supporting project implementation.	Financial Political Strategic Operational	I=4 P=1 Risk Level: Medium	MARA attaches great importance to the implementation of the project and provides relevant technical, resource, capital, and institutional support. The relevant project provinces will also promise the project supporting funds in the early stage of project implementation. During the project design and implementation process, MARA will continuously promote the establishment of a diversified financing model to ensure that the project supporting funds are in place. In case this problem will occur, the reallocation of budget will be done to support the implementation of affected activities. This may entail the delivery of alternative outputs that are still contributing to the achievement of the relevant project outcome. Constant follow-up with the pertinent co-financers will be conducted either to secure the committed co-financing or negotiate the amount of co-financing.	PMU MARA
16	Rural communities may not support the project implementation promptly and sufficiently.	Organizational Operational	I=2 P=3 Risk Level: Low	Project includes strong promotional campaigns to target rural communities/towns, as well as outreach to relevant local governments to ensure their support. Liaison by capable project team personnel will further ensure support. Integration of productive use income generation opportunities will increase community interest in systems. If the project partners in the rural communities become	PMU MARA NDRC

				remiss in their obligations and commitments to the project implementation, follow-up discussions between MARA/NDRC, and relevant local government leaders will be carried out to determine and resolve any issue.	
17	The private sector in the rural areas will not be interested in investing in green development technologies.	Financial Operational Strategic	I=4 P=1 Risk Level: Low	The project design, particularly the investment activities, will be coordinated with the relevant private sector entities in the target rural communities to involve them right away in the project planning, and get their inputs and comments in the design of the project activities. If this happens, the PMU, with the guidance of NDRC/UNDP will carry out direct consultations with target private sector entities to determine their concerns and explore ways to fulfill their initial commitments and ensure end-users better understanding about the use and benefits of EE/RE technology applications.	PMU MARA NDRC MOF
18	Recommended policies may not be approved by the relevant authorities or may be approved but not effectively enforced.	Political	I=3 P=3 Risk Level: Medium	The project will include the piloting of the application of the support policies for green town and village development to gauge the effectiveness of said policies. This will help guide the relevant government authorities in the finalization, approval, and effective enforcement of such policies. In case this happens, MARA/NDRC will facilitate discussions with project stakeholders and relevant government authorities through the project steering committee (PSC) to come up with decisions on expediting the approval, or reformulation, of the recommended policies/regulations.	PMU MARA NDRC MOF
19	Sustaining the outcomes and benefits of GEF investment on the activities implemented will not be fully sustained.	Technical Operational	I = 4 P = 2 Risk Level: Medium	Sustainable follow-up plan for the replication of the demonstrated applicable and feasible EE/RE technologies associated with green town and village development in other provinces. The plan must be supported (financed) by the local governments. In case, despite the measures, sustainability of project outcomes and benefits is seen to be in jeopardy, MARA/NDRC, the project team, and the project steering committee will meet to come up with alternative measures to ensure sustainability.	PMU MARA NDRC

20	Risks of non-availability of: (a) technical expertise and consequential changes in the project's schedule of activities; and (b) expected financing from the government and co-financing from all stakeholders during times of pandemic. *(Please refer to Annex G of the EZCERTV CEO Endorsement Request Document regarding addressing Covid-19 issues)*	Environmental	I = 4 P = 1 Moderate	MARA will make use of its wide network of both external and internal expertise (individuals or groups of individuals) who they work with on project design, implementation, and management. A clear anticipation and understanding of the type and quality expertise that are needed, their locations and when these will be required will help potential alternatives in case these will be necessary. With such an asset, the chances of this risk from happening is minimized, and impacts such as delays in the implementation of scheduled/planned project activities can be prevented. MARA (REEA) and the PMO will: (a) closely monitor the public health situation in the project sites; and also (b) regularly update its list of technical experts that can be tapped to worked on the project implementation promptly and quickly or with minimal delays. If this risk cannot be prevented, there is also the option, subject to the approval of the GEFSec for UNDP China to assist in the procurement of internal or external experts who will work on the affected project activities. If need be, the affected activities may also be modified to allow expeditious implementation and completion. Regarding the potential risks of the non-availability in a timely manner of the committed financing and co-financing during pandemic situations, During the implementation process, MARA will continuously promote the establishment of a diversified financing model to ensure that the project supporting funds are in place. In case this problem will occur, the reallocation of budgets will be done to support the implementation of affected activities. This may also entail the delivery of alternative outputs that are still contributing to the	MARA PMO	No Change

				achievement of the relevant project outcome. Constant follow-up with the pertinent co-financers will be conducted either to secure the committed co-financing or renegotiate the amount of co-financing.		
21	RE-based energy generation (power and non-power purposes) and EE system installations can be seriously affected by adverse climate-related events. *(Please refer to Annex H of the EZCERTV CEO Endorsement Request Document regarding the project's Climate Risk Assessment)*	Environmental	I = 4 P = 1 Risk Level: Low	It is already common in international design and engineering practices, as well as in the construction/installation of RE-based energy generation units to follow proper engineering and construction design and construction that ensure not only structural integrity but also climate resilience. This applies also in the procurement, design/engineering, installation, and operation of the pertinent installations[31]. Depending on the extent of the impacts of the adverse climate –related events, appropriate modifications in the installations (and budget) will be done. Potential reduction in the number of installations, or replacement with alternative demos will be done while considering the need to ensure the resulting interventions are still contributing to the realization of the project outcomes.	PMU	

[31] The design and construction/installation of the physical infrastructures that will be installed will be based on the technical and structural specifications that major bilateral and multi-lateral donors require for the infrastructure projects that they are funding in China.

Annex 6: Overview of Project Staff and Technical Consultancies

Consultant	Time Input	Tasks, Inputs and Outputs
		For Project Management
Local / National contracting		
Project Manager /Coordinator *Rate: $3,750/month*	*60 months / over 5 years*	*The Project Manager (PM), together with the Chief Technical Advisor will be responsible for the overall management of the project, including the mobilization of all project inputs, supervision over project staff, consultants, and sub-contractors.* *Duties and Responsibilities* • *Manage the overall conduct of the project.* • *Plan the activities of the project and monitor progress against the approved workplan.* • *Execute activities by managing personnel, goods and services, training, and low-value grants, including drafting terms of reference and work specifications, and overseeing all contractors' work.* • *Monitor events as determined in the project monitoring plan, and update the plan as required.* • *Provide support for completion of assessments required by UNDP, spot checks and audits.* • *Manage requests for the provision of UNDP financial resources through funding advances, direct payments or reimbursement using the FACE form.* • *Monitor financial resources and accounting to ensure the accuracy and reliability of financial reports.* • *Monitor progress, watch for plan deviations, and make course corrections when needed within project board-agreed tolerances to achieve results.* • *Ensure that changes are controlled, and problems addressed.* • *Perform regular progress reporting to the project board as agreed with the board, including measures to address challenges and opportunities.* • *Prepare and submit financial reports to UNDP on a quarterly basis.* • *Manage and monitor the project risks – including social and environmental risks - initially identified and submit new risks to the Project Board for consideration and decision on possible actions if required; update the status of these risks by maintaining the project risks log;* • *Capture lessons learned during project implementation.* • *Prepare revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.* • *Prepare the inception report no later than one month after the inception workshop.* • *Ensure that the indicators included in the project results framework are monitored annually in advance of the GEF PIR submission deadline so that progress can be reported in the GEF PIR.* • *Prepare the GEF PIR;* • *Assess major and minor amendments to the project within the parameters set by UNDP-GEF;* • *Monitor implementation plans including the gender action plan, stakeholder engagement plan, and any environmental and social management plans;* • *Monitor and track progress against the GEF Core indicators.* • *Support the Mid-term review and Terminal Evaluation process.*

Consultant	Time Input	Tasks, Inputs and Outputs
		• *Add technical tasks, as necessary.*
Project Finance Officer *Rate: $2,600/month*	*60 months / over 5 years*	*Duties and Responsibilities* • *Keep records of project funds and expenditures, and ensure all project-related financial documentation are well maintained and readily available when required by the Project Manager;* • *Review project expenditures and ensure that project funds are used in compliance with the Project Document and GoC financial rules and procedures;* • *Validate and certify FACE forms before submission to UNDP;* • *Provide necessary financial information as and when required for project management decisions;* • *Provide necessary financial information during project audit(s);* • *Review annual budgets and project expenditure reports, and notify the Project Manager if there are any discrepancies or issues;* • *Consolidate financial progress reports submitted by the responsible parties for implementation of project activities;* • *Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports.*
For Technical Assistance		
Outcome 1		
Local / National contracting		
Project Gender Officer *Rate: $1,250/week*	*12 weeks / over 5 years*	*Duties and Responsibilities* • Monitor progress in implementation of the project Gender Action Plan ensuring that targets are fully met, and the reporting requirements are fulfilled; • Oversee/develop/coordinate implementation of all gender-related work; • Review the Gender Action Plan annually, and update and revise corresponding management plans as necessary; • Work with the M&E officer and Safeguards Officer to ensure reporting, monitoring and evaluation fully address the gender issues of the project;
Project Monitoring and Evaluation Officer *Rate: $1,250/week*	*24 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor identified project risks as implementation progresses and participate in the production of progress reports ensuring that they meet the necessary reporting requirements and standards;* • *Oversee/develop/coordinate the implementation of the stakeholder engagement plan;* • *Ensure project risks are mitigated and managed throughout project implementation;* • *Ensure that all stakeholders participate to the project activities implementation according to the engagement plan, especially for women, youth, and marginalized groups;* • *Ensure project's M&E meets the requirements of the Government, the UNDP Country Office, and UNDP-GEF; develop project-specific M&E tools as necessary;* • *Visit project sites as and when required to appraise project progress on the ground and validate written progress reports.*
Project Safeguards Officer	*12 weeks / over 5 years*	*Duties and Responsibilities* • *Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDPs SES policy is fully met,*

Consultant	Time Input	Tasks, Inputs and Outputs
Rate: $1,250/week		*and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all safeguard related plans;* • *Ensure social and environmental grievances are managed effectively and transparently;* • *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* • *Ensure full disclosure with concerned stakeholders;* • *Ensure environmental and social risks are identified, avoided, mitigated, and managed throughout project implementation;* • *Work with the M&E officer to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project.*
Outcome 2		
International / Regional and global contracting		
Chief Technical Advisor *Rate: $700/day*	*280 days / over 5 years*	*Duties and Responsibilities* • *Assist the PM and PMO in the specific issues and requirements in project implementation that will be requested to achieve project objectives.* • *Assist in the development of relevant Terms of References for all consultants and contractors in coordination with the PM.* • *Advise on scheduling, procurement, and monitoring issues.* • *Provide written advice and comment on policy, regulatory, institutional, financial, technical, and awareness issues.* • *Advise on establishing and managing monitoring and evaluation programs for all aspects of the project following the Log Frame of the project and on the preparation and finalization of the Annual Project Report/Project Implementation Review as required by the UNDP/GEF M&E system.* • *Advise the UNDP, PM and PMO and independent evaluators in the conduct of the Mid-term Review (MTR) and Terminal Evaluation (TE), exit strategy and the project closure process and other UNDP-GEF requirements.* • *Provide support and inputs to the PM for the preparation of the project final report (including the best practices, lessons learnt etc.).* • *Any other relevant tasks as assigned by the PM.*
Outcome 3		
Local / National contracting		
Knowledge Management Officer *Rate: $1,250/month*	*60 months / over 5 years*	*Duties and Responsibilities* • *Develop a communication strategy.* • *Develop a knowledge management plan for the project.* • *Lead the implementation of the approved communications strategy action plan.* • *Raise awareness of the project with target audiences for all project components.* • *Build on the project's digital presence through social media.* • *Produce quarterly newsletters or infographic for the project and disseminates to all project partners.* • *Draft media advisories and press releases*

Consultant	Time Input	Tasks, Inputs and Outputs
		• *Liaise with media for press releases and events, establish media connections and aim to get maximum media attention for press releases, special events, press interview either through broadcast or print media.* • *Compile contents for all planned communication efforts* • *Generate ideas and lead development and design of various publicity materials, including web content, brochures, and short stories.* • *Assist in collecting 'most significant change' stories to support annual reporting and editing project specific and related news updates.* • *Regularly update on the impact of the communication activities, including media outreach and online efforts* • *Share communication products and updates with UNDP CO, for regional and global visibility of the project.* *Any other tasks related to communication activities in the project*

Annex 7: Partners Capacity Assessment Tool and HACT assessment

Partners Capacity Assessment Tool can be accessed Here
HACT assessment can be accessed Here

Annex 8: GEF Budget Template

The GEF Budget Template is provided below and submitted as a supplementary annex

Expenditure Category	Detailed Description	Component (USDeq.)						Total (USD eq.)	Responsible Entity
		Component 1	*Component 2*	*Component 3*	*Sub-Total*	*M&E*	*PMC*		(Executing Entity receiving funds from the GEF Agency)[1]
Equipment	Equipment and Furniture: US$3,087,500 US$2,000,000 for activities under Output 2.1: provide the equipment for demos to facilities the application of RE technology in villages. US$1,000,000 for activities under output 2.2: provide the equipment for demos to facilities the application of EE technology in villages. US$87,500 for the activities under output 2.3: provide the equipment for replication sites		3,087,500		3,087,500			3,087,500	MARA
Equipment	Information Technology Equipment: US$36,617 US$15,000 for office equipment (computers, printers, copier, etc.); US$15,000 for Communications and conferencing equipment (digital camera, overhead projector, and screen, etc.) and US$6,617 for supplies and maintenances.		36,617		36,617			36,617	MARA
Contractual Services – Individual	• Contractual services – Implementing Partner: US$355,000 in total (US$225,000 for a full time project manager and US$130,000 for the project finance officer for 50 months to support all components.)				-		355,000	355,000	MARA
Contractual Services – Company	Contractual Services-Companies: US$550,000: All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company): • Sub-contract 1 – US$40,000: Design and evaluate the 2 training sessions in Activity 1.2.2 (administration of training programs are under a separate budget, see note 6 below) • Sub-contract 2 – US$70,000: Develop guidelines and standards on the application of RE/EE/EC technologies and measures in rural areas • Sub-contract 3 – US$70,000: Standardization of	550,000			550,000			550,000	MARA

	RE/EE/EC technology specs and MEPS, testing and certification procedures for local and imported technologies, and design and operation of RE/EE/EC projects • Sub-contract 4 – US$120,000: Assess rural energy status in the selected project rural areas (US$15,000 x 8 = US$120,000) • Sub-contract 5 – US$30,000: Design and evaluate the training sessions in Activity 1.4.2 (administration of training programs are under a separate budget, see note 6 below) • Sub-contract 6 – US$160,000: Prepare energy-integrated development plans in the selected project rural areas (US$20,000 x 8 = US$160,000) • Sub-contract 7 – US$60,000: Develop guidelines for the promotion of zero-carbon villages and towns development								
Contractual Services – Company	Contractual Services-Companies: US$2,698,850: All sub-contract budgets include: national and international experts, Equipment for Demos Replication, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company): • Sub-contract 1 – US$160,000: Develop engineering designs and implementation plans for the RE and EE demos in selected rural areas (Activity 2.1.2 and Activity 2.2.1). US$20,000 x 8 = US$160,000 • Sub-contract 2 – US$240,000: Implementation of the RE and EE demos in selected rural areas (Activity 2.1.3 and Activity 2.2.2). US$30,000 x 8 = US$240,000 • Sub-contract 3 – US$120,000: Design of M&E systems of the RE and EE demos in selected rural areas (Activity 2.1.4 and Activity 2.2.3). US$15,000 x 8 = US$120,000 • Sub-contract 4 – US$40,000: Design and implement a promotional program for rural stakeholders on technical advantages of RE technologies measures (Activity 2.1.5) • Sub-contract 5 – US$30,000: Design and evaluate the training sessions in Activity 2.1.6 (administration of training programs are under a separate budget, see note 14 below) • Sub-contract 6 – US$40,000: Design and implement a promotional program for rural stakeholders on technical advantages of EE/EC technologies measures (Activity 2.2.4) • Sub-contract 7 – US$30,000: Design and evaluate the		2,673,850		2,673,850			2,673,850	MARA

	training sessions in Activity 2.2.5 (administration of training programs are under a separate budget, see note 14 below) • Sub-contract 8 – US$30,000: Design and implement a promotional program to improve use and sales of high EE appliances and equipment in rural areas (Activity 2.2.6) • Sub-contract 9 – US$50,000: Design an integrated energy management system for rural areas (Activity 2.2.7) • Sub-contract 10 – US$105,000: Prepare and disseminate case studies based on the demos (Activity 2.3.1)): US$10,000 x 8 = US$80,000; and prepare lessons learned: US$25,000 (part of M&E) • Sub-contract 11 – US$500,000: Develop implementation plans for demos replication (Activity 3.3.2). US$10,000 x 50 = US$500,000 • Sub-contract 12 – US$1,353,850: Procure Equipment for the replication of a demo project at a town level								
Contractual Services – Company	Contractual Services-Companies: US$430,000: All sub-contract budgets include: national and international experts, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company): • Sub-contract 1 – US$20,000: Assess awareness levels of relevant stakeholders about zero-carbon rural development (Activity 3.1.1) • Sub-contract 2 – US$30,000: Design and evaluate the training sessions in Activity 3.1.2 (administration of training programs are under a separate budget, see note 20 below) • Sub-contract 3 – US$30,000: Design and evaluate the training sessions in Activity 3.1.3 (administration of training programs are under a separate budget, see note 20 below) • Sub-contract 4 – US$50,000: Develop a project management and information communication platform for zero-carbon rural development (Activity 3.1.4) • Sub-contract 5 – US$30,000: Establish a social participation mechanism to strengthen stakeholders' capacity on zero-carbon rural development (Activity 3.3.1) • Sub-contract 6 – US$30,000: Develop a plan to publicize EZCERTV project (Activity 3.3.2) • Sub-contract 7 – US$30,000: Implement the promotional strategy to promote zero-carbon			430,000	430,000			430,000	MARA

	development to rural communities (Activity 3.3.3) • Sub-contract 8 – US$40,000: Design and implement promotional initiatives to involve women and marginalized groups in zero-carbon rural development (Activity 3.3.4) • Sub-contract 9 – US$40,000: Facilitate the establishment of a private sector-led service industry for RE/EE technologies in rural areas (Activity 3.4.1) • Sub-contract 10 – US$30,000: Design and evaluate the training sessions in Activity 3.4.3 (administration of training programs are under a separate budget, see note 20 below) • Sub-contract 11 – US$60,000: Design and evaluate the training sessions and the promotional program in Activity 3.5.2 (administration of training programs are under a separate budget, see note 20 below) • Sub-contract 12 – US$40,000: Design and implement promotional initiatives to enhance green financing to support zero-carbon rural development (Activity 3.5.3)								
Contractual Services – Company	Contractual Services-Companies: US$2,698,850: All sub-contract budgets include: national and international experts, Equipment for Demos Replication, travel, meetings, workshops and DSA for the subcontractor to undertake the activities and contribute to the outputs (multiple activities could be conducted by one company): • Sub-contract 1 – US$160,000: Develop engineering designs and implementation plans for the RE and EE demos in selected rural areas (Activity 2.1.2 and Activity 2.2.1). US$20,000 x 8 = US$160,000 • Sub-contract 2 – US$240,000: Implementation of the RE and EE demos in selected rural areas (Activity 2.1.3 and Activity 2.2.2). US$30,000 x 8 = US$240,000 • Sub-contract 3 – US$120,000: Design of M&E systems of the RE and EE demos in selected rural areas (Activity 2.1.4 and Activity 2.2.3). US$15,000 x 8 = US$120,000 • Sub-contract 4 – US$40,000: Design and implement a promotional program for rural stakeholders on technical advantages of RE technologies measures (Activity 2.1.5) • Sub-contract 5 – US$30,000: Design and evaluate the training sessions in Activity 2.1.6 (administration of training programs are under a separate budget, see note 14 below) • Sub-contract 6 – US$40,000: Design and implement a promotional program for rural stakeholders on technical advantages of EE/EC technologies measures				-	25,000		25,000	MARA

	(Activity 2.2.4) • Sub-contract 7 – US$30,000: Design and evaluate the training sessions in Activity 2.2.5 (administration of training programs are under a separate budget, see note 14 below) • Sub-contract 8 – US$30,000: Design and implement a promotional program to improve use and sales of high EE appliances and equipment in rural areas (Activity 2.2.6) • Sub-contract 9 – US$50,000: Design an integrated energy management system for rural areas (Activity 2.2.7) • Sub-contract 10 – US$105,000: Prepare and disseminate case studies based on the demos (Activity 2.3.1)): US$10,000 x 8 = US$80,000; and prepare lessons learned: US$25,000 (part of M&E) • Sub-contract 11 – US$500,000: Develop implementation plans for demos replication (Activity 3.3.2). US$10,000 x 50 = US$500,000 • Sub-contract 12 – US$1,353,850: Procure Equipment for the replication of a demo project at a town level								
International Consultants	International consultants: US$147,000 (including professional fees, travel expenses and DSA) for 210 working days at US$700 per working day (multiple activities could be conducted by one consultant): • 180 working days in total for activities under Output 1.1, and specifically: 30 working days for Activity 1.1.1; 30 working days for Activity 1.1.2; 40 working days for Activity 1.1.3; 40 working days for Activity 1.1.4; 40 working days for Activity 1.1.5; (US$126,000) • 30 working days for Activity 1.2.1; (US$21,000)	147,000			147,000			147,000	MARA
International Consultants	International consultants: US$301,000 (including professional fees, travel expenses and DSA) for 330 working days at US$700 per working day (including the Chief Technical Advisor), and both Mid-Term Review and Terminal Evaluation: • 50 working days for activities under Output 2.1, and specifically: 50 working days for Activity 2.1.1; (US$35,000) • 280 working days for the CTA, and specifically: 40 working days in year 1, and 60 working days for year 2 through year 5; (US$196,000) • US$35,000 for the MTR and US$35,000 for the TE (part of M&E)		231,000		231,000			231,000	MARA
International Consultants	International consultants: US$112,000 (including professional fees, travel expenses and DSA) for 160 working days at US$700 per working day (multiple			112,000	112,000			112,000	MARA

	activities could be conducted by one consultant): • 80 working days in total for activities under Output 3.2, and specifically: 40 working days for Activity 3.2.1; 40 working days for Activity 3.2.2; (US$56,000) • 40 working days for Activity 3.4.2; (US$28,000) • 40 working days for Activity 3.5.1; (US$28,000)								
International Consultants	International consultants: US$301,000 (including professional fees, travel expenses and DSA) for 330 working days at US$700 per working day (including the Chief Technical Advisor), and both Mid-Term Review and Terminal Evaluation: • 50 working days for activities under Output 2.1, and specifically: 50 working days for Activity 2.1.1; (US$35,000) • 280 working days for the CTA, and specifically: 40 working days in year 1, and 60 working days for year 2 through year 5; (US$196,000) • US$35,000 for the MTR and US$35,000 for the TE (part of M&E)				-	70,000		70,000	MARA
Local Consultants	National consultants: US$235,000 (including professional fees, travel expenses and DSA) for 500 working days at US$350 per working day (multiple activities could be conducted by one consultant), and local consultants for monitoring activities)multiple activities could be conducted by one consultant): • 130 working days for activities under Output 1.1, and specifically: 20 working days for Activity 1.1.1; 20 working days for Activity 1.1.2; 30 working days for Activity 1.1.3; 30 working days for Activity 1.1.4; 30 working days for Activity 1.1.5; (US$45,500) • 20 working days for activities under Output 1.2, and specifically: 20 working days for Activity 1.2.1; (US$7,000) • 250 days for a local RE/EE/EC technology procurement specialist to support all stakeholders, consultants, both international and local, and service companies, for all EZCERTV activities to facilitate and guarantee uniformity of RE/EE/EC technologies, measures, guidelines and standards for all selected locations as well as replication sites; (US$87,500) • 100 days for a national expert to liaise with national and local government to facilitate the approval and enforcement of all formulated policies and regulations in all EZCERTV activities; (US$35,000) • Monitoring of project risks (US$15,000); Gender action plan monitoring (US$15,000), Social and environmental safeguard monitoring (US$15,000), and stakeholder engagement plan monitoring	175,000			175,000			175,000	MARA

	(US$15,000). (M&E activities)								
Local Consultants	National consultants: US$124,500 (including professional fees, travel expenses and DSA) for 270 working days at US$350 per working day, and both Mid-Term Review and Terminal Evaluation: • 30 working days for activities Activity 2.1.1; (US$10,500) • 240 working days for 8 local consultants to support for demos replication and scale-up in each provinces (Activity 2.3.2). US$350 x 30 x 8 = US$84,000 • US$15,000 for the MTR and US$15,000 for the TE (part of M&E)		94,500		94,500			94,500	MARA
Local Consultants	National consultants: US$103,000 (including professional fees, travel expenses and DSA) for 80 working days at US$350 per working day (multiple activities could be conducted by one consultant), a project knowledge management officer: • 40 working days in total for activities under Output 3.2, and specifically: 20 working days for Activity 3.2.1; 20 working days for Activity 3.2.2; (US$14,000) • 20 working days for activities 3.4.2; (US$7,000) • 20 working days for activities 3.5.1; (US$7,000) • 60 months for the project knowledge management officer, US$1,250 per month x 60 months = US$75,000: part–time staff responsible for communicating with all stakeholders, support the project dissemination and knowledge products development			103,000	103,000			103,000	MARA
Local Consultants	National consultants: US$235,000 (including professional fees, travel expenses and DSA) for 500 working days at US$350 per working day (multiple activities could be conducted by one consultant), and local consultants for monitoring activities)multiple activities could be conducted by one consultant): • 130 working days for activities under Output 1.1, and specifically: 20 working days for Activity 1.1.1; 20 working days for Activity 1.1.2; 30 working days for Activity 1.1.3; 30 working days for Activity 1.1.4; 30 working days for Activity 1.1.5; (US$45,500) • 20 working days for activities under Output 1.2, and specifically: 20 working days for Activity 1.2.1; (US$7,000) • 250 days for a local RE/EE/EC technology procurement specialist to support all stakeholders, consultants, both international and local, and service companies, for all EZCERTV activities to facilitate and guarantee uniformity of RE/EE/EC technologies,				-	115,000		115,000	MARA

	measures, guidelines and standards for all selected locations as well as replication sites; (US$87,500) • 100 days for a national expert to liaise with national and local government to facilitate the approval and enforcement of all formulated policies and regulations in all EZCERTV activities; (US$35,000) • Monitoring of project risks (US$15,000); Gender action plan monitoring (US$15,000), Social and environmental safeguard monitoring (US$15,000), and stakeholder engagement plan monitoring (US$15,000). (M&E activities) National consultants: US$124,500 (including professional fees, travel expenses and DSA) for 270 working days at US$350 per working day, and both Mid-Term Review and Terminal Evaluation: • 30 working days for activities Activity 2.1.1; (US$10,500) • 240 working days for 8 local consultants to support for demos replication and scale-up in each provinces (Activity 2.3.2). US$350 x 30 x 8 = US$84,000 • US$15,000 for the MTR and US$15,000 for the TE (part of M&E) • US$25,000 for GEF core indicators monitoring before MTR and TE (part of M&E)								
Trainings, Workshops, Meetings	Training, Workshops and Conferences: US$176,000 • Workshops for Formulation, recommendation of new and improved policies and regulations on zero carbon development in rural areas (Activities 1.1.3 and 1.1.4), 2 sessions: US$20,000 x 2 = US$41,000 • Capacity building program for local governments on integrated RE resource planning and analysis, including a post-completion evaluation of the training program (Activity 1.2.2). 2 sessions US$30000 x 2 = US$60,000 • Capacity building program for rural governments on monitoring, recording, analysis and management of energy generation and utilization data, including a post-completion evaluation of the training program (Activity 1.4.2). US$40,000 • Project inception workshop: US$20,000 (part of M&E); and Annual PSC meeting US$3,000 x 5 = US$15,000 (M&E)	141,000			141,000			141,000	MARA
Trainings, Workshops, Meetings	Training, Workshops and Conferences: US$60,000 • Capacity building program to facilitate the establishment of a rural RE service industry (Activity 2.1.6). US$30,000 • Capacity building program to facilitate the establishment of a rural EE/EC service industry		60,000		60,000			60,000	MARA

	(Activity 2.2.5). US$30,000								
Trainings, Workshops, Meetings	Training, Workshops and Conferences: US$315,600 • Capacity building program for local government and officials and industry management teams about zero-carbon rural development (Activity 3.1.2). US$35,600 • Capacity building program for the private sector, NGOs, and rural communities about zero-carbon rural development (Activity 3.1.3). US$30,000 • US$120,000 for knowledge exchange programs with a minimum of 4 "Belt and Road" countries on the development of zero-carbon villages and towns. Each exchange program will include training programs and information sharing initiatives (Activity 3.2.2) • US$60,000 to conduct international bilateral and multilateral cooperation programs on zero-carbon rural development, training the people from" Belt and Road "countries (Activity 3.3.3). US$30,000 x 2 = US$60,000 • Capacity building program for the private sector to enable them to procure RE/EE technologies for rural development (Activity 3.4.3). US$30,000 • Capacity building and promotional program to enhance private sector participation (Activity 3.5.2). US$40,000			315,600	315,600			315,600	MARA
Trainings, Workshops, Meetings	Training, Workshops and Conferences: US$175,000 • Workshops for Formulation, recommendation of new and improved policies and regulations on zero carbon development in rural areas (Activities 1.1.3 and 1.1.4), 2 sessions: US$20,000 x 2 = US$40,000 • Capacity building program for local governments on integrated RE resource planning and analysis, including a post-completion evaluation of the training program (Activity 1.2.2). 2 sessions US$30000 x 2 = US$60,000 • Capacity building program for rural governments on monitoring, recording, analysis and management of energy generation and utilization data, including a post-completion evaluation of the training program (Activity 1.4.2). US$40,000 • Project inception workshop: US$20,000 (part of M&E); and Annual PSC meeting US$3,000 x 5 = US$15,000 (M&E)				-	35,000		35,000	MARA
Travel	Travel: US$50,000: the travel and DSA expenses for 2 PMO officers participating to each of the 3 training sessions (2 sessions for Activity 1.2.2 and 1 session for Activity 1.4.2) and to the 2 workshops (Activities 1.1.3 and 1.1.4), and for 8 stakeholders from the selected	50,000			50,000			50,000	MARA

	rural locations participating to the same 3 training sessions, and 2 workshops (50 travels at US$1,000 each), (The other participants to the 3 training sessions and 2 workshops will be connected via videoconference)								
Travel	Travel: US$30,000: the travel and DSA expenses for 2 PMO officers participating to each of the two training programs (Activities 2.1.6 and 2.2.5) and for 8 stakeholders from the selected rural locations participating to the same 2 training programs (a total of 20 travels at US$1,500 each). (The other participants to both training programs will be connected via videoconference)		30,000		30,000			30,000	MARA
Travel	Travel: US$50,000: the travel and DSA expenses for 2 PMO officers participating to each of the 4 training programs (Activities 3.1.2, Activity 3.1.3, Activity 3.4.3, and Activity 3.5.2), and for 8 stakeholders from the selected rural locations participating to the same 4 training programs (a total of 40 travels at US$1,250 each). (The other participants to the 4 training programs will be connected via videoconference)			50,000	50,000			50,000	MARA
Travel	• Travel: US$16,000: for Project Management Office (PMO) travels cost and DSA for 2 officers to each of the 8 selected locations (US$1,000 per person per location).				-		16,000	16,000	MARA
Office Supplies	• Budget set up for office supplies and stationery used in PMU US$ 4,353				-		4,353	4,353	MARA
Other Operating Costs	• Professional services: US$25,000 for project audit				-		25,000	25,000	MARA
Grand Total		1,063,000	6,213,467	1,010,600	8,287,067	245,000	400,353	8,932,420	

Annex 9: Stakeholder Engagement Plan

(The Stakeholder Engagement Plan is shown below and also submitted as a supplementary annex)

This annex provides strategic guidance for how the projects key stakeholders will be engaged in the project during its implementations. The stakeholder engagement plan may be further elaborated or revised during the projects inception period and will be finally approved at the project inception workshop. During the PPG process, a stakeholder analysis was undertaken to identify key stakeholders, assess their interests in the project and define their roles and responsibilities in its implementation. Overall, support for the project and involvement in it was secured from a diverse group of stakeholders, whose roles are summarized in **Table 4 (ProDoc, p. 47)**.

The roles of key stakeholders in the project' oversight and management, including the project steering committee, is provided in the **Governance and Management Arrangements** section of the Project Document (i.e. Project Board members, Project Management Office), and are only briefly mentioned here.

Table 1 below summarize the main engagements of the key stakeholders in the implementation of the project. It also shows the capacity building needs of each stakeholder as identified during the project development phase.

Table 1: Stakeholder Engagement and Training and Capacity Building Needs of the Stakeholder group

Stakeholder	Engagement in the Project	Training and Capacity Building Needs
Ministry of Finance (MoF)	As the GEF Operational Focal Point MOF will receive the GEF grant for the current project on behalf of the Chinese Government. MOF will be a core member of the Project Steering Committee. MOF will have a core role in providing guidance, coordination and supervision on the disbursement and expense of GEF grant, performance evaluation, summary, and promotion of project results. The project team will provide project documentation and reporting to MOF through UNDP.	With MOF being responsible for public financing, including the fund allocations to the different ministries, projects and programs financed under these ministries, there is a need to sensitize relevant staff and departments within MOF on, particularly the impacts of zero-carbon township and RE/EE/EC on the local, provincial, and national economy. This need is equally relevant for the provincial and county financial departments, as a better understanding of the economic environment impact that RE/EE/EC can have if not addressed adequately would/could raise the importance of zero-carbon township and the funds needed for it.
United Nations Development Programme (UNDP)	As the GEF Agency for the project it is overall responsible for the project oversight. UNDP will ensure project quality control to ensure that the Implementing Partner achieve the set project objective and outcomes. As part of its oversight work, UNDP will ensure that the project implementation and management follows UNDP/GEF rules and regulations. UNDP will be a core member of the Project Steering Committee. The project Implementing Partner will provide project	

Stakeholder	Engagement in the Project	Training and Capacity Building Needs
	documentation and reporting to UNDP as outlined in the project document or otherwise agreed.	
Ministry of Agriculture and Rural Areas (MARA)	MARA is the project Implementing Partner and chairs the Project Steering Committee. MARA will, through the PMO that it established within its Department of Science, Technology and Education, lead and supervise the overall implementation of the project and ensure coordination between the relevant government agencies involved in the project at both national and local levels). MARA will coordinate and supervise the use of GEF funds allocated each year by the UNDP and carry out regular reporting of achievements in accordance with UNDP rules and procedures.	Within MARA there is a need for raising the awareness of their staff particularly on the importance of the application of economically feasible RE/EE/EC and zero-carbon technologies. In addition, more general awareness on IAS needs to be raised, and trainings related to the implementation of developed/revised regulations and guidelines on RE/EE/EC technology applications will also be needed.
Rural Energy & Environment Agency (REEA), MARA	Entrusted by DSTE[32], REEA is responsible for the daily management of PMO and GEF fund management. REEA will coordinate and supervise the use of GEF funds allocated by UNDP and prepare regular reporting of achievements in accordance with UNDP rules and procedures.	The staff of REEA and in particular staff in the Project's Management Office will be participating in as many of the project developed trainings as possible to increase the PMOs knowledge but also to provide feedback on the developed training materials.
Ministry Ecology and Environment (MEE)	MEE is the focal point agency on climate change in China. It will be a core member of the project steering committee。MEE will provide advice and guidance on the implementation of the project, especially in supporting the improvement of policies, regulations, standards, and systems related to climate change in rural areas.	Within MEE there is a need for further understanding of the significance of rural renewable energy applications to China's achievement of nationally determined contributions; understand the policies and regulations of renewable energy development and strengthen support for the development of rural renewable energy.
National Development and Reform Commission (NDRC)	NDRC is an important member of PSC and will provide support for RE/EE/EC related policy research and technical improvement. NDRC will integrate relevant resources, promote the application of renewable energy and energy efficiency technologies in rural areas, and provide professional training for relevant stakeholders as needed, and support for the marketization and scale promotion of the project.	NDRC needs to know more about the situation regarding RE/EE/EC application in rural areas and survey the barriers to the development of RE/EE/EC and cooperated with MARA to revise the related policies
All-China Women's Federation (ACWF)	ACWF is a member of the project's steering committee. In addition ACWF, through its local chapters at provincial and local level will be a key project stakeholder with which the project will work together to ensure that the mainstreaming of women into the project activities and their implementation is optimized. ACWF and its local entities will, as part of the project's stakeholder engagement, be involved in	Although not specifically target towards ACWF, their representatives will be invited to a participate in the project training activities as part of the project overall awareness raising initiatives. This will also enable ACWF to provide feedback on the trainings with the aim to improve the gender aspect of said trainings.

[32] This is MARA's Department of Science Technology and Education. It will establish the EZCERTV PMO, and REEA will be responsible for the daily management of PMO on behalf of the DSTE.

Stakeholder	Engagement in the Project	Training and Capacity Building Needs
	development both the at the National and Provincial levels ,	
Provincial Department of Agriculture and rural affairs (DARA) in 8 provinces	As the focal point in each demo province, Provincial Department of Agriculture and Rural Affairs (DARA) will be responsible for supervision and guidance of project implementation in the province, assisting the government at the county level in project coordination and resource integration. Each provincial DARA will be engaged in most of the project activities at provincial and country level and will play a specific role in the Provincial Coordination Group. In addition, the staff from the DARA will take part in the project's capacity building engagements through the provisions of project developed trainings.	Provincial Department of Agriculture and rural affairs will be the target for a range of project training activities including, but not limited to, trainings on GEF project management, RE/EE/EC related policy and technology, project M&E and other requirements. Relevant staff will also receive more practical on the ground trainings
Universities and research institutions	The scientific research institutions will provide input and guidance as well as expertise to a range of the project activities. This will include support to the evaluation of energy resource in the target villages, development of the technology demo plan, guide the instruction of demoes, and replication sites. Broadly speaking the research institutions will provide important technical support for the application of RE/EE/EC, as well as assist government departments on policy research, technical promotion and training of grassroots agricultural technicians, preparation of publicity information materials.	No specific project related trainings are envisaged.
Local energy-businesses, farmers cooperatives and other enterprises in the projects targeted counties	Local energy-businesses, farmers cooperatives and other enterprises will be part of the projects direct beneficiaries as they will be the main target for the on-the-ground training on and implementation of the project promoted RE/EE/EC techniques. They will also be one of the main stakeholder groups which will be engaged in the development of the zero-carbon villages and provide the energy management services for villagers.	These entities are direct beneficiaries of the project as they will be the main target for the on-the-ground training activities. They will require training in the management, installation, operation, and maintenance of facilities/systems that apply the new low/zero carbon energy technologies.
Financial institutions	Financial institutions will be the central participants of the project particularly in the development and implementation of a green financial model for the application of zero-carbon energy technologies. They will provide investment and financing advisory services for the project sites and related energy service enterprises, and provide finance support for energy companies if possible, so as to promote the promotion of zero-carbon energy technology in rural areas.	The management staff of financial institution lack knowledge about zero-carbon energy technology applications in rural area and the performance of energy management services. Training on these areas is necessary for the staff of the financial institutions.
Civil society groups and	Where identified, the national and local NGO/CSOs will be engaged to the maximum extent possible and	Training needs not identified, but it is anticipated that training on general issues

Stakeholder	Engagement in the Project	Training and Capacity Building Needs
NGOs (CSO/NGO)	particularly in collaboration with local NGO/CSO would be sought to have them directly involved in the project's social public awareness raising activities. The local NGO/CSO including the ACWF will receive training on renewable energy and energy efficiency technology applications in rural areas to enable them to better engage and interact with the communities they serve. They are also expected to assist in the replication and scale-up of the demos that will be carried out under the project.	related to renewable energy, energy saving and the marketing of related technologies, as a minimum will be provided.
Local communities	Local farmers will be part of the project's direct beneficiaries as they will be the main target for the on-the-ground training on and implementation of the project promoted zero-carbon energy technologies/techniques. They will also be one of the main stakeholder groups that will be engaged in the local energy demand assessment and project activities monitoring efforts, also the end-user of RE /EC/EE.	Farmers and the farming communities require training on the implementation of, the project promoted zero-carbon energy technologies/techniques, as well as enhance their awareness about the application of low/zero carbon energy technologies in rural areas.

Based on the stakeholder analysis (ProDoc, p. 38) and the identified roles of each stakeholder, the following table summarizes the stakeholder plan, showing the means of engagement and engagement schedule.

Table 2: Stakeholder Plan

Project Stakeholder	Stakeholder Interest	Roles and Responsibilities in Implementation	Means of Engagement in the Project	Engagement Schedule during Implementation
GEF	GHG emission reduction	Project financing	Annual reporting	Yearly
UNDP	Sustainable Development	None. As GEF Agency, role is limited to Project oversight.	Regular consultation and follow-up meetings with implementing partner	Weekly
MARA	Rural energy industry development and rural revitalization	MARA is the project Implementing Partner and chairs the Project Steering Committee (PSC). MARA is responsible for the organization and implementation of the project .	Project activity monitoring and reporting; daily project management, coordination with project partners, QPR and PIR reporting.	Daily
NDRC	Renewable energy industry development	PSC member , Provide support on renewable energy and energy efficiency	PSC Meeting	Annual
			Implementation of Components 1 & 2	Quarterly

Project Stakeholder	Stakeholder Interest	Roles and Responsibilities in Implementation	Means of Engagement in the Project	Engagement Schedule during Implementation
		technologies and policies in rural area	activities.	
Ministry Ecology and Environment (MEE)	GHG emission reduction	PSC members provide policy support on climate change and GHG monitoring	PSC Meeting	As scheduled
All-China Women's Federation (ACWF)	Gender Mainstreaming and women economic empowerment	Member of PSC, In addition ACWF, through its local chapters at provincial and local level will be a key project stakeholder with which the project will work together to ensure that the mainstreaming of women into the project activities and their implementation is optimized.	PSC Meeting	As scheduled
Provincial Department of Agriculture and rural affairs (DARA) in 8 provinces	low-carbon transformation in for Rural energy industry, ecological livable in rural township	Will establish local PMO, and support implementation of demo and replication projects in the Province	Project activity monitoring and reporting	Quarterly
Government in county level	Comprehensive development on economic and social, rural ecology livable, and low carbon transformation	Organize the implementation of project in target county	Project activity monitoring and reporting	Quarterly
Universities and research institutions	Application models of Zero-carbon energy technology, monitoring and evaluation methods of zero-carbon township, and the best practices study	Technical Working Group, demo activity	Project activity monitoring and reporting	Quarterly
Local energy-businesses, farmers cooperatives	EE and EC promotion in rural villages, improve the energy service ability.	Demo activity, Provide technical expertise in RE/EE/EC design, installation,	Implementation of Components 2 and 3 activities.	As scheduled

Project Stakeholder	Stakeholder Interest	Roles and Responsibilities in Implementation	Means of Engagement in the Project	Engagement Schedule during Implementation
and other enterprises in the projects targeted counties		diagnoses, maintenance, and training		
Financial institutions	Shorten the investment return cycle in rural energy sector.	Assess the financial ability of energy companies, develop the Green financial mechanism. The staff of financial institutions will be trained.	Implementation of Component 3 activities.	As scheduled
Local communities	have clean and comfortable living environment and have training on RE/EE/EC	The end-user of RE/EE/EC technology will be trained.	Project activity monitoring and reporting.	Quarterly

Table 3 below shows the specific stakeholders that will be involved in the implementation of the activities that will deliver each major output of the different project components.

Table 3: Specific Stakeholders Involved in Project Activities Implementation

Components	Project Outputs	Project Stakeholders
1: Policy Formulation and Institutional Mechanisms on Zero Carbon Town and Village Development	**1.1:** Formulated, approved, and enforced policies on the promotion and support of zero carbon town and village development in line with China's rural revitalization strategy and goal of achieving carbon neutrality before 2060.	MARA, DARA, NDRC, Partner Universities and Research Institutes
	1.2: Established and operational institutional framework to promote and support zero-carbon town and village development.	MARA, DARA, MEE, Provincial & County Governments
	1.3: Formulated, approved, and enforced technical, management and operational standards in rural renewable energy (RE) development and utilization, energy storage, energy conservation and energy efficiency (EC&EE) and comprehensive energy management.	MARA, DARA, NDRC, MEE, Provincial & County Governments
	1.4: Completed assessment of renewable energy resource endowment, economic and social development level, and energy use trends and patterns in rural areas of selected provinces; and formulated energy-integrated development plans of pilot towns in selected 8 provinces.	MARA, DARA, NDRC, MEE, Provincial & County Governments, Partner Universities and Research Institutes
2: Integrated Renewable Energy and Energy	**2.1:** Completed and operational zero carbon technology demonstration involving the installation of about 167.2 MW of RE-based power generation capacity in 120 rural villages and 4 towns for energy production and utilization	DARA, Local Governments, Partner Private Sector Entities, Demo Towns/Villages, Community

Components	Project Outputs	Project Stakeholders
Efficiency Technologies Applications for Zero Carbon Town and Village Development	showcasing the cost-effective application of new RE delivery mechanisms decentralized RE-based energy generation and distribution (through mini/micro-grids) and energy storage.	Groups
	2.2: Completed and operational demonstrations of the cost-effective applications of EC&EE technologies and comprehensive energy management systems in selected energy end-use sectors in support of rural socio-economic development. Estimated annual energy savings of about 6,487.0 TJ/year in 4 towns and 120 villages.	DARA, Local Governments, Partner Private Sector Entities, Demo Towns/Villages, Community Groups
	2.3: Developed, published, and disseminated zero carbon town and village development guidelines to replicate and scale-up the impact of zero-carbon villages and towns.	MARA/DARA, Local Governments, Partner Private Sector Entities, Demo Towns/Villages, ACWF, NGOs/CSOs
3: Facilitating Rural Zero Carbon Development and Lifestyle Popularization and Knowledge Dissemination	**3.1:** Completed information, communication, and education (ICE) program on zero carbon development in rural towns and villages in China for local authorities, service companies, rural energy cooperatives, public institutions, private entrepreneurs, farmers, and the public.	MARA/DARA, Local Governments, Private Sector Entities, Demo Towns/Villages, ACWF, NGOs/CSOs
	3.2: Strengthened international cooperation and experience sharing in zero-carbon towns and villages development and green energy transformation with countries along the "Belt and Road", to support these countries in achieving their nationally determined contributions (NDC) commitments to the "Paris Agreement".	MARA/DARA, Local Governments, NDRC, Private Sector Entities, Demo Towns/Villages
	3.3: Established social mechanism for the promotion of zero-carbon town and village development and the sustainability of renewable energy industries.	MARA/DARA, NDRC, Local Governments, Private Sector Entities, Demo Towns/Villages, ACWF, NGOs/CSOs
	3.4: Established and operational market-oriented mechanism for the enhanced development and utilization of rural RE resources, EC&EE improvement, and comprehensive energy management systems for supporting sustainable socio-economic development, and creation of zero carbon towns and villages in rural China.	MARA/DARA, Local Governments, Private Sector Entities, Demo Towns/Villages, Financial Institutions, ACWFs, NGOs/CSOs
	3.5: Established and operational investment and financing mechanisms for supporting the commercial viability and operation of zero-carbon towns and villages and the development of rural renewable energy industries.	MARA/DARA, MOF, Financial Institutions, Private Sector Entities,

Table 4 below shows the list of specific stakeholders that will be involved in the implementation of the demonstration activities.

Table 4: Specific Stakeholders Involved in the Demonstration Activities Implementation

No.	Province	Demonstration	Key Implementers
1	Heilongjiang	Bundled Straws-Fired Centralized Heating System	Government: Department of Agricultural and Rural Affairs (Heilongjiang Province); People's Government of

No.	Province	Demonstration	Key Implementers
			Hailun City Private Sector: Hailun Limin Energy-saving Boiler Manufacturing Co., Ltd.
2	Ningxia Hui Autonomous Region	Biomass Fuel Pellets Production and Use for Household Heating & Cooking	Government: Department of Agricultural and Rural Affairs (Ningxia Hui Autonomous Region); People's Government of Qingtongxia City Private Sector: Ningxia Runwell Energy & Environment Engineering Co., Ltd.
3	Hebei	Biomethane Use for Clean Heating and Cooking in Rural Households	Government: Department of Agricultural and Rural Affairs (Hebei Province); People's Government of Anping County Private Sector: Hebei Jing'an Bioenergy Technology Co., Ltd.
4	Yunnan	Piped Biogas for Household Cooking and Tobacco Flue-Curing	Government: Department of Agricultural and Rural Affairs (Yunnan Province); People's Government of Tonghai County Private Sector: Yunan Fu Hui Technology Co., Ltd.
5	Beijing	Distributed Photovoltaic Power Generation and Application of a SMART Energy Management Service Platform	Government: Yanqing District People's Government of Beijing Municipality
6	Liaoning	Rural Residential Buildings Exterior Renovation	Government: Department of Agricultural and Rural Affairs (Liaoning Province); People's Government of Chaoyang County Private Sector: Chaoyang Jiajiadian Farm Co., Ltd.
7	Hubei	Distributed Biogas Generation, Supply and Utilization System	Government: Department of Agricultural and Rural Affairs (Hubei Province); People's Government of Songzi City Private Sector: Songzi Senchun Agricultural Development Co., Ltd.
8	Shanxi	Integrated Solar PV-Storage-Direct-Flexible (PSDF) Technology coupled with Rural Houses Retrofits	Government: Department of Agricultural and Rural Affairs (Shanxi Province); People's Government of Ruicheng County Private Sector: Shanxi Golden Cooperative DC Power Distribution Engineering Technology Co., Ltd.

Table 5 below shows the list of specific stakeholders that are expected to sustain/maintain and enhance the widespread use of the tools, frameworks, policies/regulations, capacity built, etc. from the EZCERTV Project, as well to facilitate further replication/scaling up of the zero/low carbon energy technologies that were demonstrated/replicated under the project. These are all expected from each stakeholder after the completion of the EZCERTV Project.

Table 5: Expected Future Roles of Stakeholders after Completion of EZCERTV Project

Stakeholder	Future Roles in Sustaining Zero Carbon Rural Development in China
Ministry of Finance (MoF)	Continue to provide consistent guidance, coordination in the financial planning of future zero/low carbon energy technology applications in the rural towns and villages of the

Stakeholder	Future Roles in Sustaining Zero Carbon Rural Development in China
	country, in line with the GOCs rural revitalization program, or other relevant rural development programs in the country by the mid-2020s.
Ministry of Agriculture and Rural Areas (MARA)	Continue in its mandate regarding agricultural development in the country, and in particular rural development but specifically promoting and supporting the enabling and enforcement of national and provincial policy, regulatory and institutional frameworks that are supportive of low/zero carbon development in rural towns and villages in the country. Its provincial units (DARAs) are also expected to support and facilitate the development, financing, and implementation of planned and future zero carbon development initiatives in the rural areas of the country.
Ministry Ecology and Environment (MEE)	As the focal point agency on climate change in China, the MEE is expected to continue supporting the promotion of the widespread applications of zero/low carbon energy technologies, as well as zero carbon rural development initiatives of the EZCERTV Project. Particularly, the MEE is expected to continue providing advice and guidance on the improvement of policies, regulations, standards, and systems related to zero carbon rural development as this impacts on climate change in rural areas.
National Development and Reform Commission (NDRC)	NDRC is expected to follow through in supporting zero/low carbon energy technology applications that were introduced, showcased, promoted, and facilitated through the EZCERTV Project through (1) relevant implementation/enforcement of RE/EE/EC related policy research and technical improvement; (2) integration of relevant resources; (3) further promotion of the application of RE/EE/EC technologies in rural areas; (4) provision of professional training for rural sector stakeholders; and (5) Support for the marketization and scale promotion of zero carbon energy technologies not only in rural areas but also in urban areas (where feasible and applicable).
Universities and research institutions	From the results of the various GEF-funded studies and GOC-funded research and development activities that will be carried out under the EZCERTV Project, the universities and research institutions that will be involved in the project are expected to follow through with other continuing education work for the rural areas particularly technical support for the application of RE/EE/EC, as well as assist government agencies on enhancing policies, technology applications, training of rural technicians, preparation of publicity information materials.
Local energy-businesses, farmers cooperatives and other enterprises in the projects targeted counties	Local energy-businesses, farmers cooperatives and other enterprises that will be actively involved in the implementation of the EZCERTV Project are all expected to follow through with their own investments in the development and implementation of zero/low carbon energy technology application projects.
Financial institutions	Financial institutions are expected to develop and implement green financing schemes for financially supporting projects on the application of zero-carbon energy technologies. They are expected to provide investment and financing advisory services for interested rural energy users (e.g., households, farms, rural industries, rural energy companies, etc.) for further promoting zero-carbon energy technology in rural areas.
Civil society groups and NGOs (CSO/NGO)	As players and beneficiaries of the EZCERTV Project the CSOs/NGOs (e.g., ACWF) are expected to follow through in social public awareness raising activities after the completion of the EZCERTV Project. They are also expected promote and campaign for the replication and scale-up of the demos that will be carried out under the project. The ACWF through its local chapters at provincial and local levels are also expected to continue ensuring gender equity is mainstreamed in all subsequent zero carbon rural development activities in the rural areas.
Local communities	Local communities are expected to continue after the completion of the EZCERTV project

Stakeholder	Future Roles in Sustaining Zero Carbon Rural Development in China
	the implementation and promotion of the application of zero-carbon energy technologies/techniques in their respective localities.
United Nations Development Programme (UNDP)	Continue to support the efforts of the GOC in the promotion and application of zero/low carbon energy technologies in rural communities, including support to zero carbon rural development and achieving China's climate change mitigation objectives and targets.

Annex 11: Gender Analysis and Gender Action Plan

(The Gender Analysis and Gender Action Plan is shown below and also submitted as a supplementary annex)

Gender equity has been mainstreamed on a number of components of the EZCERTV Project and a range of activities have been designed and planned to ensure this. Table 1 below shows the success indicators and targets for these activities and a gender action plan with specific activities is listed in Table 2.

Based on the initial surveys and assessment of potential gender issues of the project (as mentioned in the SESP), the EZCERTV Project as a whole does not have too much of a negative impact on gender equity, and in some ways has a positive effect. The potential negative impacts include potential occurrence of discriminations against women based on gender, especially regarding participation in the design and implementation of the project activities or access to opportunities and benefits. Another is the uncertainty about women's labor inputs (e.g., demonstration activities such as those on the application of technologies utilizing biomass fuels, biogas, etc.), might potentially increase women's labor inputs for household energy access. As a key measure to mitigate these impacts, the implementation of the EZCERTV Project needs to ensure capacity building for gender equality, knowledge sharing, broad participation, and leadership of women in the process, and enhanced monitoring, reporting, and tracking of the activities that have potential influence on, and impacts on, gender equity.

Since the implementation of zero-carbon energy technology application demonstrations in selected rural villages and towns usually inevitably involves substitution of energy supply resources and forms (i.e., applications of RE and EE technologies), the process can have a positive impact as well. On the one hand, the substitution of traditional energy sources can reduce the labor input for traditional biomass energy collection such as straw and fuelwood, which has largely involved women; on the other hand, the construction and operation of new energy supply systems (e.g., piped biogas/biomethane, pelletized straw fuel) creates many new jobs. These jobs also have a positive impact on women. But this process requires professional employability development interventions to ensure that women also benefit from such opportunities.

The training, capacity building, and access to employment interventions will improve the number of qualified women in rural China. The monitoring and evaluation of the project activities will include tracking a number of human development indicators. Among these is on gender equity. For example, the number of women trained, participating, and benefiting from new employment during the course of the EZCERTV Project implementation.

Table 1: Gender-related Indicators and Targets

Indicators	Means of Verification
Number of staff (**and 30% women)** from national and local governments with enhanced capacity in zero carbon development, such as in low/zero carbon energy technology project planning, procurement, financial management, community development, and safeguards.	Survey of MARA and provincial DARAs that are involved in the project.
Number of local project developers, equipment suppliers and vendors **(30% females)** trained on the design, installation, operation, and business models for the applications of RE and EE technologies.	Documentation on the training program including the profiles of the trainees; Documentation on the training course

Indicators	Means of Verification
	implementation including post training evaluation.
Number of village level promoters, technicians, administrators trained on relevant subjects for successful management and operation of the RE and EE technology systems **(The female participation rate for training and publicity at the project implementation site is not less than 40%)**	Survey of town and village technicians and administrators. Documentation on the training program including the profiles of the trainees; Documentation on the training course implementation including post training evaluation.
Number of trained people that are applying the knowledge and skills they learned from the training in their current jobs. Number of trained people that are gainfully employed utilizing the knowledge and skills they learned from the training in newly created jobs. (The percentage of women in both aspects is **at least 30%**).	Survey of trainees (survey conducted a year after the training program)
Number of new employment opportunities created by the applications of RE and EE technologies to bring about zero carbon rural development (**with 40% of women**)	Survey of employment and workforce in each participating provinces
Number of households (**and number of female-headed HH**) provided with RE and EE technologies services during the EZCERTV Project implementation period.	EZCERTV Project activity reports and monitoring reports
Number of households (and number of female-headed HH) that have applied new RE and/or EE technology applications on their own.	Household energy surveys in each participating provinces.

The targets for the above stated indicators will be set during the inception phase of the EZCERTV Project. The implementing partner (Ministry of Agriculture and Rural Affairs) will carry out surveys to monitor the achievement of the set targets. The surveys will cover energy project developers, equipment suppliers, low/zero carbon development project promoters, technicians, and administrators, as well as the energy consumers such as the rural households. The end-user awareness data shall be collected through surveys and focus group discussions at the beginning and final year of the project. Surveys will include aspects of gender equity and gender equity mainstreaming in all aspects of low/zero carbon town and village development in rural China. The surveys will also enable timely detection of any gender-specific issues cropping up so that they are addressed promptly.

Table 2: EZCERTV Gender Action Plan

Principles of Gender Equity	Gender specific activities	Indicators to be included in M&E	Timeline	Possible Output/Impact
Capacity building, knowledge sharing and exchange	Ensure women representation in all capacity building workshops and trainings, as well as at any working group or committee for project implementation and monitoring. Identify design features that impact positively on women and implement them and remove those that negatively impact the project.	Number of staff (and % women) from national and local governments with enhanced capacity in zero carbon development, such as in low/zero carbon energy technology project planning, procurement, financial management, community development, and safeguards.	Year 1,2	A well represented governing structure that involves all groups of society is in place and each participate meaningfully in the implementation of the EZCERTV Project. This also applies in the design and implementation of the projects on the replication and/or scale-up of the low/zero carbon energy technology application demonstrations. All bottlenecks that prevent the full and meaningful participation of women are identified and removed.
	Support women and girls (or children) to voice issues and concerns about decisions and process of the project that affect their lives.	The female participation rate for training, publicity, and consultation	Year 1,2,3	Concerns of vulnerable groups, including women and girls, are addressed and their voices are integrated through participatory approaches during the implementation of the project activities. This also applies all future low/zero carbon rural town and village development projects and programs in the country.
	Ensure access to loan financing by women entrepreneurs and vulnerable groups.	Number of local project developers, equipment suppliers and vendors (% females) trained on the design, installation, operation, and business models for the applications of RE and EE technologies.	Year 1,2	All vulnerable groups' needs, including women's, that are incorporated into the project will be ensured at all the phases of the project implementation to ensure economic sustainability.
	Creation of a safe and inclusive environment for the vulnerable groups and women to enhance their participation.	Whether the training materials and project documents are developed with inclusive language.	Year 1,2	Significant women participation and representation are encouraged during capacity building trainings or workshops. Widely available training materials and project documents including written text as well as audio and visual developed with

Principles of Gender Equity	Gender specific activities	Indicators to be included in M&E	Timeline	Possible Output/Impact
				inclusive language (e.g. ethnic minority language) and appropriate illustrations without gender bias and stereotype.
	Improve information and knowledge access such as operation of RE/EE systems, financial services, market information, agricultural information, and techniques for the community (both males and females) by mobile, internet, and TV.	Number of trained people that are applying the knowledge and skills they learned from the training in their current jobs. Number of trained people that are gainfully employed utilizing the knowledge and skills they learned from the training in newly created jobs. Number of households (and number of female-headed HH) that have applied new RE and/or EE technology applications on their own.	Year 1,2,3,4,5	Wider dissemination of information and knowledge related to renewable energy technologies and zero carbon development in rural China
Gender-balanced participation and female leadership	Facilitation of full representation of vulnerable groups including women, and their meaningful participation in projects and programs. Ensure adequate female representation is invited for surveys, focus groups and any training and workshops for the project. Both the project expert group and the project execution group contain a certain number of female members.	Number of staff (and % women) from national and local governments with enhanced capacity in zero carbon development, such as in low/zero carbon energy technology project planning, procurement, financial management, community development, and safeguards. The female participation rate for training, publicity, and consultation.	Year 1,2,3	This ensures that a well-represented expert and managing group is in place, as well as broad participation of vulnerable groups throughout project implementation.
	Supporting women to save their time in participating in other productive activities such as collection of traditional energy sources.	Time spent by women participating in energy-related household activities. Number of households (and number of female-headed HH)	Year 3,4,5	For women who will be involved in the project implementation, their time commitment to household chores may be reduced. Women and children's income and well-

Principles of Gender Equity	Gender specific activities	Indicators to be included in M&E	Timeline	Possible Output/Impact
	Ensuring women not add more time and efforts in doing household chores with RE or EE substitutions.	that have applied new RE and/or EE technology applications on their own.		being improve due to new employment opportunities
	Improving women's access to new jobs created by RE or EE services. Ensure the share of female labor force in new jobs created as a result of changing energy supply patterns during the construction of Zero Carbon Energy in Rural Towns and Villages	Number of new employment opportunities created by the applications of RE and EE technologies to bring about zero carbon rural development (with % of women)	Year 2,3,4,5	Women's time commitment to household chores may be reduced. Women and children's income and well-being improve due to new employment opportunities. The energy sector has sufficient women working in it
	Increasing the number of women receiving technical training and their participation in projects Strengthening the technical capacity of women and girls for RE/EE by providing training on maintenance and repair. Creation, or facilitation of the creation, of opportunities to empower and capacitate women including training needs are identified and tailor-made to suit projects being implemented. Ensure equitable allocation of resources for the implementation of appropriate implementation measures.	Number of village level promoters, technicians, administrators, end-users trained on relevant subjects for successful management and operation of the RE and EE technology systems (The female participation rate for training and publicity at the project implementation site)	Year 1,2	Both men and women are familiar with and skilled in the operation and application of the RE/EE services designed for the project. Broad participation of vulnerable groups throughout project implementation.
Gender Responsive Policy Coherence	Strengthening sensitization of project stakeholders with regards to gender equality. Invest in developing competency and knowledge on gender mainstreaming for Government's counterparts, community leaders	Whether gender equality mainstreamed in project activities, in particular in capacity-building training and workshops	Year 1,2,3	Gender responsive policies are consistent and gender considerations are treated equally in all modules of the project and for all stakeholders. Targeted and trained stakeholders are well oriented on gender issues and more confident to mainstream gender throughout

Principles of Gender Equity	Gender specific activities	Indicators to be included in M&E	Timeline	Possible Output/Impact
	and partner agency as well as UNDP's project staff. Integration of gender session in each capacity-building training or workshop.			the whole project circle.
	Strengthening the regulation and supervision role of women's organizations in the project design and implementation processes. Invite representatives of women's organizations to participate in surveys and workshops related to project design and integrate their feedback into the project document. Strengthen the catalytic role of women's organizations in mobilizing women and girls in training such as RE/EE training.	Number of times women's organizations have been involved in project-related survey, consulting, training, workshop, etc.	Year 1,2,3	Significant women participation and representation are encouraged during capacity building trainings or workshops. In particular, vulnerable groups are identified and involved in project activities.
Monitoring, tracking, and reporting of policy implementation	Designation and appointment of project staff to implement and monitor the gender mainstreaming activities accordingly and their performance appraisal is evaluated by accomplishment on the assigned gender tasks.		Year 1,2,3,4,5	
	Facilitating appropriate adjustments to activities and implementation approaches and accountability to gender equality.		Year 1,2,3,4,5	

Annex 12: Procurement Plan

The Procurement Plan can be accessed Here and also submitted as a supplementary annex

Annex 13A Description of Demonstration

It can be accessed Here and submitted as a supplementary annex

Annex 13B – GHG Emission Reduction Estimates

It can be accessed Here and submitted as a supplementary annex

Annex 14: Co-Financing Letters

It can be accessed Here and submitted as a supplementary annex

Annex 15: GEF Checklist for Audit Risk Review

It can be accessed Here and submitted as a supplementary annex